

Wardle Editorial Operating System (WEOS)

Part I — Vision & Editorial Philosophy

1. Purpose

The Wardle Editorial Operating System (WEOS) is the clinical editorial foundation that governs the creation, review, publication, and continuous improvement of diagnostic knowledge within Wardle.

WEOS exists to ensure that every educational artifact exposed to learners—from a single clue in a clinical case to a complex network of diagnostic relationships—is clinically defensible, educationally intentional, internally consistent, and governed through transparent editorial processes.

Wardle is not a content management system.

Wardle is not an AI writing platform.

Wardle is not a collection of isolated clinical cases.

Wardle is an operating system for governing diagnostic knowledge.

Every diagnosis, case, teaching objective, clinical pearl, differential diagnosis, reasoning pathway, and knowledge relationship forms part of a living editorial ecosystem that evolves through continuous review rather than one-time publication.

The purpose of WEOS is therefore not merely to store medical content, but to provide the structures, workflows, and governance required to build and maintain a trustworthy body of clinical knowledge at institutional scale.

2. Editorial Philosophy

Medical education is fundamentally different from conventional publishing.

Clinical knowledge cannot be considered complete simply because it has been written, nor can educational quality be measured by the quantity of available content.

Every published statement influences how future clinicians recognise disease, interpret evidence, construct differential diagnoses, and make decisions that ultimately affect patient care.

For this reason, WEOS treats every editorial artifact as a clinical decision rather than a document.

A teaching objective represents an educational commitment.

A diagnostic clue represents an instructional decision.

A knowledge relationship represents a clinical assertion.

A clinical pearl represents an expert recommendation.

Publication therefore signifies institutional confidence in the educational validity of these decisions rather than completion of a writing task.

Editorial work is consequently viewed as an ongoing process of refinement.

Knowledge is expected to evolve.

Evidence changes.

Clinical practice develops.

Educational priorities shift.

The editorial system must therefore remain capable of continuously improving existing knowledge while preserving accountability for every published decision.

3. Core Editorial Principles

The design of WEOS is guided by several enduring principles.

3.1 Knowledge Before Content

WEOS governs knowledge rather than documents.

Individual pieces of content exist only to communicate clinical knowledge.

Knowledge therefore remains the primary asset of the system, while cases, explanations, objectives, and educational materials are different expressions of that knowledge.

3.2 Editorial Decisions Must Be Explicit

Every meaningful change to educational knowledge should correspond to an identifiable editorial decision.

The system should never silently transform educational content.

Editors should always understand:

- what changed
- why it changed
- who approved it
- what evidence supports it
- what educational impact the change produces

Editorial governance depends upon transparent decision-making rather than hidden automation.

3.3 Clinical Accuracy Is Necessary but Not Sufficient

Correct medical facts alone do not produce effective education.

Every artifact should also be evaluated according to its educational quality.

WEOS therefore governs:

- clinical correctness
- diagnostic reasoning
- educational usefulness
- learner progression
- curriculum alignment
- consistency with surrounding knowledge

Publication requires all of these dimensions to be considered together.

3.4 Every Artifact Has a Purpose

No editorial artifact exists in isolation.

Every artifact should answer a clearly defined educational question.

Examples include:

- Why should this teaching objective exist?
- Which misconception does this clinical pearl address?
- Which diagnostic uncertainty does this graph relationship resolve?
- Which reasoning step does this clue teach?
- Which learning outcome does this case support?

Artifacts without educational purpose create complexity without improving learning.

3.5 Human Judgment Governs Clinical Knowledge

Artificial intelligence can assist editorial work but cannot replace clinical responsibility.

The responsibility for educational quality, clinical accuracy, and publication always remains with qualified human reviewers.

AI may recommend.

AI may draft.

AI may analyse.

AI may identify inconsistencies.

Only human editorial governance establishes institutional trust.

3.6 Continuous Improvement Over Static Publication

Publication is not the end of editorial work.

Published knowledge should continuously evolve as:

- evidence changes
- guidelines change
- learner performance identifies weaknesses
- reviewers improve explanations
- new diagnostic relationships emerge
- curriculum priorities evolve

Every published artifact should therefore remain reviewable throughout its lifetime.

3.7 Editorial Consistency Is a System Property

Consistency should not depend upon individual reviewers remembering previous decisions.

The editorial system itself should promote consistency by exposing:

- conflicting knowledge
- duplicated teaching
- inconsistent terminology
- contradictory reasoning

- overlapping educational objectives
- missing educational coverage

Editorial quality should emerge from the system as much as from individual expertise.

4. Editorial Operating Model

WEOS views editorial work as a governed progression of clinical knowledge.

Rather than treating publication as a single event, knowledge moves through a sequence of increasingly trustworthy states.

Knowledge begins as an idea.

It develops into structured educational content.

It undergoes expert review.

It becomes institutional knowledge.

It continues to evolve throughout its published life.

The operating model therefore emphasises progression rather than completion.

Every editorial activity contributes to increasing confidence in the quality, educational value, and clinical reliability of knowledge.

This progression applies equally to diagnoses, cases, teaching objectives, clinical pearls, reasoning pathways, educational resources, and future editorial artifacts.

5. Scope of the Editorial System

WEOS governs every educational artifact that contributes to diagnostic learning.

These include, but are not limited to:

- Diagnoses
- Clinical cases
- Diagnostic clues
- Teaching objectives
- Teaching rules
- Clinical pearls

- Clinical pitfalls
- Differential diagnoses
- Diagnostic reasoning pathways
- Knowledge relationships
- Educational summaries
- Investigation guidance
- Management guidance
- Recall prompts
- Scoring systems
- References
- AI-generated editorial drafts
- Editorial reviews
- Publication decisions

The operating system intentionally separates the governance of these artifacts from the technology used to store or present them.

The editorial model should remain valid regardless of future implementation details.

6. Definition of Editorial Success

WEOS does not measure success by the number of published cases.

Instead, editorial success is defined by the quality of the governed knowledge.

An effective editorial system should consistently produce knowledge that is:

- Clinically accurate
- Educationally purposeful
- Internally consistent
- Evidence-informed
- Transparently governed
- Continuously improvable
- Explainable to reviewers
- Trustworthy to learners
- Scalable across specialties
- Sustainable for long-term maintenance

These characteristics form the standard against which every future capability within WEOS should be evaluated.

Whenever architectural decisions are made, the preferred solution should be the one that most effectively strengthens these characteristics.

Closing Principle

Wardle does not manage medical content.

Wardle governs clinical knowledge.

Every architectural decision within the Wardle Editorial Operating System should reinforce that principle.

Part II — The Editorial Domain Model

1. Purpose

The Editorial Domain Model defines the knowledge entities governed by the Wardle Editorial Operating System (WEOS).

These entities are not implementation objects, database tables, or software components. They are editorial concepts that represent how clinical knowledge is created, evaluated, maintained, and published.

Every capability within WEOS operates upon one or more of these entities.

The domain model establishes:

- what exists within the editorial system
- why it exists
- how entities relate to one another
- how knowledge progresses through editorial governance
- where editorial decisions occur

All future implementations should preserve these concepts regardless of the underlying technology.

2. Editorial Philosophy of the Domain

Medical knowledge is not published as isolated documents.

Instead, it exists as an interconnected network of educational artifacts that collectively teach diagnostic reasoning.

A diagnosis is not merely accompanied by a case.

It is explained through objectives.

It is reinforced through teaching rules.

It is clarified through clinical pearls.

It is distinguished through differential diagnoses.

It is connected through knowledge relationships.

It is evaluated through evidence.

It is continuously improved through editorial review.

WEOS therefore models editorial knowledge as an ecosystem rather than a hierarchy of documents.

3. Editorial Artifact Hierarchy

The editorial system governs a collection of related artifacts.

These artifacts exist at different levels of abstraction.

```
Clinical Knowledge
|
|— Diagnosis
|   |— Metadata
|   |— Educational Framework
|   |— Clinical Cases
|   |— Knowledge Relationships
|   |— Educational Content
|   |— Evidence
|   |— Reviews
|   └— Publication
|
└— Editorial Governance
```

The Diagnosis is the primary unit of editorial governance.

All other artifacts either describe, teach, validate, or govern the diagnosis.

4. Primary Editorial Entities

4.1 Diagnosis

The Diagnosis is the central knowledge entity within WEOS.

Every editorial activity ultimately contributes to improving one or more diagnoses.

A diagnosis defines the educational identity around which all associated knowledge is organised.

A diagnosis is responsible for:

- clinical identity
- taxonomy
- metadata
- educational scope
- publication status
- knowledge relationships
- associated educational artifacts

A diagnosis is never considered "finished."

It continuously evolves throughout its editorial lifetime.

4.2 Clinical Case

Clinical cases are educational expressions of a diagnosis.

Their purpose is not to document patient encounters but to teach diagnostic reasoning.

A case should progressively guide learners toward a diagnosis while reinforcing important discriminators, avoiding premature diagnostic disclosure, and supporting educational objectives.

Cases exist to teach.

They are not the diagnosis itself.

Multiple cases may exist for the same diagnosis.

Different cases may target different learner levels, clinical settings, or reasoning challenges while contributing to the same body of governed knowledge.

4.3 Educational Framework

Every diagnosis possesses an educational framework that defines what learners should understand after studying the diagnosis.

The educational framework includes:

- teaching objectives
- teaching rules
- clinical pearls
- clinical pitfalls
- recall prompts
- key concepts
- educational priorities

This framework defines the educational intent of the diagnosis before individual cases are written.

Cases should reinforce the educational framework rather than create it.

4.4 Diagnostic Knowledge Relationships

Clinical knowledge does not exist independently.

Diagnoses relate to one another through clinically meaningful relationships.

Examples include:

- differential diagnoses
- diagnostic mimics
- discriminating features
- supporting relationships
- prerequisite concepts
- shared mechanisms
- complication relationships

These relationships collectively form the diagnostic knowledge graph.

The knowledge graph is an editorial artifact requiring continuous governance rather than an automatically generated structure.

4.5 Educational Content

Educational content explains the diagnosis outside the context of gameplay.

It transforms diagnostic reasoning into structured learning.

Examples include:

- clinical summaries
- diagnostic reasoning explanations
- investigation guidance
- management principles
- examination findings
- pathophysiology
- complications
- references

Educational content should remain consistent with both clinical evidence and the educational framework.

4.6 Evidence

Evidence supports editorial confidence.

Evidence does not exist merely to satisfy citation requirements.

Its purpose is to justify educational decisions.

Evidence may support:

- clinical statements
- diagnostic relationships
- teaching rules
- management recommendations
- investigation guidance
- educational claims

The absence of supporting evidence does not necessarily imply incorrectness but always reduces editorial confidence.

4.7 Editorial Review

Editorial review transforms authored content into governed knowledge.

Review is a continuous process rather than a single approval event.

Editorial review evaluates:

- clinical correctness

- educational quality
- reasoning integrity
- internal consistency
- evidence sufficiency
- learner impact

Every review produces explicit editorial decisions that contribute to the governance history of an artifact.

4.8 Publication

Publication represents institutional confidence in an editorial artifact.

Publication is not merely a visibility state.

It signifies that the artifact has satisfied the required editorial standards for learner exposure.

Published knowledge remains subject to future review, improvement, and retirement.

5. Relationships Between Entities

The domain model intentionally separates educational concepts while recognising that they influence one another.

Examples include:

- Cases teach Diagnoses.
- Objectives define what Cases should teach.
- Teaching Rules reinforce Objectives.
- Clinical Pearls enrich understanding of Diagnoses.
- Differential Relationships distinguish Diagnoses.
- Evidence supports Educational Content.
- Reviews govern every artifact.
- Publication exposes governed knowledge to learners.

No artifact exists independently.

Every editorial artifact contributes to a larger network of diagnostic understanding.

6. Editorial Dependencies

Certain editorial entities depend upon others before meaningful work can begin.

For example:

A diagnosis requires metadata before educational generation.

Educational objectives should exist before comprehensive case authoring.

Teaching rules should exist before evaluating educational coverage.

Knowledge relationships should inform differential reasoning.

Cases should reinforce established educational intent.

Publication depends upon satisfactory governance of all required artifacts.

These dependencies define the logical progression of editorial work while remaining flexible enough to support iterative refinement.

7. Editorial Identity

Every editorial artifact possesses an independent identity.

An artifact may evolve through multiple revisions while remaining the same conceptual entity.

For example:

A clinical pearl may undergo numerous revisions.

A teaching objective may be rewritten.

A diagnostic relationship may gain stronger supporting evidence.

These changes improve the artifact without replacing its identity.

Editorial governance therefore focuses upon preserving the continuity of knowledge while allowing continuous refinement.

8. Artifact Independence

Editorial artifacts are intentionally modular.

A clinical pearl should not require rewriting a case.

Updating investigation guidance should not invalidate teaching objectives.

Improving a differential relationship should automatically strengthen future educational content without requiring manual duplication.

This modularity enables scalable editorial maintenance while preserving consistency throughout the knowledge ecosystem.

9. Domain Model Principles

The Editorial Domain Model is governed by the following principles.

Every diagnosis is the centre of a governed body of knowledge.

Cases are educational expressions rather than isolated documents.

Educational intent precedes educational delivery.

Knowledge relationships are first-class editorial artifacts.

Evidence strengthens editorial confidence.

Review governs all knowledge.

Publication represents institutional trust.

Knowledge remains continuously evolvable throughout its lifetime.

These principles define the conceptual structure upon which every subsequent component of the Wardle Editorial Operating System is built.

Closing Principle

The Editorial Domain Model describes what WEOS governs.

Subsequent sections describe how that knowledge is created, reviewed, improved, and entrusted to learners.

Part III — Editorial Capabilities

1. Purpose

The Editorial Domain Model defines what exists within the Wardle Editorial Operating System.

Editorial Capabilities define what the system must enable editors to accomplish.

Capabilities describe enduring editorial functions rather than software features.

They remain valid regardless of changes in user interface, implementation technology, or workflow design.

Every capability represents a core responsibility of WEOS in governing clinical knowledge.

No implementation should exist without supporting at least one editorial capability.

2. Capability Philosophy

WEOS is not organised around pages or administrative tools.

It is organised around editorial work.

Editors do not think in terms of viewing database entities.

Editors think in terms of completing meaningful editorial tasks.

Examples include:

- Create a diagnosis.
- Review a clinical case.
- Improve diagnostic reasoning.
- Approve a teaching objective.
- Validate a graph relationship.
- Publish educational knowledge.

Editorial capabilities therefore describe the work performed by clinicians rather than the software used to perform it.

3. Capability Categories

WEOS capabilities are organised into six major domains.

These domains collectively describe the complete lifecycle of clinical knowledge.

Knowledge Creation

↓

Knowledge Development

↓

Editorial Review

↓

Knowledge Governance

↓

Publication

↓

Continuous Improvement

Each domain contains multiple editorial capabilities.

4. Knowledge Creation

Knowledge creation establishes new editorial artifacts.

Creation may occur through human authoring, AI-assisted generation, or structured editorial workflows.

Knowledge creation does not imply publication.

Every newly created artifact enters editorial governance.

Core capabilities include:

- Diagnosis registration
- Metadata authoring
- Educational scaffold generation
- Clinical case authoring
- Educational content authoring
- Knowledge relationship creation
- Reference acquisition
- AI-assisted draft generation

The purpose of creation is to establish candidate knowledge for editorial evaluation.

5. Knowledge Development

Development improves existing knowledge.

Unlike creation, development assumes that an editorial artifact already exists.

Development capabilities include:

- Revising educational content
- Improving clinical reasoning
- Refining clue progression
- Expanding differential diagnoses
- Improving teaching objectives
- Updating teaching rules
- Strengthening clinical pearls
- Revising management guidance
- Expanding investigation guidance
- Improving educational consistency

Knowledge development is expected to continue throughout the lifetime of every artifact.

6. Editorial Review

Editorial review evaluates whether an artifact satisfies clinical and educational expectations.

Review does not create knowledge.

Review establishes confidence.

Review capabilities include:

- Clinical correctness review
- Educational quality review
- Diagnostic reasoning review
- Knowledge graph review
- Differential review
- Teaching objective review
- Teaching rule review
- Clinical pearl review
- Evidence review
- AI draft review
- Publication readiness review

Every review produces an explicit editorial decision.

Review never silently modifies knowledge.

7. Knowledge Governance

Governance maintains the integrity of the editorial ecosystem.

Governance differs from review.

Review evaluates individual artifacts.

Governance evaluates the relationships between artifacts.

Governance capabilities include:

- Detecting contradictory knowledge
- Identifying duplicate teaching
- Maintaining terminology consistency
- Managing knowledge relationships
- Validating curriculum coverage
- Monitoring evidence sufficiency
- Managing publication dependencies
- Maintaining editorial standards
- Recording governance decisions
- Managing lifecycle transitions

Governance ensures that independently reviewed artifacts collectively form a coherent educational system.

8. Publication

Publication transforms governed knowledge into learner-facing knowledge.

Publication capabilities include:

- Publication readiness assessment
- Dependency validation
- Final editorial approval
- Version publication
- Publication scheduling
- Learner release
- Publication withdrawal
- Publication supersession
- Publication history

Publication represents institutional trust rather than technical deployment.

9. Continuous Improvement

Clinical knowledge is expected to evolve.

Continuous improvement capabilities ensure that publication is never considered the end of editorial work.

These capabilities include:

- Evidence refresh
- Guideline monitoring
- Educational quality improvement
- Learner feedback integration
- Curriculum evolution
- Diagnostic graph refinement
- AI-assisted quality analysis
- Editorial analytics
- Knowledge retirement
- Knowledge replacement

Continuous improvement transforms WEOS from a publishing system into a living knowledge system.

10. Artificial Intelligence Capabilities

Artificial intelligence functions as an editorial assistant rather than an editorial authority.

AI capabilities include:

Knowledge Generation

Generating candidate educational artifacts.

Knowledge Analysis

Detecting inconsistencies, omissions, duplication, and opportunities for improvement.

Knowledge Recommendation

Suggesting educational improvements without modifying governed knowledge.

Knowledge Quality Assessment

Providing confidence estimates and quality projections for editorial consideration.

Editorial Assistance

Supporting editors during authoring, review, and governance activities.

AI capabilities always operate within human editorial governance.

AI contributes recommendations.

Editors make decisions.

11. Editorial Capability Dependencies

Capabilities are interconnected.

Certain capabilities depend upon the successful completion of others.

For example:

A clinical case cannot be meaningfully reviewed before it exists.

Publication cannot occur before governance.

Governance depends upon review.

Review depends upon authored knowledge.

Knowledge creation depends upon diagnosis registration.

These dependencies define the progression of editorial work while allowing iterative refinement where appropriate.

12. Capability Principles

Every editorial capability should satisfy the following principles.

Capability Independence

Capabilities should remain conceptually independent even when sharing implementation.

Capability Composability

Capabilities should combine to form larger editorial workflows without duplication.

Capability Explainability

Editors should understand why each capability exists.

Capability Auditability

Every meaningful editorial action should produce an auditable outcome.

Capability Extensibility

New capabilities should integrate naturally into the editorial operating model without requiring architectural redesign.

13. Capability Maturity

Capabilities mature over time.

Within WEOS, every capability progresses through four stages.

Foundational

The capability exists conceptually.

Operational

Editors can actively perform the capability.

Governed

The capability participates fully in editorial governance.

Intelligent

The capability proactively assists editors through recommendations, dependency analysis, and quality assessment.

The long-term objective of WEOS is not merely operational capability.

It is editorial intelligence.

14. The Capability Architecture

Editorial capabilities operate upon the Editorial Domain Model to continuously improve governed clinical knowledge.

Editorial Domain

↓

Editorial Capability

↓

Editorial Decision

↓

Governed Knowledge

↓

Learner Experience

↓

Editorial Feedback

↓

Knowledge Improvement

This progression represents the operational heart of WEOS.

Closing Principle

The purpose of WEOS is not to provide software features.

Its purpose is to enable the continuous creation, governance, publication, and improvement of trustworthy clinical knowledge.

Every implementation decision should strengthen one or more editorial capabilities.

Capabilities—not user interface components—define the enduring architecture of the Wardle Editorial Operating System.

Part IV — Knowledge Evolution & Editorial Lifecycles

1. Purpose

Clinical knowledge is not static.

Medical evidence evolves.

Educational priorities change.

Diagnostic reasoning improves.

Clinical guidelines are revised.

New misconceptions emerge.

The purpose of WEOS is therefore not simply to publish educational artifacts, but to continuously govern their evolution throughout their lifetime.

Every editorial artifact managed by WEOS progresses through a defined lifecycle.

These lifecycles establish how knowledge is created, improved, reviewed, trusted, maintained, and, when necessary, retired.

A lifecycle is not merely a workflow.

It represents the changing degree of editorial confidence placed in an artifact over time.

2. Knowledge Evolution Philosophy

Knowledge should never be considered complete.

Publication is not the conclusion of editorial work.

Rather, publication represents the point at which institutional confidence is sufficient for learner exposure.

After publication, knowledge remains subject to:

- evidence updates
- educational improvement
- curriculum changes
- reviewer feedback
- learner performance
- AI-assisted analysis
- clinical practice changes

WEOS therefore governs living knowledge rather than static publications.

3. Universal Editorial Lifecycle

Every governed editorial artifact follows the same conceptual progression.

Concept

↓

Draft

↓

Editorial Review

↓

Governance

↓

Approved

↓

Published

↓

Continuous Improvement

↓

Retired or Superseded

Not every artifact requires every stage.

However, every artifact progresses by increasing editorial confidence rather than increasing completeness.

4. Editorial Confidence

The purpose of a lifecycle is to represent confidence.

Each transition increases confidence through explicit editorial decisions.

Confidence is established by evaluating:

- clinical correctness
- educational quality
- evidence sufficiency
- internal consistency
- curriculum suitability
- governance requirements

Confidence is never inferred from artifact age or publication history.

It is earned through review.

5. Diagnosis Lifecycle

The diagnosis is the highest-level governed entity within WEOS.

Its lifecycle coordinates every subordinate editorial artifact.

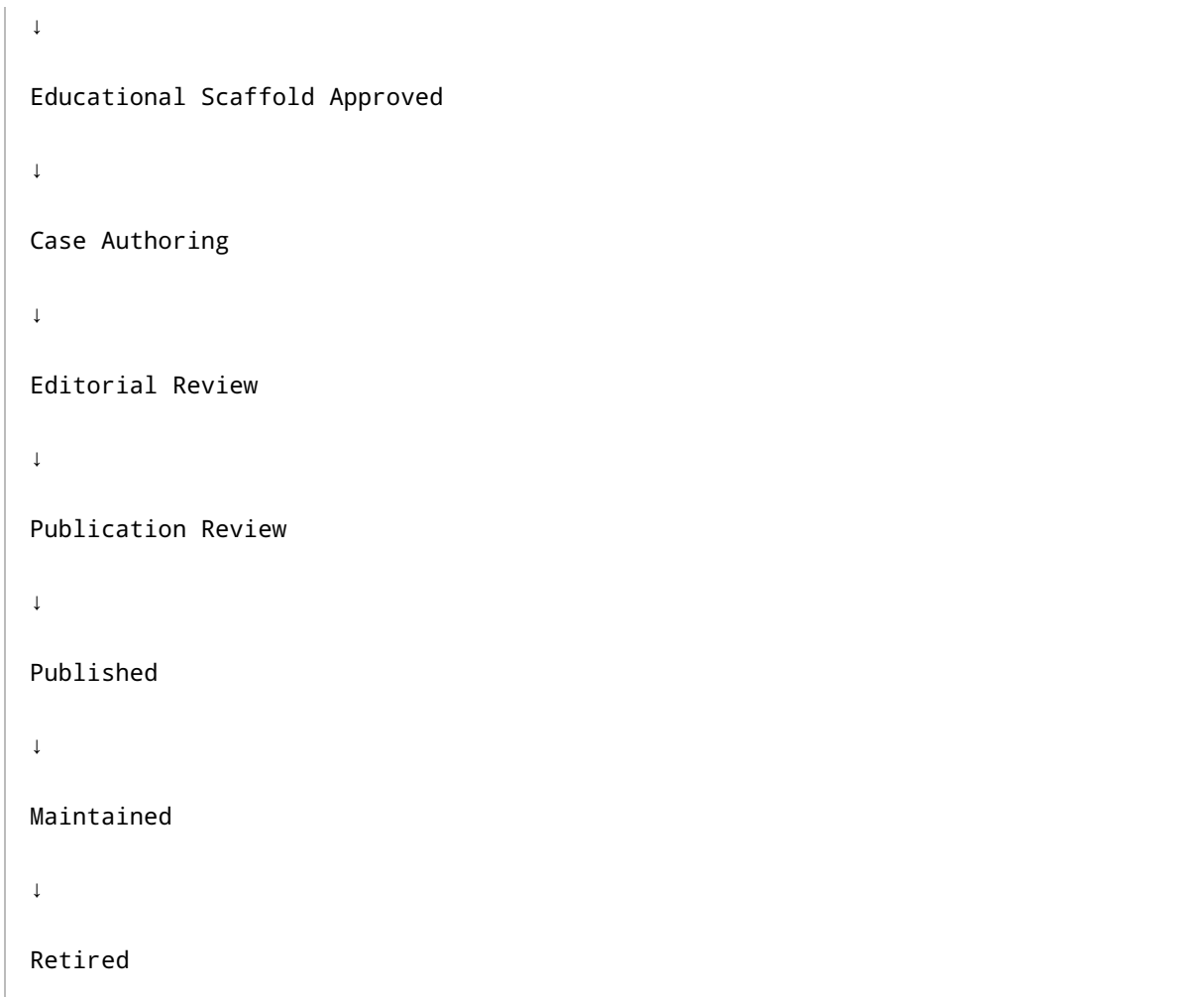
Registry Entry

↓

Metadata Complete

↓

Educational Scaffold Required



A diagnosis may remain in any stage indefinitely until sufficient editorial confidence is achieved.

Publication never bypasses governance.

6. Educational Scaffold Lifecycle

Before comprehensive educational content can exist, every diagnosis requires an educational scaffold.

The scaffold defines the educational intent of the diagnosis.

The scaffold typically includes:

- learning objectives
- teaching rules
- differential framework
- diagnostic discriminators

- educational priorities
- clinical pearls
- pitfalls

Lifecycle:

Not Started

↓

AI or Human Draft

↓

Editorial Review

↓

Approved

↓

Available for Case Development

↓

Maintained

The scaffold precedes large-scale case generation because it establishes what future cases are intended to teach.

7. Clinical Case Lifecycle

Clinical cases represent educational implementations of the scaffold.

Their lifecycle reflects progressive refinement of diagnostic reasoning.

Draft

↓

Clue Review



A case may remain approved without being published if broader diagnosis-level publication requirements have not yet been satisfied.

8. Knowledge Relationship Lifecycle

Knowledge relationships evolve independently of cases.

Relationships include:

- differential diagnoses
- discriminators
- supporting relationships
- prerequisite concepts
- related diagnoses

Lifecycle:

Suggested

↓

Evidence Assessment

↓

Editorial Review

↓

Approved

↓

Knowledge Graph

↓

Continuous Validation

Relationships may be strengthened, weakened, or removed as evidence evolves.

9. Educational Content Lifecycle

Educational content supports the diagnosis outside gameplay.

Lifecycle:

Draft

↓

Review

↓

Approved

↓

Published

↓

Evidence Refresh

↓

Revision

↓

Republished

Educational revisions should preserve historical accountability while continuously improving learner understanding.

10. AI Draft Lifecycle

Artificial intelligence generates candidate knowledge rather than governed knowledge.

AI-generated artifacts never bypass editorial review.

Lifecycle:

Generation

↓

Pending Human Review

↓

Approved

or

Rejected

or

Returned for Revision

↓

Governed Artifact

Approval transforms an AI draft into an editorial artifact.

The draft itself remains part of the editorial history.

11. Editorial Review Lifecycle

Review itself is a governed process.

Lifecycle:

Review Requested

↓

In Progress

↓

Evidence Requested

↓

Decision Recorded

↓

Closed

↓

Reopened (if necessary)

Editorial reviews are historical records rather than transient activities.

12. Publication Lifecycle

Publication represents institutional trust.

Lifecycle:

Publication Assessment

↓

Publication Approved

↓

Scheduled

↓

Released

↓

Maintained

↓

Withdrawn

or

Superseded

Publication status should never replace editorial status.

An artifact may remain editorially approved while awaiting publication.

13. Continuous Maintenance

Maintenance is an explicit lifecycle stage.

Published knowledge should be continuously monitored for:

- outdated evidence
- inconsistent terminology
- educational weaknesses
- learner misconceptions
- duplicated teaching
- missing differentials

- evolving clinical guidance

Maintenance transforms publication into an ongoing editorial responsibility.

14. Retirement

Not all knowledge remains educationally valuable forever.

Artifacts may eventually be retired due to:

- replacement
- obsolescence
- curriculum redesign
- merged diagnoses
- deprecated clinical practice
- editorial restructuring

Retirement preserves historical accountability while preventing obsolete knowledge from influencing future learners.

15. Lifecycle Independence

Every editorial artifact possesses its own lifecycle.

A diagnosis may be published while new cases remain under review.

A teaching objective may be revised without affecting approved graph relationships.

A clinical pearl may be updated independently of case revisions.

This independence enables continuous improvement without requiring wholesale republication.

16. Lifecycle Dependencies

Although lifecycles are independent, they remain logically connected.

Examples include:

- Educational scaffolds support case development.
- Cases reinforce learning objectives.
- Knowledge relationships strengthen differential reasoning.

- Educational content explains published knowledge.
- Evidence increases editorial confidence.
- Publication depends upon sufficient governance across all required artifacts.

Dependencies coordinate editorial progression without preventing iterative improvement.

17. Lifecycle Principles

The lifecycle architecture of WEOS is governed by the following principles.

Every artifact has a lifecycle.

Every transition represents increased editorial confidence.

Publication is a governance milestone rather than a terminal state.

Maintenance is a permanent editorial responsibility.

Knowledge remains continuously evolvable.

Editorial history is preserved.

Independent artifacts evolve independently.

Dependencies guide progression without restricting improvement.

Closing Principle

The purpose of editorial lifecycles is not to move content through software states.

Their purpose is to govern the continuous evolution of trustworthy clinical knowledge.

Every transition within WEOS should represent a deliberate increase in institutional confidence, ensuring that medical knowledge becomes progressively more reliable, more educationally valuable, and more clinically defensible over time.

Part V — Knowledge Production & Editorial Intelligence

1. Purpose

The purpose of the Wardle Editorial Operating System is not simply to govern existing knowledge.

It must also provide a structured, transparent, and clinically safe mechanism through which new knowledge enters the editorial ecosystem.

Knowledge production encompasses every process responsible for introducing new editorial artifacts into WEOS.

These artifacts may originate from human authors, artificial intelligence, imported clinical guidance, existing editorial content, or future knowledge sources.

Regardless of origin, no artifact enters the governed knowledge base directly.

Every artifact begins life as candidate knowledge.

Only after appropriate editorial governance does candidate knowledge become trusted institutional knowledge.

2. Philosophy of Knowledge Production

Clinical knowledge should never appear spontaneously within the editorial system.

Every editorial artifact must have a clearly understood origin, purpose, and intended educational contribution.

The source of knowledge influences editorial confidence but never replaces editorial governance.

A diagnosis written by an experienced clinician, an AI-generated teaching objective, or an imported clinical guideline all enter WEOS through the same philosophical principle:

Knowledge is proposed before it is trusted.

Editorial confidence is earned through governance rather than inherited from the source.

3. Editorial Intelligence

Editorial Intelligence is the decision-making layer responsible for understanding the editorial state of the knowledge ecosystem and determining what work should occur next.

Editorial Intelligence is distinct from Artificial Intelligence.

Artificial Intelligence refers to computational models capable of generating or analysing information.

Editorial Intelligence refers to the governance framework that evaluates editorial completeness, detects deficiencies, orchestrates production workflows, and recommends editorial actions.

Artificial Intelligence may contribute to Editorial Intelligence, but Editorial Intelligence remains a broader architectural concept that includes deterministic rules, editorial policies, governance logic, quality models, dependency analysis, and human judgment.

Editorial Intelligence exists to answer questions such as:

- What knowledge is missing?
- Which diagnoses require development?
- Which educational objectives remain uncovered?
- Which cases should be generated next?
- Which graph relationships require review?
- Which artifacts have become outdated?
- Which published knowledge requires maintenance?

The objective of Editorial Intelligence is not automation.

Its objective is editorial awareness.

4. Knowledge Producers

Knowledge may originate from multiple producers.

WEOS recognises the origin of knowledge independently from its editorial status.

Knowledge producers include, but are not limited to:

- Human authors
- Artificial Intelligence
- Clinical guideline imports
- Editorial migrations
- Knowledge graph extraction
- Curriculum imports

- Evidence extraction systems
- Future automated editorial systems

Every artifact should preserve its production history.

The producer of an artifact contributes provenance, not authority.

Authority is established through editorial governance.

5. Candidate Knowledge

Before becoming governed knowledge, every editorial artifact exists as Candidate Knowledge.

Candidate Knowledge represents proposed educational material awaiting editorial evaluation.

Examples include:

- proposed teaching objectives
- proposed teaching rules
- proposed clinical pearls
- proposed pitfalls
- proposed diagnostic relationships
- proposed educational summaries
- proposed clinical cases
- proposed recall prompts
- proposed investigation guidance
- proposed management guidance

Candidate Knowledge is intentionally separated from governed editorial content.

This separation protects learners from unreviewed educational material while allowing extensive experimentation during editorial development.

6. Draft Artifacts

Candidate Knowledge is represented through Draft Artifacts.

Drafts are temporary editorial objects that exist solely to facilitate review.

Drafts are not educational resources.

They are editorial proposals.

Draft artifacts may include:

- Diagnosis Draft
- Educational Scaffold Draft
- Teaching Objective Draft
- Teaching Rule Draft
- Clinical Pearl Draft
- Pitfall Draft
- Differential Draft
- Knowledge Relationship Draft
- Education Section Draft
- Clinical Case Draft
- Clue Draft
- Explanation Draft
- Recall Prompt Draft

Every draft possesses:

- editorial identity
- producer
- generation history
- editorial rationale
- review status
- revision history
- governance history

Approval transforms a draft into a governed editorial artifact while preserving its editorial history.

7. Educational Scaffold Generation

Educational development begins with the educational scaffold.

The scaffold defines the educational intent of a diagnosis before comprehensive content development begins.

The scaffold commonly establishes:

- educational objectives
- teaching priorities
- diagnostic discriminators
- differential framework
- clinical pearls
- pitfalls
- recall priorities
- expected learner outcomes

Cases should emerge from the scaffold.

They should not define it.

This principle ensures that every case contributes intentionally to diagnosis-level educational goals.

8. Dependency-Driven Generation

Knowledge production within WEOS is governed by editorial dependencies rather than arbitrary generation requests.

Generation occurs because prerequisite editorial structures indicate that new work is required.

For example:

A diagnosis with complete metadata but no educational scaffold should recommend scaffold generation.

A diagnosis with an approved scaffold but no cases should recommend case generation.

A diagnosis with cases but incomplete educational explanations should recommend education generation.

A diagnosis with established reasoning but missing graph relationships should recommend relationship generation.

Generation therefore responds to editorial needs rather than isolated user commands.

9. Editorial Suggestions

Not every recommendation should immediately produce new knowledge.

Editorial Intelligence frequently produces Suggestions rather than Drafts.

Suggestions identify opportunities for improvement without creating candidate artifacts.

Examples include:

- suggested teaching objective
- suggested differential diagnosis
- suggested graph relationship
- suggested clue refinement
- suggested investigation
- suggested management update

- suggested educational gap
- suggested duplicate teaching
- suggested unsupported claim review

Suggestions represent editorial opportunities.

Editors decide whether those opportunities should become candidate knowledge.

10. Generation Policies

Each editorial artifact follows an explicit generation policy.

Generation policies define:

- whether AI may generate the artifact
- whether manual authoring is supported
- whether regeneration is permitted
- whether evidence is required before review
- which reviewer role is required
- which governance pathway applies

Generation policy is independent of implementation.

It reflects institutional editorial standards rather than technical capability.

11. Artificial Intelligence Within WEOS

Artificial Intelligence functions as an editorial contributor.

It performs work that assists human editors without replacing editorial judgment.

AI may contribute by:

- drafting educational content
- generating clinical cases
- proposing teaching objectives
- suggesting graph relationships
- identifying educational gaps
- detecting duplicated teaching
- analysing reasoning quality
- recommending curriculum improvements
- identifying inconsistencies
- estimating editorial confidence

Artificial Intelligence never establishes institutional trust.

Editorial trust is created exclusively through governed human decisions.

12. Human Editorial Responsibility

Clinical knowledge ultimately remains a human responsibility.

Editors determine:

- clinical correctness
- educational value
- evidence sufficiency
- publication suitability
- learner safety
- institutional confidence

Artificial Intelligence accelerates editorial work.

Editors govern editorial knowledge.

This distinction remains fundamental to every future capability within WEOS.

13. Knowledge Production Lifecycle

Knowledge production follows a universal progression.

Editorial Need Identified

↓

Generation Recommended

↓

Candidate Knowledge Created

↓

Editorial Review

↓

Governance

↓

Approved Editorial Artifact

↓

Publication Eligibility

↓

Published Knowledge

This lifecycle applies regardless of the producer.

14. Knowledge Provenance

Every editorial artifact should preserve complete provenance.

Provenance establishes transparency throughout the editorial lifecycle.

The provenance of an artifact should include:

- origin
- producer
- generation method
- creation time
- supporting context
- review history
- governance decisions
- publication history

Provenance enables explainability without implying correctness.

An artifact is trusted because it has been governed, not because of where it originated.

15. Principles of Knowledge Production

Knowledge production within WEOS is governed by the following principles.

Every editorial artifact begins as candidate knowledge.

Knowledge origin never replaces editorial governance.

Editorial Intelligence orchestrates production.

Artificial Intelligence assists editorial work but does not govern it.

Educational scaffolds precede comprehensive content generation.

Generation should follow editorial dependencies.

Editorial suggestions identify work before creating knowledge.

Every artifact preserves provenance.

Editorial trust is earned through review.

Institutional confidence emerges from governance rather than automation.

Closing Principle

The purpose of Knowledge Production is not to generate more educational content.

Its purpose is to introduce clinically meaningful, educationally intentional, and transparently governed candidate knowledge into the editorial ecosystem.

Within WEOS, knowledge is not trusted because it has been created.

Knowledge is trusted because it has been governed.

Part VI — Editorial Intelligence & Decision Architecture

1. Purpose

The purpose of the Wardle Editorial Operating System is not merely to store, generate, or review clinical knowledge.

It must also interpret the editorial state of that knowledge.

Editorial Intelligence is the reasoning layer that transforms governed information into meaningful editorial decisions.

Rather than presenting editors with isolated facts, Editorial Intelligence synthesises knowledge across the editorial ecosystem to answer practical questions such as:

- What requires attention?
- What is safe to approve?
- What prevents publication?
- What should be generated next?
- Which diagnosis is incomplete?
- Which evidence is insufficient?
- Which educational objectives remain uncovered?

Editorial Intelligence exists to reduce editorial complexity without reducing editorial transparency.

2. Editorial Intelligence

Editorial Intelligence is the governed reasoning layer of WEOS.

It continuously interprets editorial knowledge, evaluates dependencies, applies governance policies, and produces recommendations that assist human decision-making.

Editorial Intelligence does not replace editors.

It supports editors by transforming fragmented editorial information into coherent understanding.

Editorial Intelligence combines:

- editorial policies

- lifecycle state
- dependency analysis
- evidence assessment
- governance rules
- quality models
- knowledge relationships
- AI recommendations
- human decisions

to produce meaningful editorial guidance.

3. Editorial Questions

Every capability within WEOS ultimately exists to answer an editorial question.

Examples include:

- Is this diagnosis ready for scaffold generation?
- Is this teaching objective clinically appropriate?
- Is this graph relationship defensible?
- Is this case educationally safe?
- Is learner exposure appropriate?
- Does this diagnosis require another case?
- Which evidence remains unsupported?
- Which differentials are insufficiently represented?

Editorial Intelligence exists to answer these questions consistently throughout the system.

4. From Data to Decisions

WEOS distinguishes between information and decisions.

Information describes the editorial state.

Decisions interpret that state.

For example:

Information may indicate:

- three unsupported claims
- one unresolved differential
- incomplete educational objectives

- pending AI review

Editorial Intelligence transforms this information into a decision.

For example:

"This diagnosis is not yet ready for publication because unresolved educational and evidence deficiencies remain."

The purpose of Editorial Intelligence is therefore interpretation rather than presentation.

5. Editorial Assessments

Editorial Intelligence evaluates artifacts through structured assessments.

An assessment represents an editorial interpretation rather than a stored property.

Examples include:

- Clinical Assessment
- Educational Assessment
- Evidence Assessment
- Reasoning Assessment
- Knowledge Relationship Assessment
- Curriculum Assessment
- Publication Assessment

Assessments remain explainable.

Editors should always understand why an assessment reached its conclusion.

6. Editorial Decisions

Every governed artifact eventually reaches an editorial decision.

Editorial decisions are explicit, auditable outcomes produced through governance.

Typical decisions include:

- Approve
- Reject
- Request Revision
- Request Evidence

- Recommend Regeneration
- Defer
- Publish
- Withdraw
- Supersede

Every decision records:

- rationale
- supporting assessments
- responsible reviewer
- affected artifacts
- governance history

Editorial decisions are institutional decisions rather than user interface events.

7. Editorial Recommendations

Editorial Intelligence may recommend work without making editorial decisions.

Recommendations assist editors by identifying opportunities for improvement.

Examples include:

- generate another case
- strengthen differential coverage
- revise clue progression
- expand investigation guidance
- review unsupported claims
- add graph relationships
- improve recall prompts
- update references

Recommendations guide editorial effort while preserving human authority.

8. Editorial Readiness

Readiness is a computed interpretation of editorial maturity.

Readiness is never manually assigned.

Instead, Editorial Intelligence continuously evaluates whether an artifact satisfies the requirements for progressing to its next lifecycle stage.

Examples include:

Generation Readiness

Editorial Review Readiness

Publication Readiness

Maintenance Readiness

Retirement Readiness

Readiness should always explain itself.

Editors should understand:

- why an artifact is ready
- why it is not ready
- what remains outstanding
- what action should occur next

9. Editorial Dependencies

Knowledge within WEOS is interconnected.

Editorial Intelligence continuously evaluates dependencies between artifacts.

Examples include:

Educational objectives supporting clinical cases.

Teaching rules supporting learning outcomes.

Graph relationships strengthening differential reasoning.

Evidence supporting educational content.

Cases reinforcing curriculum coverage.

Publication depending upon governance completeness.

Dependency analysis ensures that editorial progression occurs logically while remaining flexible enough to support iterative improvement.

10. Editorial Confidence

Editorial confidence represents the degree of institutional trust placed in an artifact.

Confidence is not equivalent to certainty.

Instead, it reflects the cumulative outcome of editorial governance.

Confidence is strengthened through:

- review
- evidence
- consistency
- educational quality
- reasoning integrity
- governance history

Confidence decreases when:

- evidence becomes outdated
- inconsistencies emerge
- educational weaknesses are identified
- clinical guidance changes
- governance deficiencies remain unresolved

Editorial confidence continuously evolves throughout an artifact's lifecycle.

11. Editorial Explainability

Every recommendation, assessment, and decision produced by Editorial Intelligence should be explainable.

Editors should never encounter unexplained conclusions.

Every editorial conclusion should answer:

- What was evaluated?
- Which policies were applied?
- Which evidence influenced the result?
- Which dependencies contributed?
- Which recommendations were generated?
- What actions remain available?

Explainability strengthens institutional trust by making governance transparent.

12. Editorial Decision Patterns

Although different artifacts undergo different review processes, every editorial decision follows the same conceptual pattern.

Editorial Artifact

↓

Current Editorial State

↓

Editorial Assessments

↓

Dependency Analysis

↓

Evidence Evaluation

↓

Governance Policies

↓

Editorial Intelligence

↓

Recommendation

↓

Human Editorial Decision

↓

Governed Outcome

This pattern provides consistency across every editorial capability within WEOS.

13. Editorial Work Prioritisation

Editorial Intelligence continuously evaluates the editorial ecosystem to determine where effort should be directed.

Prioritisation considers:

- learner impact
- publication blockers
- missing educational coverage
- unresolved governance issues
- evidence deficiencies
- incomplete diagnoses
- curriculum importance
- maintenance requirements

Prioritisation assists editors without replacing editorial judgement.

14. Editorial Intelligence Principles

Editorial Intelligence operates according to the following principles.

Interpretation is more valuable than presentation.

Recommendations should reduce editorial complexity.

Decisions should always be explainable.

Readiness is computed rather than assigned.

Dependencies should guide editorial progression.

Recommendations assist but never replace governance.

Human editors remain responsible for institutional trust.

Editorial Intelligence exists to improve judgement rather than automate it.

15. Relationship to Artificial Intelligence

Artificial Intelligence contributes information.

Editorial Intelligence governs decisions.

Artificial Intelligence may identify missing differentials.

Editorial Intelligence determines whether those differentials should enter editorial review.

Artificial Intelligence may generate a clinical pearl.

Editorial Intelligence determines whether the draft satisfies generation policies and requires governance.

This separation ensures that the architecture remains stable regardless of future advances in artificial intelligence.

Editorial Intelligence is therefore an enduring architectural capability rather than a specific technological implementation.

Closing Principle

Editorial Intelligence is the reasoning engine of the Wardle Editorial Operating System.

Its purpose is not to replace human editors, but to continuously interpret the evolving state of clinical knowledge, reveal what matters, explain why it matters, and support consistent, transparent, and defensible editorial decisions.

Within WEOS, information becomes knowledge through governance.

Knowledge becomes trusted through editorial decisions.

Editorial Intelligence exists to make those decisions understandable, consistent, and scalable.

Part VII — Editorial Workspace Architecture

1. Purpose

The Editorial Workspace is the primary environment through which editors interact with the Wardle Editorial Operating System (WEOS).

Its purpose is not to expose stored information.

Its purpose is to support editorial thinking.

The workspace should continuously assist editors in understanding the current state of governed clinical knowledge, identifying what requires attention, completing editorial work, and making safe, explainable decisions.

The workspace therefore exists to support editorial judgement rather than administrative navigation.

2. Workspace Philosophy

The Editorial Workspace is designed around editorial work rather than data structures.

Editors should never need to understand how information is stored internally.

Instead, the workspace should present knowledge according to the editorial questions being answered.

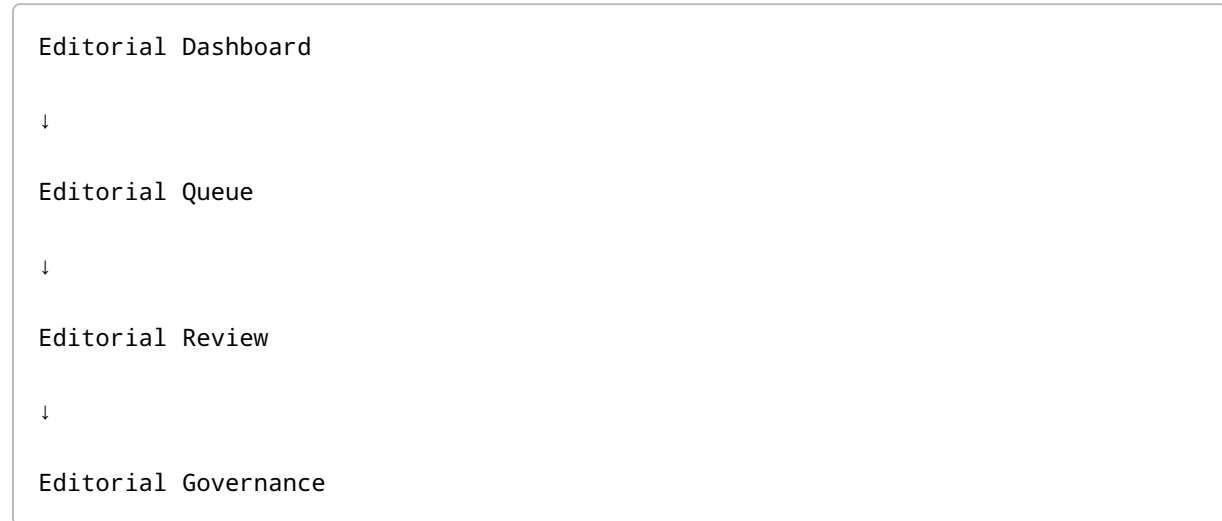
Examples include:

- What requires my attention?
- Is this diagnosis educationally complete?
- What should be generated next?
- Can this artifact be approved?
- What prevents publication?
- Which knowledge requires maintenance?

The organisation of the workspace should therefore reflect editorial reasoning rather than database organisation.

3. Workspace Model

The Editorial Workspace consists of four complementary environments.



Each environment serves a distinct purpose while sharing the same governed knowledge model.

Editors move between environments as editorial work progresses.

4. Editorial Dashboard

The Dashboard provides situational awareness.

Its purpose is to answer:

"What is the current state of the editorial ecosystem?"

The Dashboard should summarise:

- diagnosis completeness
- editorial readiness
- publication readiness
- governance status
- educational coverage
- outstanding work
- maintenance requirements

The Dashboard is observational.

It supports understanding rather than decision-making.

5. Editorial Queue

The Queue represents work awaiting editorial attention.

Unlike the Dashboard, the Queue is action-oriented.

The Queue should organise work according to editorial priorities rather than artifact type.

Examples include:

- diagnoses requiring scaffolding
- AI drafts awaiting review
- cases awaiting approval
- graph relationships requiring governance
- unsupported claims
- publication blockers
- evidence updates
- maintenance tasks

The Queue answers:

"What should I work on next?"

6. Editorial Review

Editorial Review is where candidate knowledge becomes governed knowledge.

Review is centred upon a single editorial task.

Examples include:

- reviewing a case
- reviewing a teaching objective
- reviewing a graph relationship
- reviewing educational content
- reviewing a clinical pearl

Every review should provide the complete context required for safe editorial decision-making.

Editors should not need to navigate between multiple locations to understand an artifact.

7. Editorial Governance

Governance evaluates the editorial system rather than individual artifacts.

Governance focuses upon:

- consistency
- completeness
- evidence
- relationships
- curriculum alignment
- editorial quality
- publication standards

Governance provides confidence that independently reviewed artifacts collectively form a coherent educational system.

8. Editorial Work Packets

Editorial work is performed through Work Packets.

A Work Packet is the smallest complete unit of editorial work.

A packet contains everything required to complete one editorial decision.

Examples include:

- Case Review Packet
- Teaching Rule Review Packet
- Clinical Pearl Review Packet
- Differential Review Packet
- Knowledge Relationship Review Packet
- Educational Section Review Packet
- Publication Packet

A Work Packet should answer:

- What is being reviewed?
- Why does it exist?
- What supports it?
- What remains unresolved?
- Which decision is required?
- What happens next?

Editors should complete work from within the packet without unnecessary context switching.

9. Workspace Layers

The Editorial Workspace presents information through progressively deeper layers.

Editorial Awareness

↓

Editorial Work

↓

Editorial Decision

↓

Editorial Detail

↓

Editorial History

Editors should begin with broad situational awareness and progressively reveal detail only as required.

The workspace should minimise cognitive load while preserving transparency.

10. Progressive Disclosure

Not every editorial detail deserves equal prominence.

The workspace should progressively reveal information according to editorial importance.

High-level decisions should appear before implementation detail.

Examples include:

- publication recommendation before validation logs
- governance blockers before audit metadata
- educational deficiencies before database identifiers
- evidence status before raw references

Editors should first understand what requires action before investigating why.

11. Editorial Context

Every editorial task should preserve context.

Editors should always understand:

- the diagnosis
- the educational purpose
- related artifacts
- lifecycle stage
- dependencies
- previous decisions
- current recommendations

Context should accompany editorial work rather than requiring repeated navigation.

12. Workspace Navigation

Navigation exists to support editorial thinking rather than expose application structure.

Movement throughout the workspace should follow editorial progression.

Examples include:

Diagnosis

↓

Educational Scaffold

↓

Clinical Case

↓

Evidence

↓

Review



Publication

rather than arbitrary navigation between unrelated administrative pages.

13. Workspace Consistency

Every editorial surface should present information using a consistent conceptual structure.

Editors should encounter familiar patterns regardless of artifact type.

Every review surface should communicate:

- identity
- editorial purpose
- current state
- supporting evidence
- dependencies
- recommendations
- available decisions
- governance history

Consistency reduces cognitive effort and improves editorial confidence.

14. Workspace Intelligence

The Editorial Workspace should continuously expose the output of Editorial Intelligence.

Rather than merely displaying stored information, the workspace should communicate:

- recommended work
- publication readiness
- generation readiness
- governance concerns
- educational gaps
- dependency conflicts
- maintenance priorities

Editors should spend their effort making decisions rather than searching for information.

15. Workspace Principles

The Editorial Workspace operates according to the following principles.

Editorial work should determine workspace organisation.

Context should remain visible throughout editorial tasks.

Work Packets should contain complete decision context.

Navigation should follow editorial progression.

Progressive disclosure should reduce cognitive load.

Recommendations should precede raw information.

Governance should remain transparent.

Editorial Intelligence should guide attention without replacing judgement.

Every editorial decision should be explainable.

The workspace should make editorial work easier, not merely possible.

Closing Principle

The Editorial Workspace is not an administrative interface.

It is the operational environment through which governed clinical knowledge is created, evaluated, maintained, and entrusted to learners.

Its success is measured not by the quantity of information displayed, but by its ability to help editors make safe, consistent, efficient, and clinically defensible decisions.

Wardle Editorial Operating System (WEOS)

Part VIII - Institutional Editorial Governance

WEOS-ARCH-008 | Version 0.1 | Draft for approval

1. Purpose

The Wardle Editorial Operating System governs clinical knowledge on behalf of learners, educators, reviewers, and institutions.

Clinical knowledge does not become trustworthy merely because it has been written, generated, reviewed, or technically published.

It becomes institutionally trusted when qualified people exercise defined authority through transparent, accountable, and reproducible editorial governance.

The purpose of Institutional Editorial Governance is to define how responsibility, authority, accountability, and oversight operate throughout WEOS.

This chapter establishes:

- who contributes editorial knowledge;
- who may evaluate it;
- who may make editorial decisions;
- how authority is assigned;
- how disagreements are resolved;
- how conflicts of interest are managed;
- how evidence and high-risk content receive appropriate oversight;
- how publication authority is exercised;
- how governance remains accountable;
- how editorial governance scales from individual editors to institutional and multi-institutional structures.

Institutional Editorial Governance does not define user-interface permissions or specific organisational job titles.

It defines the enduring authority model that every implementation and editorial organisation must preserve.

2. Philosophy of Institutional Governance

WEOS treats editorial governance as an institutional responsibility rather than an individual activity.

An individual editor may possess expertise.

A reviewer may possess experience.

A specialty board may possess collective authority.

However, expertise alone does not establish institutional trust.

Institutional trust emerges when expertise is exercised through:

- defined responsibilities;
- appropriate scope of authority;
- transparent assessment;
- explicit decisions;
- preserved rationale;
- evidence-informed judgement;
- accountable publication;
- continuing oversight.

Governance should protect the editorial system from:

- unreviewed clinical assertions;
- informal approval;
- authority based solely on system access;
- publication without accountability;
- concentration of incompatible responsibilities;
- hidden conflicts of interest;
- inconsistent specialty standards;
- unresolved reviewer disagreement;
- unsupported exceptions;
- loss of institutional memory.

Institutional governance strengthens clinical judgement by placing it within a durable framework of responsibility.

3. Institutional Trust

Institutional trust represents the confidence that an organisation places in governed clinical knowledge.

It is not equivalent to confidence in a particular author, editor, artificial-intelligence system, or external source.

Institutional trust is created through the combined effect of:

- stable editorial identity;
- appropriate clinical scope;
- evidence;
- qualified review;
- educational evaluation;
- governance decisions;
- dependency assessment;

- publication authority;
- maintenance;
- preserved history.

Institutional trust belongs to the governed artifact and its approved version.

It does not transfer automatically from the reputation of its creator.

Institutional trust is earned through governance rather than inherited from origin.

4. Governance Scope

Institutional Editorial Governance operates across complementary levels.

4.1 Artifact Governance

Artifact Governance evaluates an individual editorial artifact.

It asks whether the artifact is clinically defensible, educationally appropriate, evidence-supported, correctly scoped, and eligible to progress.

4.2 Diagnosis Governance

Diagnosis Governance evaluates the complete editorial foundation of a diagnosis, including identity, metadata, educational intent, cases, education, relationships, evidence, dependencies, and maintenance.

4.3 Specialty Governance

Specialty Governance evaluates knowledge within the clinical standards and educational priorities of a specialty or subspecialty.

4.4 Curriculum Governance

Curriculum Governance evaluates whether independently governed artifacts collectively form an appropriate programme of learning.

4.5 Institutional Governance

Institutional Governance oversees authority, policies, quality, evidence, publication controls, maintenance, auditability, and organisational accountability.

An artifact may be clinically acceptable while its diagnosis remains incomplete.

A diagnosis may be well developed while specialty-level inconsistencies remain unresolved.

Institutional governance ensures that trust is evaluated at every relevant level.

5. Editorial Authority

Editorial authority is the recognised ability to make a governed decision within a defined scope.

Authority should reflect:

- professional competence;
- editorial responsibility;
- artifact risk;
- decision consequence;
- clinical scope;
- educational scope;
- institutional mandate.

Authority must not be inferred solely from technical access.

The ability to change a database record, apply a migration, or operate an administrative interface does not confer clinical or publication authority.

Editorial authority should always answer:

- Who may make this decision?
- Which artifact or domain does the authority cover?
- Which conditions limit the authority?
- Which decisions require additional review?
- Who remains accountable for the outcome?
- How is the decision recorded?

6. Editorial Roles

WEOS recognises editorial roles according to responsibility rather than software permission.

6.1 Contributor

A Contributor introduces candidate knowledge, identified errors, references, feedback, or maintenance concerns.

Contribution establishes provenance but not authority.

6.2 Editorial Author

An Editorial Author develops candidate artifacts according to approved educational intent and responds to review findings.

Authorship does not automatically confer approval authority.

6.3 Clinical Reviewer

A Clinical Reviewer evaluates factual correctness, diagnostic interpretation, investigations, management, safety, severity, population applicability, and risk of harmful misunderstanding.

6.4 Educational Reviewer

An Educational Reviewer evaluates learning purpose, reasoning progression, clue sequencing, learner level, misconception prevention, educational usefulness, and curriculum alignment.

6.5 Knowledge Governance Reviewer

A Knowledge Governance Reviewer evaluates structured relationships, graph facts, reasoning paths, differential mappings, contradiction, duplication, provenance, and downstream consequences.

6.6 Evidence Reviewer

An Evidence Reviewer evaluates source authority, recency, applicability, methodological quality, claim alignment, and conflicting evidence.

6.7 Specialty Reviewer

A Specialty Reviewer provides deeper authority for specialist diagnostic nuance, advanced investigations, high-risk management, uncommon disease, or evolving guidance.

6.8 Senior Editor

A Senior Editor resolves complex decisions, coordinates multidisciplinary review, supervises specialty governance, and authorises high-impact editorial outcomes within their mandate.

6.9 Publication Authority

The Publication Authority determines whether governed knowledge may become learner-facing or institutionally operational.

6.10 Editorial Operations Lead

The Editorial Operations Lead assigns work, monitors queues, coordinates reviewers, tracks maintenance, and escalates operational risk without silently exercising clinical authority.

6.11 System Administrator

A System Administrator manages access, migrations, restoration, and technical operation.

Administrative capability does not create editorial authority.

7. Role Qualification and Scope

Editorial authority should be based upon demonstrable suitability for the decision being made.

Suitability may include:

- clinical training;
- specialty experience;
- educational experience;
- editorial competence;
- familiarity with WEOS standards;

- evidence-appraisal competence;
- knowledge-governance competence;
- institutional appointment;
- continuing participation in quality review.

Qualification should be proportionate to risk.

Where an artifact requires several forms of authority, governance should combine them rather than assume one reviewer possesses every necessary perspective.

8. Editorial Ownership

Every actively governed artifact should have identifiable editorial ownership.

Ownership establishes responsibility for continued stewardship rather than unrestricted decision authority.

Ownership may exist at artifact, diagnosis, specialty, curriculum, publication, and maintenance levels.

Ownership should answer:

- Who ensures work progresses?
- Who monitors unresolved concerns?
- Who coordinates required reviewers?
- Who responds when evidence changes?
- Who ensures published knowledge remains maintained?

Ownership must remain transferable without losing governance history.

9. Assignment of Editorial Work

Editorial work should be assigned according to competence, artifact type, specialty, risk, learner impact, urgency, independence, and availability.

Assignment creates responsibility to review or act.

It does not predetermine the decision.

Reviewers should remain free to approve, reject, request revision, request evidence, defer, escalate, declare insufficient expertise, or declare a conflict of interest.

10. Separation of Duties

Certain responsibilities should remain separate where combining them would weaken governance.

Separation may require that:

- an author does not provide the only approval of their own work;
- technical application does not substitute for editorial approval;
- publication is authorised separately from generation;
- high-risk clinical content receives independent review;
- disputed decisions are adjudicated independently;
- emergency actions receive retrospective review;
- migrations affecting governed knowledge are authorised before execution.

Where small teams require overlapping roles, governance should compensate through disclosure, additional review, preserved rationale, audit, and restricted publication authority.

11. Assessment and Decision Authority

Editorial assessments and editorial decisions are distinct.

An assessment interprets quality or state.

A decision changes or confirms governed standing.

Decision authority should be proportionate to consequence.

11.1 Ordinary Editorial Decisions

Examples include routine wording revision, duplicate-candidate rejection, low-risk Learning Goal approval, or straightforward alias resolution.

11.2 Elevated Editorial Decisions

Examples include Clinical Case approval, Teaching Relationship activation, Diagnosis Education approval, or governed-artifact supersession.

11.3 High-Risk Editorial Decisions

Examples include emergency management, drug doses, serious-diagnosis exclusion claims, pregnancy or paediatric treatment, and cross-specialty guidance.

11.4 Institutional Decisions

Examples include publication, withdrawal, governance exceptions, diagnosis retirement, and adoption of institutional standards.

12. Editorial Disagreement

Disagreement is an expected feature of responsible clinical governance.

WEOS should not conceal disagreement through silent editing or informal compromise.

A disagreement record should identify:

- disputed issue;
- affected artifact;
- competing interpretations;
- evidence considered;
- clinical and educational consequences;
- reviewers involved;
- unresolved questions;
- required escalation.

The objective is not forced consensus.

The objective is a defensible institutional decision.

13. Escalation and Adjudication

Escalation occurs when ordinary review cannot safely or appropriately resolve an issue.

It may be required because of reviewer disagreement, insufficient expertise, high risk, contradictory evidence, cross-specialty impact, unresolved conflict, or publication urgency.

Adjudication should preserve:

- competing positions;
- evidence considered;
- adjudicating authority;
- final rationale;
- conditions;
- future review obligations.

Adjudication should not erase earlier disagreement.

14. Conflict of Interest

A conflict of interest exists when judgement may be influenced, or reasonably perceived to be influenced, by interests outside the editorial merits of the artifact.

Possible responses include:

- disclosure;
- restricted participation;
- independent review;
- removal from final authority;
- reassignment;
- additional approval.

Undeclared conflicts weaken institutional trust even where the decision is clinically correct.

15. Evidence Oversight

Clinical evidence supports editorial decisions but does not replace them.

Evidence Oversight should ensure that:

- claims are linked to appropriate support;
- sources apply to the stated population and context;
- evidence remains sufficiently current;
- conflicting evidence is represented honestly;
- high-risk claims receive proportionate scrutiny;
- retracted or superseded evidence is identified;
- evidence changes trigger maintenance.

A reference may be credible but not applicable.

Evidence contributes to institutional confidence without independently determining publication.

16. Governance of High-Risk Knowledge

High-risk knowledge may include:

- drug doses;
- emergency treatment;
- procedural guidance;
- contraindications;
- antimicrobial recommendations;
- paediatric and pregnancy management;
- critical-care thresholds;
- diagnostic exclusion statements;
- escalation criteria;
- scoring-system cut-offs;
- guideline-dependent management.

High-risk governance may require stronger evidence, specialty review, independent review, dual approval, explicit scope, stronger validation, shorter maintenance intervals, and defined emergency correction pathways.

Risk should be determined by the consequence of error rather than artifact type alone.

17. Publication Authority

Publication represents an institutional decision to permit governed knowledge to influence learners or institutional activity.

Publication authority must remain separate from authorship, generation, validation, technical deployment, scheduling, and operational availability.

A Publication Authority may publish, schedule, defer, restrict, withdraw, supersede, or decline publication.

The absence of an automated blocker does not constitute publication approval.

18. Withdrawal, Supersession, and Retirement

Published knowledge may require withdrawal, supersession, or retirement.

Withdrawal removes knowledge from active exposure because continued use is inappropriate.

Supersession replaces it with a preferred successor while preserving history.

Retirement ends active editorial use when the artifact or diagnosis no longer serves an appropriate purpose.

Removal from active use must not erase institutional history.

19. Emergency Editorial Governance

Emergency action may be required where learner-facing knowledge presents a credible and significant risk.

Possible actions include immediate withdrawal, temporary suppression, scheduling suspension, disabling playability, restricting generation, or urgent correction.

Every emergency action should preserve authority, risk, time, action, expected duration, and required follow-up.

Urgency may shorten the pathway.

It must not remove accountability.

20. Governance Committees and Editorial Boards

As WEOS grows, governance may be distributed across specialty boards, clinical-safety panels, educational-quality committees, knowledge-governance boards, publication committees, curriculum committees, and evidence groups.

Every governance body should possess:

- defined scope;

- membership requirements;
- decision authority;
- conflict rules;
- preserved decisions;
- escalation pathways;
- review responsibilities.

Committees should not become symbolic approval layers.

21. Specialty Governance

Specialty Governance may define specialty ownership, terminology, reviewer requirements, high-risk categories, evidence expectations, maintenance priorities, and curriculum expectations.

Specialty decisions should remain visible to the wider ecosystem.

Cross-specialty knowledge should support joint review rather than arbitrary ownership.

22. Curriculum Governance

Curriculum Governance evaluates whether the knowledge ecosystem provides coherent learning.

It should consider diagnosis representation, Learning Goal coverage, important mimics, case diversity, learner progression, population representation, specialty balance, and maintenance.

Curriculum governance should not be reduced to counting cases.

Curriculum confidence emerges from educational coherence rather than content volume.

23. Relationship to Editorial Intelligence

Editorial Intelligence may identify blockers, recommend reviewers, detect conflicts, surface evidence deficiencies, prioritise maintenance, calculate readiness, and recommend escalation.

Editorial Intelligence improves governance awareness.

It does not possess governance authority.

A recommendation to approve is not approval.

A readiness result is not publication.

24. Relationship to Artificial Intelligence

Artificial Intelligence may generate, summarise, compare, detect, propose, and prepare context.

It must not independently establish clinical authority, accept evidence, adjudicate disagreement, approve high-risk knowledge, publish, withdraw, or grant exceptions.

AI may assist the decision process.

Human editorial governance remains responsible for the decision.

25. Governance Records

Every material governance action should produce a durable record preserving artifact identity, version, question, actor, authority, evidence, findings, rationale, decision, state change, unresolved concerns, and required follow-up.

A status without rationale is insufficient governance.

A decision without an identifiable decision-maker is insufficient governance.

A changed artifact without version history is insufficient governance.

26. Governance Exceptions

Exceptions should remain rare, explicit, limited, and reviewable.

An exception should identify the requirement, reason, authority, scope, duration, risk, compensating controls, follow-up, and expiry.

Temporary necessity should not become permanent policy through repeated informal use.

27. Governance Quality Assurance

Governance quality may be assessed through completeness of review records, consistency of decisions, evidence sufficiency, disagreement, post-publication correction, unresolved decision age, unauthorised action, emergency-action frequency, overdue maintenance, and separation-of-duty compliance.

WEOS should improve the system producing decisions rather than only correcting individual defects.

28. External Review and Accreditation

External review should be able to determine which standards govern knowledge, who possesses authority, how decisions are recorded, how evidence is assessed, how publication occurs, how errors are corrected, and how AI contributions are governed.

External endorsement may strengthen confidence.

It does not replace internal accountability.

29. Governance Across Institutional Scale

The same principles should remain valid as WEOS grows.



Distributed authority should remain transparent and accountable.

Knowledge approved by one institution should not automatically become approved by every participating institution.

30. Governance Continuity

Governance must remain resilient when people, organisations, and technologies change.

Continuity requires stable identities, transferable ownership, documented authority, preserved decisions, succession, access continuity, restoration capability, maintained standards, and periodic revalidation.

Institutional knowledge should not depend upon the memory of one editor.

31. Principles of Institutional Editorial Governance

Authority Must Be Explicit. Authority should not be inferred solely from authorship, seniority, reputation, or technical access.

Responsibility Must Be Identifiable. Every material decision should have an accountable human decision-maker.

Expertise Must Be Appropriately Scoped. Reviewers should act within their competence and mandate.

Approval Must Apply to a Version. Approval must not apply ambiguously to mutable knowledge.

Publication Requires Separate Authority. Approved knowledge must not become learner-facing without an explicit Publication Decision.

High-Risk Knowledge Requires Stronger Governance. Governance intensity should reflect the consequence of error.

Disagreement Must Remain Visible. Material disagreement should be preserved through resolution.

Conflicts Must Be Managed Transparently. Undisclosed conflicts weaken confidence.

Evidence Supports but Does Not Govern. Sources inform decisions but do not independently create trust.

Technical Capability Does Not Confer Clinical Authority. Administrative and engineering access remain distinct from editorial authority.

Governance Must Remain Auditable. Decisions require context, rationale, responsibility, and history.

Institutional Memory Must Outlive Individuals. Governance should remain durable through personnel and technology change.

32. Relationship to the Wider WEOS Architecture

The Editorial Domain Model defines what is governed.

Editorial Capabilities define what work must be possible.

Editorial Lifecycles define how confidence changes.

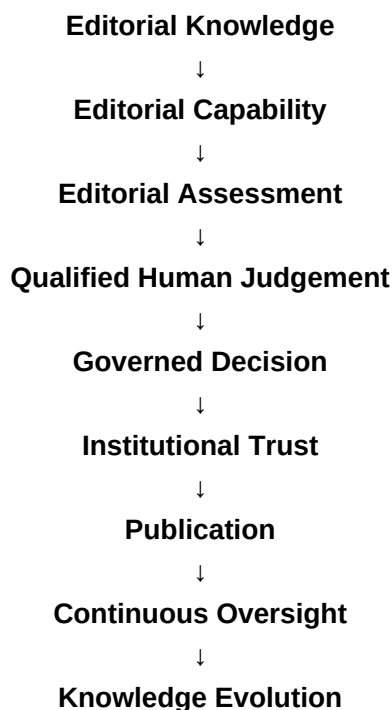
Knowledge Production defines how candidate knowledge enters.

Editorial Intelligence interprets what requires attention.

The Editorial Workspace supports completion of work.

Institutional Editorial Governance defines who may exercise authority, how decisions become trusted, and how accountability is preserved.

The Evolution of the Editorial Knowledge Ecosystem describes how these structures grow.



Closing Principle

Clinical knowledge does not become trustworthy because it was created by an expert, generated by an advanced system, supported by a reference, or made available through software.

It becomes trustworthy when appropriate expertise is exercised through transparent, accountable, evidence-informed, and continuously maintained institutional governance.

People contribute knowledge.

Reviewers evaluate knowledge.

Editorial Intelligence supports judgement.

Governance establishes authority.

Institutions assume responsibility.

Publication expresses trust.

Accountability preserves that trust over time.

Part IX — Evolution of the Editorial Knowledge Ecosystem

1. Purpose

The Wardle Editorial Operating System (WEOS) is designed as an evolving knowledge governance platform.

It is not intended to reach a final state.

Medicine evolves.

Educational practice evolves.

Clinical evidence evolves.

Artificial intelligence evolves.

Editorial governance must therefore evolve alongside them.

The purpose of this chapter is to define how WEOS grows while preserving the architectural principles established throughout this document.

Future evolution should increase the sophistication of editorial governance without changing its philosophical foundations.

2. Philosophy of Evolution

Growth within WEOS should never be measured by the quantity of educational content.

Instead, progress should be measured by the maturity with which clinical knowledge is governed.

As the system evolves it should become:

- more clinically reliable
- more educationally intentional
- more internally consistent
- more explainable
- more scalable
- more maintainable

Evolution therefore represents increasing editorial maturity rather than increasing software complexity.

3. Evolution Through Capability

The evolution of WEOS occurs through progressively richer editorial capabilities.

Each new capability should strengthen one or more existing principles rather than introduce isolated functionality.

Capabilities should evolve in four stages.

Foundational

The editorial concept exists.

Operational

Editors can perform the work.

Governed

Editorial policies consistently govern the work.

Intelligent

Editorial Intelligence proactively assists editors while preserving human authority.

The objective is not automation.

The objective is progressively better editorial judgement.

4. Evolution of Knowledge

Knowledge itself evolves continuously.

Examples include:

- revised evidence
- new diagnostic criteria
- changing terminology
- emerging diseases
- evolving investigations

- changing management recommendations
- improved educational strategies

WEOS should support continuous refinement without disrupting institutional confidence.

Historical governance should always remain preserved.

5. Evolution of Editorial Artifacts

Every governed artifact should be capable of continuous refinement.

A diagnosis may acquire:

- additional cases
- improved objectives
- stronger teaching rules
- refined graph relationships
- improved educational content
- updated evidence

The system should encourage incremental improvement rather than periodic reconstruction.

6. Evolution of Editorial Intelligence

Editorial Intelligence is expected to become increasingly sophisticated throughout the lifetime of WEOS.

Future improvements may include:

- improved dependency reasoning
- richer recommendation models
- stronger quality assessment
- better curriculum analysis
- more comprehensive contradiction detection
- increasingly explainable recommendations
- improved maintenance prioritisation

These improvements strengthen editorial judgement without altering the governance philosophy.

7. Evolution of Artificial Intelligence

Artificial Intelligence will continue to improve.

WEOS should benefit from these advances without becoming dependent upon any specific model, provider, or technology.

Future AI systems may become capable of:

- producing higher quality drafts
- generating richer educational explanations
- identifying increasingly subtle inconsistencies
- proposing stronger knowledge relationships
- analysing curriculum coverage
- assisting reviewers during governance

Regardless of technological advancement, AI remains an editorial contributor rather than an editorial authority.

Human governance remains unchanged.

8. Evolution of Governance

Institutional governance should become progressively more comprehensive.

Examples include:

- richer publication standards
- stronger evidence governance
- improved quality assurance
- expanded curriculum governance
- enhanced specialty governance
- multidisciplinary editorial review
- institutional accreditation support

Governance should expand without compromising transparency or explainability.

9. Evolution of the Editorial Workspace

The Editorial Workspace should evolve according to editorial needs rather than technological trends.

Future workspace improvements should reduce editorial effort while increasing editorial understanding.

As Editorial Intelligence becomes more sophisticated, the workspace should progressively shift from displaying information to supporting judgement.

Editors should spend less time locating knowledge and more time governing it.

10. Institutional Evolution

WEOS is intended to support increasingly sophisticated editorial organisations.

Examples include:

Single editor



Editorial team



Specialty editorial boards



Institutional curriculum committees



Multi-institution collaboration



International editorial networks

The underlying editorial philosophy remains unchanged regardless of organisational scale.

11. Knowledge Network Evolution

As diagnoses mature, the knowledge ecosystem becomes increasingly interconnected.

Future evolution should strengthen:

- differential reasoning
- knowledge relationships
- curriculum integration
- educational continuity
- longitudinal learning

- reasoning pathways

The objective is to develop a coherent network of governed diagnostic knowledge rather than a collection of independent educational resources.

12. Sustainability

WEOS should remain sustainable over decades of editorial development.

Sustainability requires:

- modular editorial artifacts
- independent lifecycles
- transparent governance
- explainable decisions
- maintainable knowledge structures
- preserved editorial history

Architectural decisions should favour long-term maintainability over short-term convenience.

13. Architectural Stability

Although implementation technologies will inevitably change, the conceptual architecture of WEOS should remain stable.

Future implementations may replace:

- user interfaces
- databases
- programming languages
- artificial intelligence models
- storage technologies
- deployment platforms

None of these changes should alter:

- editorial philosophy
- domain model
- governance principles
- lifecycle model
- editorial capabilities
- institutional trust model

These concepts define WEOS independently of implementation.

14. Principles of Evolution

The continued evolution of WEOS should always preserve the following principles.

Knowledge evolves continuously.

Governance evolves continuously.

Editorial Intelligence evolves continuously.

Artificial Intelligence evolves continuously.

Institutional trust evolves continuously.

The architecture should evolve through refinement rather than replacement.

Complexity should increase only when it produces measurable editorial value.

Editorial judgement should always remain central.

15. Long-Term Vision

The long-term vision of WEOS is not simply to become a better editorial platform.

Its vision is to become a comprehensive operating system for governing diagnostic knowledge throughout its entire educational lifecycle.

The mature WEOS should enable institutions to:

- create clinical knowledge
- improve educational quality
- supervise AI-assisted authoring
- maintain evidence-informed content
- preserve institutional memory
- support curriculum development
- continuously strengthen diagnostic education

The value of WEOS ultimately lies not in the software itself, but in the quality of the knowledge it enables institutions to govern.

Closing Principle

The Wardle Editorial Operating System is intentionally designed as an evolving knowledge ecosystem.

Its architecture is founded upon enduring editorial principles rather than transient technologies.

As medicine advances, educational practice develops, and artificial intelligence improves, WEOS should continue to evolve without changing its central purpose:

To enable the transparent, explainable, continuously improving governance of trustworthy clinical knowledge.

Wardle Editorial Canon

Standards for Governed Editorial Artifacts

Part I — Canon Foundations

1. Purpose of the Canon

The Wardle Editorial Canon defines the authoritative editorial meaning, quality expectations, governance requirements, and lifecycle principles of the artifacts governed by the Wardle Editorial Operating System.

The WEOS Architecture defines the overall philosophy and conceptual structure of the editorial system.

The Canon defines how that philosophy applies to specific editorial artifacts.

It establishes:

- what each artifact means
- why each artifact exists
- what educational function it serves
- what dependencies it possesses
- how it may be created
- how it must be reviewed
- what constitutes acceptable quality
- what decisions may be made about it
- when it may influence learner-facing content
- how it is maintained throughout its lifetime

The Canon is not a database specification.

It is not an API contract.

It is not a description of the current user interface.

It is the editorial standard from which those implementations should be derived.

2. Authority of the Canon

The Canon is subordinate to the principles established in the WEOS Architecture and authoritative over artifact-specific editorial behaviour.

Where implementation and Canon diverge, the divergence should be identified explicitly.

Implementation should not silently redefine the meaning of an editorial artifact.

The Canon may therefore describe:

1. Current implementation

How the artifact presently exists within Wardle.

2. Canonical meaning

What the artifact represents within WEOS.

3. Required evolution

Changes necessary to align implementation with the canonical model.

This distinction prevents the Canon from becoming either:

- a passive description of existing code, or
- an aspirational document disconnected from the product.

The Canon connects present implementation to intended editorial architecture.

3. Scope

The Canon applies to every artifact that contributes to the creation, governance, delivery, or maintenance of diagnostic education within Wardle.

These include:

- diagnosis registry entities
- diagnosis metadata
- editorial briefs
- learning goals
- teaching rules
- diagnostic relationships
- evidence relationships
- clinical cases
- case revisions
- clues
- clue-progression assessments
- differentials
- diagnosis education
- educational sections
- graph candidates
- graph facts

- AI-generated drafts
- validation records
- editorial reviews
- publication decisions
- governance history

The Canon also applies to future artifact types introduced into WEOS.

Every new artifact should receive a Canon standard before becoming a mature editorial capability.

4. Foundational Distinctions

The Canon depends upon several distinctions that must remain explicit throughout WEOS.

4.1 Diagnosis and Case

A diagnosis is a governed clinical knowledge entity.

A clinical case is an educational expression of that diagnosis.

A case may teach one diagnosis through a particular presentation, learner level, reasoning pathway, and clue progression.

The diagnosis remains the enduring knowledge entity.

The case remains one instructional implementation.

A diagnosis may exist without a case.

A case should not exist without a resolved or explicitly governed diagnosis relationship.

4.2 Educational Intent and Learner-Facing Content

Educational intent defines what Wardle intends learners to understand.

Learner-facing content communicates that intent.

Educational intent is represented through artifacts such as:

- learning goals
- teaching rules
- required mimics
- required pitfalls

- investigation anchors
- management anchors
- generation guidance

Learner-facing content includes:

- clinical cases
- explanations
- diagnosis education
- recall prompts
- educational summaries

Content should be evaluated against approved educational intent.

Content should not define educational intent accidentally.

4.3 Candidate Knowledge and Governed Knowledge

Candidate knowledge is proposed knowledge awaiting editorial judgement.

Governed knowledge has undergone the required editorial decision process.

Examples of candidate knowledge include:

- a graph candidate
- an AI-generated clue revision
- an unresolved differential mapping
- a draft teaching relationship
- generated diagnosis education
- a proposed teaching rule

Candidate knowledge may be clinically plausible without being institutionally trusted.

Governed knowledge requires an explicit decision.

4.4 Approval and Publication

Approval and publication are separate editorial concepts.

Approval indicates that an artifact has satisfied its required editorial review.

Publication indicates that an approved artifact has been authorised for learner exposure or operational use.

An artifact may be:

- approved but not published
- published and later superseded
- approved but blocked by diagnosis-level dependencies
- editorially valid but not playable
- playable but not scheduled for release

The system must not treat approval, readiness, playability, and publication as synonyms.

4.5 Editorial Status and Operational Status

Editorial status describes governance.

Operational status describes availability or use.

Examples of editorial states include:

- draft
- needs review
- needs edit
- approved
- rejected

Examples of operational states include:

- playable
- generatable
- active
- published
- archived
- scheduled

A diagnosis may be active in the registry while still requiring editorial development.

A case may be approved while not yet ready for publication.

Diagnosis education may be approved while not yet published.

These distinctions must remain visible in both policy and implementation.

5. Universal Artifact Standard

Every governed editorial artifact should be describable through a common canonical structure.

5.1 Identity

Every artifact must possess a stable identity independent of its current wording or revision.

Identity enables:

- revision history
- governance continuity
- relationship tracking
- supersession
- auditability

A revision changes an artifact.

It does not necessarily create a new conceptual artifact.

5.2 Artifact Type

Every artifact must have a clearly defined editorial type.

The artifact type determines:

- its purpose
- its quality standard
- its lifecycle
- its generation policy
- its review requirements
- its publication behaviour

Artifact type should represent editorial meaning rather than merely storage format.

5.3 Diagnosis Context

Every diagnosis-scoped artifact must preserve its relationship to the diagnosis registry.

Where the artifact connects more than one diagnosis, the source and target roles must be explicit.

Examples include:

- differential relationships
- teaching relationships
- mimic relationships
- contradiction relationships
- graph relationships

Textual diagnosis names alone are insufficient where registry resolution is possible.

5.4 Purpose

Every artifact must have an explicit editorial or educational purpose.

Purpose answers:

- Why does this artifact exist?
- Which editorial problem does it solve?
- Which learner outcome does it support?
- Which other artifacts depend upon it?

An artifact without a defined purpose should not become governed knowledge.

5.5 Source and Provenance

Every artifact must preserve how it originated.

Possible sources include:

- manual authoring
- AI-assisted generation
- imported knowledge
- administrative revision
- restoration
- extraction from existing education
- extraction from a case
- migration from legacy content

Source establishes provenance.

It does not establish authority.

5.6 Version and Revision

Artifacts that can change materially must support version or revision history.

Revision history should preserve:

- previous content
- source of the revision
- author or generator

- creation time
- validation outcome
- review decision
- publication relationship

The current version should never erase the governed history of earlier versions.

5.7 Lifecycle State

Every artifact must have a defined lifecycle appropriate to its type.

There is no requirement that all artifacts use the same status vocabulary.

A clinical case, diagnosis education record, graph candidate, differential mapping, and teaching relationship represent different editorial processes and may therefore require different states.

However, every lifecycle must make the following conceptual distinctions possible:

- not yet governed
- awaiting review
- revision required
- accepted
- rejected
- no longer current

Where relevant, the lifecycle should also distinguish:

- validated
 - ready for publication
 - published
 - archived
 - deprecated
 - superseded
 - merged
-

5.8 Dependencies

Every artifact must identify the knowledge and decisions upon which it depends.

Dependencies may include:

- diagnosis metadata
- approved teaching rules
- educational brief

- differential resolution
- supporting evidence
- case revision
- approved graph relationship
- learning-goal coverage
- validation outcome

Dependencies should guide editorial progression.

They should not be hidden implementation assumptions.

5.9 Assessments

Artifacts may be evaluated through one or more editorial assessments.

Examples include:

- clinical correctness
- reasoning quality
- educational value
- evidence sufficiency
- clue coherence
- differential coverage
- learner safety
- publication readiness

Assessments interpret the artifact.

They do not themselves constitute approval.

5.10 Blockers and Warnings

The Canon distinguishes blockers from warnings.

A **blocker** prevents a defined lifecycle transition.

A **warning** identifies risk or incompleteness that does not necessarily prevent transition.

Every blocker should state:

- what requirement failed
- why it matters
- which transition it prevents
- what action can resolve it

Warnings should not be presented as blockers merely because they are important.

Blockers should not be hidden among general quality signals.

5.11 Editorial Decision

Every governed artifact must support explicit editorial decisions appropriate to its lifecycle.

Possible decisions include:

- approve
- reject
- request changes
- request evidence
- resolve
- defer
- merge
- supersede
- deprecate
- archive
- publish
- withdraw

A decision should preserve:

- responsible reviewer
 - decision time
 - rationale
 - affected version
 - supporting assessment
 - resulting state
-

5.12 Available Actions

The actions available for an artifact should be derived from:

- artifact type
- current state
- reviewer role
- unresolved blockers
- dependency state
- applicable policy

The presence of an action should never imply that the action is safe.

Editorial Intelligence should determine whether an action is currently permissible and explain why.

5.13 Audit History

Every meaningful editorial action should contribute to an auditable history.

Audit history should make it possible to reconstruct:

- what existed
- what changed
- who or what changed it
- why it changed
- who reviewed it
- what decision was made
- what was exposed to learners

Audit history is part of the artifact's governance, not secondary metadata.

6. Canonical Knowledge States

Although individual artifacts use different lifecycle vocabularies, WEOS recognises several universal conceptual states.

6.1 Absent

The artifact does not yet exist.

Absence is not failure by itself.

It becomes an editorial deficiency only where policy requires the artifact.

6.2 Suggested

The system or an editor has identified a potential need.

A suggestion is not yet authored content.

Examples include:

- suggested differential
- suggested graph relationship

- suggested case
 - suggested teaching objective
-

6.3 Candidate

A proposed artifact exists but has not completed governance.

Candidate status is appropriate for artifacts such as graph relationships, evidence relationships, and teaching relationships.

6.4 Draft

The artifact has authored content but remains under development.

Draft status does not imply that formal review has begun.

6.5 Generated

The artifact was produced through an automated or AI-assisted process.

Generated describes origin or production state.

It must not be interpreted as approval.

6.6 Awaiting Review

The artifact is sufficiently complete for human editorial evaluation.

Examples may be expressed in implementation as:

- review
 - needs review
 - pending review
 - review required
-

6.7 Revision Required

A reviewer or validator has determined that changes are required.

Examples may be expressed as:

- needs edit
- request changes
- failed validation
- unresolved
- ambiguous

Revision required should always include actionable rationale.

6.8 Approved

The artifact has satisfied its defined editorial review.

Approval applies to a specific artifact version.

Approval does not automatically imply publication, playability, activation, or scheduling.

6.9 Ready for Operational Use

The artifact has satisfied the dependencies required for its intended operational role.

Examples include:

- ready to publish
- playable
- generatable
- active
- ready for review at diagnosis level

Readiness must always identify the operation for which the artifact is ready.

6.10 Published

The artifact is authorised for learner-facing or institutional use.

Publication should always be traceable to an approved version and explicit publication decision.

6.11 Rejected

The artifact has been reviewed and not accepted into governed knowledge.

Rejection should preserve the artifact and decision history unless policy explicitly permits deletion.

6.12 Superseded, Deprecated, or Archived

The artifact is no longer the preferred current representation.

These states have different meanings:

- **Superseded** — replaced by a newer governed artifact.
- **Deprecated** — retained but no longer recommended for ongoing use.
- **Archived** — removed from active use while history is preserved.

The Canon should not use these terms interchangeably.

6.13 Merged

The artifact has been consolidated into another artifact.

Merged artifacts should preserve:

- original identity
 - destination artifact
 - merge rationale
 - governance decision
 - provenance contribution
-

7. Current Wardle Lifecycle Vocabulary

Current Wardle implementation already contains several distinct lifecycle models.

Clinical Cases

Cases progress through states including:

- Draft
- Validating
- Validated
- Review
- Needs Edit
- Approved
- Ready to Publish

- Published
- Rejected

This lifecycle combines editorial review, validation, publication readiness, and publication state.

The Canon should preserve the meaning of these stages while ensuring that approval and publication remain distinct.

Diagnosis Education

Diagnosis education currently supports:

- Draft
- Generated
- Needs Review
- Needs Edit
- Approved
- Published
- Rejected
- Archived

This lifecycle explicitly recognises generated content as separate from reviewed content.

Graph Candidates

Graph candidates currently support:

- Candidate
- Approved
- Rejected
- Merged

A graph candidate may be promoted into an active graph fact.

Teaching Relationships

Teaching relationships currently support:

- Candidate
- Needs Review
- Active
- Rejected
- Deprecated

The use of `Active` reflects operational graph participation and should not be treated as identical to generic approval.

Differential Resolution

Differential mappings currently support:

- Resolved
- Ambiguous
- Unresolved
- Rejected

These states describe identity resolution rather than editorial publication.

Diagnosis Editorial Onboarding

Diagnosis onboarding currently progresses through:

- New
- Rules Started
- Brief Started
- Education Started
- Case Started
- Ready for Review
- Complete

This is a diagnosis-development progression, not the lifecycle of a single artifact.

The Canon should retain these conceptual differences rather than force every artifact into one universal enum.

8. Canonical Source Categories

WEOS recognises several broad source categories.

Human Authored

Produced directly by an editor or clinical author.

AI Assisted

Produced with meaningful AI contribution and subject to human governance.

Generated

Produced primarily through an automated generation process.

Imported

Introduced from an external structured or unstructured source.

Extracted

Derived from an existing Wardle artifact, such as graph candidates extracted from diagnosis education or case reasoning.

Administrative Revision

Produced through direct editorial repair, migration, or controlled administrative correction.

Restored

Recovered from an earlier trusted source or historical version.

Source categories should remain separate from review status and publication status.

9. Canonical Artifact Families

The Canon groups artifacts into six families.

9.1 Diagnosis Foundation Artifacts

- Diagnosis Registry Entry
- Diagnosis Metadata
- Diagnosis Editorial Brief
- Learning Goals
- Teaching Rules

9.2 Clinical Case Artifacts

- Clinical Case
- Case Revision
- Case Clue
- Case Explanation
- Case Differential
- Clue Progression Analysis
- Clue-Discriminator Annotation
- Learning-Goal Coverage
- Validation Run
- Case Review

9.3 Diagnosis Education Artifacts

- Diagnosis Education
- Summary
- Clinical Pattern
- Key Symptoms
- Key Signs
- Examination Pearls
- Scoring Systems
- Investigations
- Differentials
- Management
- Complications
- Pitfalls
- Recall Prompts
- References

9.4 Knowledge and Reasoning Artifacts

- Graph Candidate
- Graph Fact
- Teaching Relationship
- Evidence Node
- Evidence Relationship
- Reasoning Path
- Diagnostic Comparison
- Differential Link

9.5 Candidate and AI Draft Artifacts

- AI Draft Audit
- Clue Revision Draft
- Generated Education Draft
- Suggested Teaching Rule
- Suggested Relationship
- Suggested Case

9.6 Governance Artifacts

- Editorial Assessment
 - Editorial Review
 - Editorial Decision
 - Validation Result
 - Publication Decision
 - Governance History
-

10. Canonical Review Principle

Every artifact must be reviewed according to its own purpose.

A case should not be reviewed using only education-section standards.

A graph relationship should not be reviewed as though it were prose.

A teaching rule should not be assessed merely for factual correctness.

A differential mapping should not be approved solely because the text appears clinically plausible.

The Canon therefore defines artifact-specific quality standards while preserving a common governance structure.

11. Canonical Publication Principle

Not every artifact is directly learner-facing.

Artifacts may be classified as:

Learner-Facing

Directly visible to learners.

Examples include:

- cases
- explanations
- diagnosis education
- recall prompts

Learner-Influencing

Not necessarily visible but materially shaping learner-facing content.

Examples include:

- teaching rules
- learning goals
- reasoning paths
- graph relationships
- generation guidance

Editorial-Only

Used to govern or assess content.

Examples include:

- validation runs
- draft audits
- review records
- unsupported-claim signals
- readiness assessments

Publication policy must account for all three categories.

Editorial-only artifacts may block publication despite never being shown to learners.

12. Normative Statements

The following statements apply throughout the Canon.

- Every governed artifact **MUST** possess a stable editorial identity.
 - Every artifact **MUST** have a defined purpose.
 - Every artifact **MUST** preserve its source or provenance.
 - Every artifact **MUST** have an appropriate lifecycle.
 - Every material revision **MUST** preserve history.
 - Every approval **MUST** apply to an identifiable artifact version.
 - Every editorial decision **MUST** preserve reviewer accountability and rationale.
 - AI-generated content **MUST NOT** become governed knowledge without human editorial review.
 - Approval **MUST NOT** be treated as synonymous with publication.
 - Publication **MUST NOT** occur without an explicit publication decision.
 - Candidate knowledge **MUST NOT** be presented as institutionally trusted knowledge.
 - Rejected and superseded artifacts **SHOULD** remain historically traceable.
 - Lifecycle states **MUST** reflect the meaning of the artifact rather than convenience of implementation.
 - Blockers **MUST** identify the transition they prevent.
 - Readiness conclusions **MUST** be explainable.
 - Learner-facing artifacts **MUST** be traceable to governed editorial intent.
 - Implementations **SHOULD** reuse canonical terminology wherever doing so does not erase necessary domain distinctions.
-

13. Relationship to Subsequent Parts

The remaining sections of the Canon apply these foundations to individual artifact families.

Part II defines Diagnosis Governance standards.

Part III defines Clinical Case standards.

Part IV defines Diagnosis Education standards.

Part V defines Knowledge and Reasoning Governance standards.

Part VI defines AI Draft standards.

Part VII defines Governance Record standards.

Each artifact chapter should use the universal standard established in this Part while specifying its own quality requirements, dependencies, decisions, and lifecycle.

Closing Principle

The purpose of the Wardle Editorial Canon is not to make every artifact identical.

Its purpose is to make every artifact understandable, governable, and trustworthy.

Artifacts differ in function, lifecycle, and educational role.

They are unified by explicit purpose, transparent provenance, versioned history, appropriate review, explainable decisions, and accountable governance.

Wardle Editorial Canon

Standards for Governed Editorial Artifacts

Part II — Diagnosis Governance

1. Purpose

Diagnosis Governance defines how a diagnosis enters, develops within, and remains governed by the Wardle Editorial Operating System.

The diagnosis is the central organising entity of WEOS.

Cases, education, teaching rules, reasoning paths, differentials, evidence relationships, graph artifacts, and publication decisions all depend upon a stable diagnosis identity.

Diagnosis Governance therefore establishes the foundation upon which every other editorial capability operates.

Its purpose is to ensure that each diagnosis is:

- clinically identifiable
- terminologically consistent
- appropriately classified
- educationally scoped
- suitable for intended Wardle use
- linked to its governed editorial artifacts
- developed through an explicit onboarding process
- protected from duplication or ambiguous identity
- continuously maintainable

Diagnosis Governance does not require every diagnosis to possess complete educational content at the moment of registration.

It requires that every diagnosis possess a trustworthy identity and a clear path toward editorial development.

2. Diagnosis as the Primary Governance Entity

A diagnosis within WEOS is not merely a label attached to cases.

It is a first-class governed clinical knowledge entity.

The diagnosis provides the enduring identity around which the editorial system organises:

- terminology
- taxonomy
- aliases
- cases
- diagnosis education
- teaching rules
- learning goals
- reasoning paths
- differential relationships
- graph candidates and facts
- evidence relationships
- AI draft history
- publication readiness
- maintenance activity

Cases may change.

Education may be revised.

Teaching relationships may be added or deprecated.

The diagnosis remains the stable conceptual centre of this knowledge.

A diagnosis should therefore possess an identity independent of any single case, title, explanation, or educational section.

3. Diagnosis Registry Entry

3.1 Purpose

The Diagnosis Registry Entry is the authoritative identity record for a diagnosis within WEOS.

It establishes which clinical concept Wardle recognises as a distinct diagnosis and provides the stable reference through which related editorial artifacts are connected.

The registry is not merely a search dictionary.

It is the identity and taxonomy foundation of the editorial system.

3.2 Canonical responsibilities

A Diagnosis Registry Entry should establish:

- canonical diagnosis identity
- preferred learner-facing label
- normalised identity for matching and deduplication
- accepted aliases
- classification and taxonomy
- editorial development status
- operational availability
- suitability for gameplay
- suitability for automated generation
- generation guidance
- governance history

The registry should answer:

- What diagnosis is this?
 - Is it distinct from other registry entries?
 - How should it be displayed?
 - How may users or systems refer to it?
 - Which specialty and clinical context does it belong to?
 - May it appear as a Wardle answer?
 - May new cases be generated for it?
 - Has its diagnosis-level editorial foundation been developed?
-

3.3 Identity standard

Every registry diagnosis must possess:

- a stable identifier
- a canonical clinical name
- a normalised canonical identity
- a display label
- a registry status
- an active or inactive operational state

The canonical name should represent the preferred clinical diagnosis rather than a descriptive case title.

Examples of acceptable canonical identities include:

- Chronic Pulmonary Aspergillosis
- Unstable Angina
- Neurogenic Bladder
- Scaphoid Fracture

Examples of unsuitable canonical identities include:

- Chest Pain in a Middle-Aged Man
- Progressive Fatigue with Hyperpigmentation
- Swollen Painful Wrist Following a Fall
- Fever, Murmur, and Splinter Haemorrhages

Descriptive phrases may be appropriate as case titles.

They should not replace diagnosis identity.

3.4 Canonical and display identity

The canonical diagnosis name and learner-facing display label may differ where clinically justified.

The canonical name exists for governance and identity.

The display label exists for clear presentation.

Differences should be deliberate rather than accidental.

The display label must not:

- introduce diagnostic ambiguity
 - use non-standard terminology without justification
 - conceal the true diagnosis behind a narrative phrase
 - conflict with accepted aliases
 - imply a subtype that the registry entry does not represent
-

3.5 Normalisation

Every diagnosis should possess a normalised identity suitable for:

- duplicate detection
- alias resolution
- case mapping
- differential mapping
- search
- migration
- graph linking

Normalisation should remove superficial variation without combining clinically distinct diagnoses.

Normalisation may address:

- letter case
- punctuation
- spacing
- common spelling variants
- accepted abbreviation formats

Normalisation must not erase meaningful clinical distinctions.

For example:

- Type 1 Diabetes Mellitus and Type 2 Diabetes Mellitus must remain distinct.
 - Crohn Disease and Ulcerative Colitis must remain distinct.
 - Stable Angina and Unstable Angina must remain distinct.
 - Acute and chronic forms should not be merged where they carry different educational meaning.
-

4. Diagnosis Aliases

4.1 Purpose

Aliases allow multiple clinically acceptable terms to resolve to one governed diagnosis identity.

Aliases support:

- learner answer matching
- case mapping
- search
- imported content
- terminology variation
- abbreviation handling
- legacy migration

Aliases do not create additional diagnoses.

They provide alternative paths to the same diagnosis.

4.2 Alias categories

The Canon recognises the following alias purposes.

Canonical

The authoritative diagnosis name.

Accepted

A clinically valid alternative term that may be accepted in search or learner answers.

Abbreviation

A recognised shortened form.

Search Only

A term useful for discovery or legacy matching but unsuitable as a preferred learner-facing answer.

4.3 Alias quality requirements

An alias should be:

- clinically equivalent to the canonical diagnosis
- unambiguous within the intended context
- appropriately classified
- free from unnecessary descriptive phrasing
- traceable to the correct registry entry

An alias must not be used to hide distinct clinical concepts within one diagnosis record.

Where two terms overlap but are not truly equivalent, they should be represented through relationships rather than aliases.

4.4 Alias governance

Alias creation should be reviewed where the term could affect:

- learner answer acceptance
- differential resolution
- graph identity
- duplicate detection
- imported content mapping

Aliases that are misleading, outdated, overly broad, or clinically unsafe should be rejected or classified as search-only.

5. Diagnosis Registry Status

The diagnosis registry lifecycle describes the standing of the diagnosis identity itself.

It does not describe the completeness of its cases or education.

5.1 Draft

The registry identity is being prepared or reviewed and should not yet participate fully in operational use.

5.2 Active

The diagnosis is recognised as a valid registry entity.

Active status does not automatically mean:

- editorial development is complete
- cases exist
- cases are playable
- education is published
- automated generation is safe

5.3 Hidden

The diagnosis remains in the registry but should not appear in ordinary discovery or selection contexts.

Hidden status may be appropriate for:

- internal mapping records
- incomplete migration targets
- restricted editorial concepts
- temporarily suppressed diagnoses

5.4 Deprecated

The registry identity should no longer be used for new editorial work.

A deprecated diagnosis may be preserved because of:

- historical cases
- legacy mappings
- audit requirements
- replacement by another registry diagnosis
- terminology evolution

Deprecation should identify the preferred replacement where one exists.

6. Operational Availability

Registry status and operational availability must remain separate.

6.1 Active flag

The active flag determines whether the diagnosis currently participates in operational registry use.

An active registry identity should not be assumed to possess complete educational governance.

6.2 Playability

Playability determines whether the diagnosis may appear as the correct answer in a learner-facing case.

A diagnosis may be non-playable because it is:

- overly broad
- primarily an alias or mapping concept
- educationally unsuitable as a final answer
- reserved for future development
- insufficiently specific
- not intended for gameplay

Playability is a diagnosis-level permission.

It does not establish that any particular case is playable.

Both the diagnosis and the case must independently satisfy their playability requirements.

6.3 Generatability

Generatability determines whether automated systems are permitted to create new case drafts for the diagnosis.

A diagnosis may be playable but not generatable.

This distinction is appropriate where:

- cases require specialist nuance
- automated generation creates a high risk of ambiguity
- the diagnosis is clinically heterogeneous

- evidence or reasoning foundations are incomplete
- generation policies prohibit automated authoring

Generatability permits draft production.

It does not permit publication.

7. Diagnosis Metadata

7.1 Purpose

Diagnosis metadata establishes the clinical, educational, and operational context of a registry diagnosis.

Metadata supports:

- curriculum organisation
- specialty ownership
- case balancing
- generation guidance
- difficulty sequencing
- search
- coverage analysis
- prioritisation
- portfolio governance

Metadata should describe the diagnosis itself rather than the characteristics of one particular case.

7.2 Core classification metadata

Diagnosis metadata may include:

- category
- specialty
- subspecialty
- body system
- organ system
- difficulty band
- rarity band
- typical clinical setting
- relevant age group
- urgency level
- ICD coding
- descriptive or compositional status
- search priority

7.3 Specialty

Specialty identifies the broad clinical discipline responsible for the diagnosis.

Examples include:

- Cardiology
- Neurology
- Endocrinology
- Respiratory Medicine
- Orthopaedics
- General Surgery
- Paediatrics
- Psychiatry

Specialty should support editorial ownership and curriculum grouping.

It should not be used as a substitute for body system or category.

7.4 Subspecialty

Subspecialty provides a more specific clinical grouping within the specialty.

Examples include:

- Electrophysiology
- Heart Failure
- Neuromuscular Medicine
- Movement Disorders
- Metabolic Bone Disease
- Urogynaecology

Subspecialty should only be used where it adds meaningful editorial organisation.

7.5 Category

Category describes the educational or disease grouping of the diagnosis.

Examples include:

- Infectious
- Autoimmune
- Metabolic

- Vascular
- Neoplastic
- Degenerative
- Traumatic

Category should support knowledge organisation across specialties.

7.6 Body and organ systems

Body system describes the broad physiological or anatomical system.

Organ system provides a more granular organ or anatomical focus where useful.

These fields should not duplicate specialty without adding meaning.

For example:

- Specialty: Neurology
 - Body system: Nervous System
 - Organ system: Peripheral Nerve
-

7.7 Difficulty band

Difficulty band describes the expected educational complexity of the diagnosis.

The current canonical bands are:

- Basic
- Intermediate
- Advanced

Difficulty should consider:

- complexity of recognition
- breadth of differentials
- reasoning depth
- required prior knowledge
- subtlety of discriminators
- management complexity

Difficulty should not be based solely on disease rarity.

7.8 Rarity band

Rarity supports balanced curriculum and case planning.

The current canonical bands are:

- Common
- Uncommon
- Rare

Rarity should not determine educational importance by itself.

Rare diagnoses may be essential where they are:

- clinically dangerous
 - frequently confused
 - important examination topics
 - necessary exclusions
 - central to escalation reasoning
-

7.9 Clinical setting

Clinical setting describes where the diagnosis is typically encountered or most appropriately taught.

Current categories include:

- Outpatient
- Emergency
- Inpatient
- Intensive Care
- Community

This metadata should guide case framing without preventing alternative presentations.

7.10 Age group

Age group identifies the population in which the diagnosis is educationally most relevant.

Current categories include:

- Paediatric
- Adult
- Geriatric
- Any

Age-group metadata should not prohibit clinically valid atypical cases.

Such cases should, however, be intentionally authored.

7.11 Urgency

Urgency describes the usual clinical priority associated with recognising the diagnosis.

Current categories include:

- Routine
- Urgent
- Emergent

Urgency should influence:

- case framing
 - escalation teaching
 - management emphasis
 - prioritisation of review
 - learner-safety requirements
-

8. Generation Guidance

8.1 Purpose

Diagnosis-level generation guidance constrains how new cases and educational content should be drafted.

It provides reusable diagnosis-specific direction rather than relying on each generation request to rediscover the clinical structure.

Generation guidance may include:

- preferred clue types
- excluded clue types
- difficulty guidance
- presentation constraints
- required discriminators
- required mimics
- common errors to avoid
- appropriate settings
- age guidance
- urgency guidance

8.2 Preferred clue types

Preferred clue types identify information domains that are educationally central to diagnosing the condition.

Examples include:

- history
- physical examination
- laboratory testing
- imaging
- pathology
- risk factors
- clinical course

Preferred clues guide generation.

They should not require every case to use the same sequence.

8.3 Excluded clue types

Excluded clue types identify information that should be avoided or de-emphasised.

This may be appropriate where a clue type:

- reveals the diagnosis too early
- is not central to clinical reasoning
- creates dependence on specialist testing
- produces repetitive cases
- does not suit the intended learner level

Exclusion should be educationally justified.

It should not merely reflect absent data in existing cases.

9. Diagnosis Editorial Onboarding

9.1 Purpose

Diagnosis Editorial Onboarding represents the development of a registry diagnosis from an identity record into an editorially usable diagnosis foundation.

Onboarding is not publication.

It is not the lifecycle of any single artifact.

It is the coordinated progress of diagnosis-level editorial development.

9.2 Onboarding stages

New

The diagnosis has entered the registry but diagnosis-level editorial development has not meaningfully begun.

Rules Started

Teaching rules have begun to define the diagnosis's educational constraints and priorities.

Brief Started

The editorial brief has begun to consolidate learning goals, required rules, mimics, pitfalls, investigations, management anchors, and generation guidance.

Education Started

Diagnosis education has entered drafting or generation.

Case Started

At least one clinical case has entered authoring or generation.

Ready for Review

The diagnosis possesses sufficient scaffold and content for coordinated diagnosis-level review.

Complete

The required onboarding components have satisfied the defined completion policy.

Completion does not necessarily mean that:

- all future cases have been written
- all graph work is finished
- maintenance is no longer required
- every artifact is published

It means the diagnosis has completed its initial editorial foundation.

9.3 Onboarding principles

Onboarding should:

- reveal missing foundational work
- guide the next appropriate editorial action
- distinguish absence from failure
- avoid treating generated drafts as approved artifacts
- remain explainable
- preserve independent artifact lifecycles
- support iterative progress

Onboarding should not become a second publication lifecycle.

10. Diagnosis Editorial Brief

10.1 Purpose

The Diagnosis Editorial Brief is the diagnosis-level specification that translates registry identity into educational intent.

It is the editorial contract for future case generation, education development, reasoning design, and graph development.

The brief answers:

- What must learners understand?
 - Which teaching rules are required?
 - Which mimics must be addressed?
 - Which pitfalls must be prevented?
 - Which investigations are educationally important?
 - Which management anchors must be taught?
 - What difficulty is appropriate?
 - How should cases be generated?
 - How should education be structured?
 - What graph development is needed?
-

10.2 Relationship to other artifacts

The brief should guide:

- teaching-rule selection
- case generation
- learning-goal coverage
- diagnosis education
- reasoning paths
- differential relationships
- graph candidates
- publication assessment

The brief does not replace these artifacts.

It coordinates them.

10.3 Summary

The brief summary should define the diagnosis's core editorial identity.

It should communicate:

- what makes the diagnosis educationally important
- the central recognition pattern
- the primary reasoning challenge
- major learner risks
- the intended scope of Wardle coverage

The summary should not duplicate the learner-facing diagnosis education summary.

It is written for editorial use.

10.4 Learning goals

Learning goals define what the learner should be able to recognise, distinguish, interpret, or act upon.

High-quality learning goals should be:

- diagnosis-specific
- clinically meaningful
- assessable through cases or recall
- linked to reasoning rather than trivia
- appropriately scoped

- capable of coverage analysis

Examples include:

- Recognise the pattern of chronic pulmonary symptoms with cavitary lung disease and fungal evidence.
- Distinguish unstable angina from myocardial infarction using symptoms, biomarkers, and electrocardiography.
- Identify neurogenic bladder in a patient with neurological dysfunction and abnormal voiding.

Learning goals should not be vague statements such as:

- Understand the diagnosis.
 - Know the symptoms.
 - Learn management.
-

10.5 Required teaching rules

The brief should identify the teaching rules that must govern downstream content.

A required teaching rule may constrain:

- acceptable presentations
- required differentials
- expected evidence
- clue timing
- educational emphasis
- case generation
- graph development

A brief should reference governed teaching rules rather than duplicating their full content.

10.6 Required mimics

Required mimics define diagnoses that should be considered in the diagnosis's educational framework.

A mimic may be required because it is:

- commonly confused with the diagnosis
- clinically dangerous to miss
- central to diagnostic discrimination
- important for examinations
- necessary for realistic case reasoning

Required mimic lists should not become indiscriminate differential catalogues.

They should prioritise educational value.

10.7 Required pitfalls

Required pitfalls identify learner errors the diagnosis must explicitly guard against.

Examples include:

- anchoring on a common diagnosis despite contradictory evidence
- over-relying on a single investigation
- confusing a complication with the primary diagnosis
- interpreting a normal early test as exclusion
- missing an atypical presentation
- exposing the answer through an overly specific early clue

Pitfalls in the editorial brief define educational requirements.

Learner-facing pitfalls are addressed separately within Diagnosis Education.

10.8 Investigation anchors

Investigation anchors define the diagnostic tests or findings that are central to teaching the diagnosis.

They may identify:

- first-line investigations
- confirmatory investigations
- characteristic findings
- important limitations
- tests that distinguish mimics
- resource-limited alternatives

Investigation anchors guide cases and education.

They should not become complete clinical protocols.

10.9 Management anchors

Management anchors define the management principles that Wardle should consistently teach.

They may include:

- immediate stabilisation priorities
- first-line treatment
- definitive management
- escalation thresholds
- contraindicated actions
- specialist referral
- prevention of complications

Management anchors should remain appropriate to the educational scope of Wardle and the target learner level.

10.10 Difficulty guidance

Difficulty guidance explains how the diagnosis should be adapted across learner levels.

It may define:

- which presentations suit basic learners
- which mimics suit intermediate cases
- which atypical patterns suit advanced cases
- how clue specificity should change
- when specialist investigations are acceptable
- how much ambiguity is educationally appropriate

Difficulty guidance should produce different reasoning challenges rather than merely more obscure facts.

10.11 Case-generation guidance

Case-generation guidance defines the boundaries within which cases should be authored.

It may describe:

- preferred presentation
- acceptable clinical settings
- required reasoning objective
- clue sequencing expectations
- prohibited early disclosures
- required mimics
- expected discriminator timing
- management or escalation focus
- desired case diversity

Generation guidance should enable variation without losing educational identity.

10.12 Education guidance

Education guidance defines the expected emphasis of learner-facing diagnosis education.

It may identify:

- sections requiring particular depth
- high-risk misconceptions
- required differentials
- essential investigations
- management priorities
- scoring systems
- examination pearls
- recall themes

Education guidance should improve coherence across independently generated sections.

10.13 Graph guidance

Graph guidance defines which relationships should be represented or explored.

It may identify:

- priority mimics
- shared presentations
- discriminator relationships
- evidence relationships
- complication relationships
- management contrasts
- recall relationships

Graph guidance should direct candidate generation.

It should not automatically establish graph facts.

11. Teaching Rules

11.1 Purpose

A Diagnosis Teaching Rule is a governed diagnosis-specific instruction that constrains or guides content generation, review, education, reasoning, or graph development.

Teaching rules convert clinical and educational judgement into reusable editorial policy.

They exist to prevent the editorial system from repeatedly rediscovering the same requirements.

11.2 Teaching-rule structure

A teaching rule should possess:

- stable identity
 - title
 - category
 - importance
 - rationale
 - acceptable manifestations
 - required differentials
 - expected evidence
 - difficulty guidance
 - timing constraints
 - applicability scope
 - lifecycle state
 - source
 - version
-

11.3 Stable key

A teaching rule should possess a stable key independent of its wording.

This allows the rule to be:

- revised
- referenced by the editorial brief
- linked to evidence
- linked to teaching relationships
- used during generation
- used during validation
- measured for coverage

Changing the title should not break the rule's identity.

11.4 Category

Teaching-rule categories should represent clinically meaningful editorial concerns.

Possible categories include:

- presentation
- discriminator
- differential
- investigation
- management
- complication
- escalation
- clue progression
- learner misconception
- generation constraint

Categories should support governance and analysis rather than visual organisation alone.

11.5 Importance

Importance identifies the consequence of failing to satisfy the rule.

A rule may be:

- essential
- high priority
- recommended
- contextual

Essential rules may block approval or publication where applicable.

Importance should not be inferred solely from display order.

11.6 Rationale

Every teaching rule should explain why it exists.

The rationale should identify:

- the clinical risk
- the educational objective
- the common error being prevented
- the reasoning value
- the governance consequence

A rule without rationale becomes difficult to review, maintain, or safely apply.

11.7 Acceptable manifestations

Acceptable manifestations describe clinically valid ways the rule may appear in content.

They provide controlled flexibility.

For example, a teaching rule about heart failure presentation may permit:

- exertional dyspnoea
- orthopnoea
- paroxysmal nocturnal dyspnoea
- peripheral oedema

The rule should not require every manifestation in every case.

11.8 Required differentials

A teaching rule may identify differentials that must be represented or considered.

This may affect:

- case explanation
- clue progression
- reasoning paths
- education
- graph development

Required differentials should resolve to registry diagnoses where possible.

11.9 Expected evidence

Expected evidence identifies findings that should support or discriminate the diagnosis.

This may include:

- symptoms
- signs
- laboratory findings
- imaging
- risk factors
- clinical course
- management response

Expected evidence can guide both generation and validation.

11.10 Avoid-too-early constraint

Some findings are diagnostically valuable but educationally harmful when introduced too early.

The avoid-too-early constraint should be used where a clue:

- reveals the diagnosis prematurely
- collapses meaningful differential reasoning
- prevents progressive uncertainty
- converts a reasoning case into recall of a pathognomonic phrase

This constraint should influence clue sequencing rather than exclude the evidence entirely.

11.11 Applicability

A teaching rule may apply to:

- diagnosis education
- case generation
- graph development
- more than one of these domains

Applicability must be explicit.

A rule intended for case generation should not automatically block education publication unless policy states that it applies there.

11.12 Teaching-rule lifecycle

Teaching rules should support at least the following conceptual states:

- proposed
- awaiting review
- active
- rejected
- deprecated

An active teaching rule is available to influence editorial work.

Deprecation should preserve history and identify replacement where applicable.

12. Learning Goals

12.1 Canonical status

Learning goals currently exist within the Diagnosis Editorial Brief and are measured through case learning-goal coverage.

They should therefore be treated as governed diagnosis-level educational requirements even where they are stored as structured brief content rather than separate first-class records.

12.2 Purpose

Learning goals define the intended educational outcomes for a diagnosis.

They establish what cases and education should collectively teach.

Learning goals support:

- case design
 - case coverage assessment
 - content generation
 - review
 - curriculum analysis
 - recall design
 - publication readiness
-

12.3 Quality requirements

A learning goal should:

- describe an observable clinical reasoning outcome
 - be specific to the diagnosis
 - be teachable through Wardle content
 - avoid vague educational verbs
 - identify meaningful learner capability
 - remain stable enough for longitudinal coverage analysis
-

12.4 Coverage

A diagnosis should not be considered educationally complete merely because learning goals exist.

Coverage should identify:

- which cases support the goal
- how strongly each case supports it
- which discriminators are covered
- which mimics are covered
- which requirements remain missing

Coverage is an assessment of implementation.

It is not the learning goal itself.

13. Diagnosis-Level Readiness

13.1 Purpose

Diagnosis-Level Readiness interprets whether the diagnosis foundation is sufficient for a defined next step.

Readiness must always be qualified.

Examples include:

- ready for scaffold generation
- ready for education generation
- ready for case generation
- ready for coordinated review
- ready for publication assessment

There is no single universal diagnosis-ready state.

13.2 Metadata readiness

A diagnosis is metadata-ready when its identity and required classification are sufficiently complete for downstream editorial development.

Metadata readiness may require:

- valid canonical identity
- resolved duplicate risk
- appropriate specialty
- appropriate taxonomy
- difficulty band
- operational eligibility

- reviewed playability
 - reviewed generatability
-

13.3 Scaffold readiness

A diagnosis is scaffold-ready when enough diagnosis context exists to generate or author its educational foundation.

This may require:

- metadata readiness
 - clear educational scope
 - no unresolved identity conflict
 - generation eligibility where AI is used
-

13.4 Case-generation readiness

A diagnosis is ready for case generation when the required educational and reasoning foundation exists.

Depending on policy, this may require:

- approved or reviewable editorial brief
- defined learning goals
- required teaching rules
- required mimics
- generation guidance
- sufficient reasoning support
- no blocking identity issue

Case-generation readiness authorises draft creation.

It does not authorise case approval or publication.

13.5 Education-generation readiness

A diagnosis is ready for education generation when its educational intent and clinical scope are sufficiently defined.

This may require:

- registry identity
- metadata

- editorial brief
 - teaching rules
 - required differentials
 - relevant investigation and management anchors
-

13.6 Ready for diagnosis review

A diagnosis is ready for coordinated review when the minimum required artifacts exist in reviewable form.

This may include:

- registry metadata
- editorial brief
- teaching rules
- diagnosis education
- at least one case or justified absence
- identified differentials
- relevant reasoning and graph context

Generated but unreviewed content may contribute to review readiness.

It must not contribute to approval.

13.7 Onboarding completion

Onboarding completion indicates that the diagnosis has established its required initial editorial foundation.

Completion policy should be explicit and configurable.

It should not be inferred merely from the presence of records.

Rejected, incomplete, or placeholder artifacts should not satisfy completion requirements.

14. Diagnosis Governance Decisions

Diagnosis Governance may produce decisions including:

- activate registry entry
- hide registry entry
- deprecate registry entry
- approve metadata
- request metadata changes

- confirm playability
- restrict playability
- confirm generatability
- restrict generatability
- approve alias
- reject alias
- merge duplicate diagnoses
- approve editorial brief
- request brief revision
- activate teaching rule
- deprecate teaching rule
- mark onboarding complete
- reopen onboarding

Each decision should preserve:

- reviewer
- rationale
- affected version or state
- resulting status
- related diagnosis where relevant
- date and time

15. Duplicate and Merge Governance

15.1 Purpose

Duplicate diagnosis entries fragment knowledge and threaten editorial consistency.

Duplicate governance should consider:

- canonical name similarity
 - alias overlap
 - clinical equivalence
 - subtype distinctions
 - case mappings
 - education
 - graph relationships
 - teaching rules
 - historical use
-

15.2 Merge principle

Diagnoses should only be merged where they represent the same clinical concept.

Similarity alone is insufficient.

The merge process should preserve:

- aliases
- case relationships
- education history
- differential mappings
- graph provenance
- teaching relationships
- audit history
- source identity

The merged diagnosis should remain historically traceable.

15.3 Non-merge examples

Diagnoses should not be merged where they differ meaningfully in:

- pathophysiology
- clinical presentation
- severity
- age group
- management
- educational reasoning
- accepted diagnostic criteria

Where related but distinct diagnoses exist, WEOS should use knowledge relationships rather than merging identity.

16. Diagnosis Maintenance

Diagnosis Governance continues after onboarding.

Maintenance may be triggered by:

- terminology changes
- revised classification
- duplicate detection

- guideline changes
- new educational priorities
- learner confusion
- graph inconsistency
- curriculum restructuring
- new case needs
- evidence changes

Maintenance may affect metadata, briefs, teaching rules, aliases, or operational permissions without requiring the diagnosis identity to be replaced.

17. Quality Standard for a Governed Diagnosis Foundation

A high-quality diagnosis foundation should be:

Clinically identifiable

The diagnosis is represented by a clear, accepted clinical identity.

Terminologically coherent

Canonical names, display labels, and aliases are deliberate and internally consistent.

Appropriately classified

Specialty, category, body system, difficulty, rarity, setting, age group, and urgency are meaningful.

Educationally scoped

The editorial brief defines what learners should understand and how content should teach it.

Governed by reusable rules

Teaching rules constrain and guide downstream content.

Operationally explicit

Playability, generatability, registry activity, and onboarding status are independently understood.

Relationship-ready

The diagnosis can participate safely in differentials, graph relationships, evidence relationships, and reasoning paths.

Continuously maintainable

Changes can occur without erasing identity or governance history.

18. Normative Statements

- Every diagnosis **MUST** possess a stable Diagnosis Registry identity.
 - Every playable case **MUST** resolve to a governed diagnosis identity.
 - Canonical diagnosis names **MUST NOT** be replaced by descriptive case titles.
 - Diagnosis aliases **MUST** represent clinically equivalent terminology.
 - Clinically distinct diagnoses **MUST NOT** be merged solely because their names or presentations are similar.
 - Registry activation **MUST NOT** be interpreted as editorial completion.
 - Playability **MUST** remain distinct from diagnosis activation.
 - Generatability **MUST** remain distinct from playability.
 - Automated generation **MUST NOT** occur for a diagnosis prohibited from generation.
 - Diagnosis metadata **SHOULD** describe the diagnosis rather than one specific case.
 - Learning goals **MUST** define meaningful educational outcomes.
 - Teaching rules **MUST** preserve stable identity and rationale.
 - The Editorial Brief **MUST** coordinate educational intent without replacing downstream artifacts.
 - Generated diagnosis artifacts **MUST NOT** count as approved merely because they exist.
 - Diagnosis readiness **MUST** identify the specific transition or operation for which the diagnosis is ready.
 - Onboarding completion **MUST NOT** replace artifact-level approval or publication governance.
 - Diagnosis merges **MUST** preserve provenance and historical relationships.
 - Diagnosis governance decisions **MUST** remain auditable.
-

Closing Principle

A diagnosis is the enduring centre of Wardle’s editorial knowledge.

Before Wardle can safely generate, review, teach, or publish a diagnosis, it must first know exactly what the diagnosis is, how it should be classified, what learners should understand about it, and which rules should govern its educational expression.

Diagnosis Governance transforms a clinical label into a stable, purposeful, and governable foundation for diagnostic education.

Wardle Editorial Canon

Standards for Governed Editorial Artifacts

Part III — Clinical Case Standards

1. Purpose

Clinical Case Standards define how cases are authored, generated, revised, evaluated, approved, made playable, published, and maintained within the Wardle Editorial Operating System.

A Wardle clinical case is not merely a collection of symptoms followed by a diagnosis.

It is a structured educational experience designed to develop diagnostic reasoning through progressive disclosure.

Every case should require the learner to:

- interpret clinical information
- maintain and refine a differential diagnosis
- recognise discriminating findings
- respond to changing diagnostic probability
- identify the most defensible diagnosis
- learn from the reasoning process after completion

The purpose of case governance is therefore not only to ensure that the final answer is correct.

It is to ensure that the path to that answer is clinically coherent, educationally intentional, appropriately difficult, and safe for learner exposure.

2. The Clinical Case as an Educational Artifact

A clinical case is an educational expression of a governed diagnosis.

It represents one possible presentation, reasoning pathway, learner level, clinical setting, and educational emphasis.

A case does not define the diagnosis.

It teaches selected aspects of the diagnosis.

Multiple cases may teach the same diagnosis while differing in:

- age group
- clinical setting
- disease stage
- presentation
- difficulty
- diagnostic mimics
- reasoning objective
- escalation requirement
- management context
- clue progression

Case diversity should increase educational coverage rather than produce superficial variation.

Changing demographic details alone does not create a meaningfully distinct case.

3. Case Purpose

Every case must possess an explicit educational purpose.

The purpose should identify:

- what reasoning skill the case teaches
- which learning goal it supports
- which differential challenge it presents
- which discriminator the learner should recognise
- which misconception it prevents
- which escalation or management decision it reinforces

A case should not exist solely because a diagnosis lacks inventory.

Case generation should respond to an identified educational need.

Examples of valid purposes include:

- teaching early recognition of bacterial meningitis before advanced neurological deterioration
 - distinguishing chronic pulmonary aspergillosis from pulmonary tuberculosis
 - recognising unstable angina despite normal cardiac biomarkers
 - identifying scaphoid fracture despite normal initial radiography
 - distinguishing Addison disease from nonspecific chronic fatigue syndromes
-

4. Case Identity

4.1 Stable case identity

Every clinical case must possess a stable identity independent of its current revision.

The case identity allows Wardle to preserve:

- revision history
- review history
- validation history
- publication history
- learner attempts
- scheduling history
- differential mappings
- AI draft history
- reasoning assessments

Editing a case should normally create a new revision rather than replace its historical identity.

4.2 Case title

The case title should support editorial identification without revealing the diagnosis to learners prematurely.

Editorial titles may be descriptive, but learner-facing titles must not expose the answer.

A case title should not:

- state the diagnosis directly before play
- contain a pathognomonic phrase that makes reasoning unnecessary
- imply an unsupported syndrome
- exaggerate the clinical presentation
- substitute narrative marketing language for clinical clarity

Editorial systems may display the diagnosis separately from the learner-facing title.

4.3 Diagnosis relationship

Every case should resolve to a governed Diagnosis Registry entry.

The relationship may initially be:

- matched
- review required
- unresolved
- requiring a new registry entry

A case with unresolved diagnosis identity should not become publishable.

The proposed diagnosis text should remain preserved even after registry mapping so that the original authorial intent remains auditable.

5. Case Source and Provenance

A case may originate through:

- AI generation
- manual authoring
- administrative editing
- restoration
- migration
- adaptation from an earlier case

Source describes origin.

It does not establish quality or authority.

Every case should preserve:

- original producer
- generation or authoring context
- diagnosis used
- relevant editorial brief
- reasoning path used
- teaching rules applied
- revision history
- validation history
- reviewer decisions

AI-generated and manually authored cases should undergo the same clinical and educational quality standards.

6. Case Revision

6.1 Purpose

A Case Revision is a versioned snapshot of the case at a particular point in its editorial history.

A revision should preserve the complete case content required to understand what was reviewed or published.

This includes:

- title
 - date or scheduling context where relevant
 - difficulty
 - history
 - symptoms
 - laboratory data
 - clues
 - explanation
 - differentials
 - diagnosis mapping
 - source
 - publication track
-

6.2 Current revision

Every actively governed case should identify one current revision.

Editorial actions such as validation, review, approval, and publication should apply to the current revision or explicitly identify another revision.

Approval should never apply ambiguously to a mutable case record.

6.3 Revision creation

A new revision should be created when a material change affects:

- clinical facts
- clue wording
- clue order
- explanation
- diagnosis

- differential list
- reasoning objective
- difficulty
- educational purpose
- management guidance
- investigation findings

Minor non-substantive corrections may be handled according to implementation policy, but no correction should erase a previously reviewed or published state.

6.4 Revision comparison

Reviewers should be able to determine:

- what changed
- why it changed
- who or what produced the change
- which previous issue it addresses
- whether the change affects reasoning
- whether revalidation is required

A revision should not be approved solely because an earlier revision was approved.

Material revisions require renewed assessment.

7. Case Structure

A Wardle case may include:

- introductory history
- symptoms
- physical examination
- vital signs
- laboratory findings
- imaging
- pathology
- clinical progression
- treatment response
- contextual clues
- final explanation
- differentials

These elements should be organised according to diagnostic reasoning rather than arbitrary data categories.

8. Case Opening

8.1 Purpose

The opening establishes the clinical problem while preserving sufficient uncertainty for reasoning.

It should communicate:

- patient context
- presenting complaint
- relevant timeline
- initial severity
- clinical setting

8.2 Quality standard

A strong opening should be:

- clinically plausible
- concise
- sufficiently informative
- not prematurely diagnostic
- appropriate to the target difficulty
- free from irrelevant narrative detail

The opening should create a diagnostic problem rather than tell a story for its own sake.

8.3 Narrative restraint

Cases should avoid excessive narrative language that does not contribute to diagnosis.

Details should exist because they:

- alter diagnostic probability
- establish severity
- clarify chronology
- distinguish a mimic
- guide investigation
- support management
- strengthen realism

Narrative engagement should emerge from clinical uncertainty, not decorative prose.

9. Clinical Clues

9.1 Definition

A clue is a discrete unit of clinical information revealed during case progression.

A clue may represent:

- history
- symptom
- examination finding
- vital sign
- laboratory result
- imaging result
- pathology result
- risk factor
- treatment response
- clinical course

Each clue should change, refine, confirm, or challenge the learner's diagnostic reasoning.

9.2 Clue necessity

Every clue should justify its inclusion.

A clue should perform at least one function:

- broaden the differential
- narrow the differential
- support the target diagnosis
- weaken a mimic
- identify severity
- reveal a complication
- trigger escalation
- guide management
- test interpretation of uncertainty

Clues that do not meaningfully affect reasoning should be removed or consolidated.

9.3 Clinical accuracy

Every clue must be clinically compatible with:

- the target diagnosis
- the case stage
- the patient context
- the severity
- the stated timeline
- other clues

A clue may be atypical where educationally intentional, but the case should explain its significance.

9.4 Clue granularity

A clue should contain one coherent unit of clinical information.

It should not combine unrelated findings simply to accelerate the case.

Overloaded clues make it difficult to assess which information changed diagnostic probability.

9.5 Clue wording

Clue wording should be:

- clinically precise
- concise
- grammatically complete
- interpretable without hidden assumptions
- free from unnecessary explanation

The clue should present the finding.

The breakdown should explain its diagnostic significance.

A clue must not include its own reasoning explanation.

10. Clue Progression

10.1 Purpose

Clue progression describes how diagnostic understanding changes as information is revealed.

A strong case should create meaningful movement from uncertainty toward justified diagnostic confidence.

The learner should not know the answer from the first clue unless the case is intentionally basic.

The learner should not remain unable to distinguish the diagnosis after the final clue.

10.2 Diagnostic states

At each clue stage, the case may be evaluated according to:

- leading diagnosis
- active differentials
- eliminated mimics
- unresolved mimics
- confidence change
- emerging discriminators
- remaining ambiguity

Clue progression should be assessed across the complete sequence rather than clue by clue in isolation.

10.3 Progressive disclosure

Clues should generally progress from:

- broad clinical presentation
- contextual refinement
- examination or physiological evidence
- focused investigation
- discriminating or confirmatory evidence

This sequence is not mandatory.

Alternative sequences are acceptable where they reflect realistic clinical reasoning.

10.4 Diagnostic lock-in

Diagnostic lock-in occurs when one clue makes the answer effectively certain.

A lock-in clue may be appropriate near the end of the case.

Premature lock-in occurs when the diagnosis becomes obvious before meaningful reasoning has occurred.

The expected lock-in point should be compatible with case difficulty.

10.5 Premature leakage

Premature leakage occurs when a clue reveals the diagnosis too early through:

- pathognomonic wording
- named diagnostic criteria
- an overly specific test
- a disease-specific treatment
- explicit specialist terminology
- a diagnosis embedded in imaging or pathology wording

A clinically accurate clue may still be educationally inappropriate if introduced too early.

10.6 Unresolved ambiguity

A case has unresolved ambiguity when the available clues support more than one diagnosis without a defensible discriminator.

Ambiguity may be educationally intentional during the case.

It should not persist at the final decision point unless the case explicitly teaches uncertainty.

11. Clue-Discriminator Annotation

11.1 Purpose

A Clue-Discriminator Annotation records how a specific clue weakens or eliminates a competing diagnosis.

It creates an explicit link between case content and differential reasoning.

11.2 Required elements

An annotation should identify:

- case
 - clue order
 - eliminated or weakened diagnosis
 - discriminator
 - reasoning
 - elimination strength
 - educational value
 - reviewer state
-

11.3 Elimination strength

Elimination strength should describe how strongly the clue affects the differential.

Possible interpretations include:

- weak
- moderate
- strong
- decisive

A finding should not be described as decisive unless it genuinely rules out or overwhelmingly disfavors the competing diagnosis in the case context.

11.4 Educational value

Educational value describes whether the discriminator is worth teaching.

A discriminator may be clinically valid but educationally weak if it is:

- excessively obscure
 - dependent on unavailable resources
 - not generalisable
 - irrelevant to the intended learner
 - unlikely to improve future reasoning
-

12. Clue Revision Draft

12.1 Purpose

A Clue Revision Draft proposes a controlled change to a case clue.

It allows generated or editor-authored improvements to be reviewed before altering the governed case revision.

12.2 Draft contents

A clue revision draft may contain:

- original clue
 - revised clue
 - proposed added clue
 - rationale
 - expected effect
 - clue position
 - source audit
 - review status
 - reviewer decision
 - application history
-

12.3 Expected effect

The expected effect should state what the change is intended to improve.

Examples include:

- delay diagnostic lock-in
 - strengthen discrimination from pulmonary tuberculosis
 - correct a mismatch between fever and cardiac findings
 - add missing escalation evidence
 - remove unsupported laboratory data
 - improve clue coherence
 - reduce ambiguity
 - increase educational value
-

12.4 Review and application

Approval of a clue revision draft does not itself modify the case unless the system explicitly applies the revision.

The system should distinguish:

- draft accepted
- draft rejected
- changes requested
- draft applied
- draft superseded

Application should create or contribute to a new case revision where the change is material.

13. Clinical Reasoning Objective

Every case should possess an identifiable reasoning objective.

Examples include:

- differential discrimination
- early-presentation recognition
- escalation recognition
- complication recognition
- management contrast
- resource-limited reasoning

The reasoning objective should determine:

- clue selection
- clue order
- required differentials
- explanation emphasis
- expected learner action
- difficulty

A case that attempts to teach every aspect of a diagnosis may become unfocused.

14. Differential Diagnoses

14.1 Purpose

Case differentials define the clinically plausible alternatives that frame the diagnostic challenge.

They should not be generic lists appended after case completion.

They should reflect the actual uncertainty created by the case.

14.2 Differential quality

A case differential should be:

- clinically plausible
 - relevant to the presented findings
 - appropriately difficult
 - educationally useful
 - resolvable using the available clues
 - mapped to a governed diagnosis where possible
-

14.3 Differential roles

A differential may serve as:

- primary mimic
- general differential
- important exclusion
- teaching differential

These roles should not be treated as equivalent.

A primary mimic usually requires explicit discriminator coverage.

An important exclusion may require escalation or safety reasoning even if less likely.

14.4 Differential restraint

A case should not include every possible diagnosis.

Excessive differential lists reduce clarity and make review difficult.

The case should prioritise alternatives that:

- are genuinely plausible
 - represent common confusion
 - are dangerous to miss
 - teach an important discriminator
 - align with the case purpose
-

15. Differential Mapping

15.1 Purpose

Differential Mapping resolves case differential text to Diagnosis Registry identities.

This supports:

- graph integration
 - duplicate prevention
 - reasoning analysis
 - coverage assessment
 - consistent terminology
 - cross-case comparison
-

15.2 Resolution states

A differential mapping may be:

- resolved
- ambiguous
- unresolved
- rejected

These states describe mapping confidence.

They do not independently determine whether the differential is clinically appropriate.

15.3 Mapping review

A reviewer should confirm:

- correct registry diagnosis

- clinical equivalence
- absence of inappropriate merging
- correct source location
- sufficient confidence
- suitable differential role

Unresolved mapping should not necessarily block early drafting.

It may block final governance where the differential is central to the case.

16. Differential Link

A Differential Link represents a governed relationship between the case and a resolved diagnosis.

It should preserve:

- case
- case revision where relevant
- linked registry diagnosis
- differential role
- source mapping
- review state
- rationale where required

Differential links allow the editorial system to distinguish authored text from governed clinical relationships.

17. Differential Elimination

A case should demonstrate how major differentials are weakened or eliminated.

For each important mimic, reviewers should be able to identify:

- which clue affected it
- what discriminator was used
- how strong the elimination was
- whether confusion remains
- whether the explanation communicates the distinction

A differential should not appear in the final explanation without meaningful connection to the case.

18. Case Learning-Goal Coverage

18.1 Purpose

Case Learning-Goal Coverage evaluates how strongly a case supports a diagnosis-level learning goal.

The case should not be considered educationally sufficient merely because it is clinically correct.

It should contribute to defined educational intent.

18.2 Coverage elements

Coverage may include:

- learning goal identity
 - coverage strength
 - covered discriminators
 - missing discriminators
 - covered mimics
 - missing mimics
 - evidence source
-

18.3 Coverage strength

Coverage strength should reflect the degree to which the learner must engage with the goal.

A learning goal is not strongly covered merely because the relevant fact appears in the explanation.

Strong coverage usually requires the learner to use the concept during case reasoning.

18.4 Missing coverage

Missing coverage is not automatically a case defect.

A single case is not expected to teach every diagnosis-level goal.

It becomes a defect where the case claims to support a goal but omits its essential reasoning requirements.

19. Escalation Coverage

19.1 Purpose

Escalation Coverage assesses whether the case teaches recognition of clinical deterioration, urgency, or the need for higher-level intervention.

It is particularly important for diagnoses where delayed escalation could cause significant harm.

19.2 Escalation types

Escalation may include:

- emergency stabilisation
 - urgent referral
 - specialist review
 - intensive monitoring
 - definitive intervention
 - treatment failure
 - complication recognition
-

19.3 Coverage requirements

Where escalation is part of the case purpose, the case should include:

- evidence that escalation is needed
- appropriate timing
- a clear reasoning pathway
- clinically correct response
- explanation of why delay is unsafe

Escalation should not be added as an unrelated management note after diagnosis.

20. Case Explanation

20.1 Purpose

The case explanation converts the completed case into structured diagnostic education.

It should explain why the diagnosis is most defensible and how the clues altered the differential.

The explanation should not merely restate the case.

20.2 Core components

A high-quality case explanation may contain:

- diagnosis
 - concise summary
 - diagnostic reasoning
 - key findings
 - differential analysis
 - clue interpretation
 - investigation interpretation
 - management implications
 - educational conclusion
-

20.3 Diagnostic reasoning

The reasoning should show:

- how the presentation was initially framed
- which diagnoses were plausible
- which clues shifted probability
- which discriminator was decisive
- why the final diagnosis is preferred
- why major mimics are less likely

Reasoning should follow the actual clue progression.

It should not invent evidence that was not present in the case.

20.4 Key findings

Key findings should be selected because they materially support the diagnosis.

They should not be an indiscriminate list of every abnormality.

Each key finding should be associated with its clinical significance.

20.5 Differential analysis

Differential analysis should compare diagnoses using case-specific evidence.

It should avoid generic textbook contrasts unrelated to the presented patient.

A high-quality comparison answers:

- Why was the alternative plausible?
 - Which finding weakens it?
 - Which finding supports the target?
 - Is the alternative excluded or merely less likely?
-

20.6 Explanation safety

An explanation must not:

- contradict the clues
 - attribute one clue's meaning to another clue
 - invent findings
 - overstate diagnostic certainty
 - claim that a finding is pathognomonic when it is not
 - provide unsafe management guidance
 - obscure uncertainty
 - rely on descriptive diagnosis wording instead of the canonical diagnosis
-

21. Evidence Relationships

Cases may support diagnosis-level evidence relationships.

A case-supported evidence relationship should identify:

- diagnosis
- evidence node
- relationship type
- strength
- discriminator weight
- reasoning summary
- supporting case
- governance state

Case evidence should reflect what the case demonstrates.

It should not be treated as external clinical evidence or guideline authority.

22. Case Validation

22.1 Purpose

Case Validation evaluates whether a case revision satisfies defined technical, clinical, reasoning, and educational requirements.

Validation supports review.

It does not replace human approval.

22.2 Validation dimensions

Validation may assess:

- required content
 - diagnosis mapping
 - clue structure
 - playability
 - clue progression
 - reasoning coverage
 - evidence consistency
 - differential quality
 - explanation completeness
 - unsupported claims
 - publication requirements
-

22.3 Validation outcomes

Validation may produce:

- passed
- failed
- error

A validation error means the assessment could not be completed.

It must not be interpreted as a passing result.

22.4 Validation history

Each validation run should preserve:

- case
- revision
- outcome
- blockers
- warnings
- validator version
- execution time
- relevant quality signals

A newer case revision should normally require new validation.

22.5 Validation limitations

Validation can identify:

- missing structures
- policy violations
- reasoning inconsistencies
- high-risk patterns
- technical defects

Validation cannot independently establish:

- full clinical correctness
 - educational excellence
 - institutional approval
 - publication authority
-

23. Reasoning Draft Validation

Cases may also undergo specialised reasoning validation.

Reasoning validation may assess:

- trust score
- trust tier
- reasoning coverage
- evidence coverage
- discriminator coverage
- unsupported claims

- hallucination risk
- blockers
- warnings
- strengths
- recommendations

Trust tiers such as high trust, review required, low trust, or blocked are assessments.

They are not approval decisions.

24. Case Review

24.1 Purpose

Case Review is the governed human evaluation of a case revision.

It determines whether the case is clinically sound, educationally valuable, and suitable for its next lifecycle transition.

24.2 Review scope

Case review should assess:

- diagnosis identity
 - clinical plausibility
 - clue accuracy
 - clue progression
 - differential quality
 - discriminator quality
 - reasoning objective
 - explanation
 - learning-goal coverage
 - escalation coverage where relevant
 - AI contribution
 - validation findings
 - learner safety
 - publication suitability
-

24.3 Review decisions

A case review may result in:

- approved
- rejected
- needs edit

The decision should apply to an identifiable revision.

24.4 Review rationale

Every non-approval decision should provide actionable rationale.

Approval should also preserve a concise basis where the case contains atypical, complex, or high-risk reasoning.

24.5 Review completeness

A reviewer should not approve a case without access to:

- current revision
- complete clue sequence
- diagnosis
- differentials
- explanation
- validation results
- unresolved blockers
- material AI drafts
- relevant teaching intent

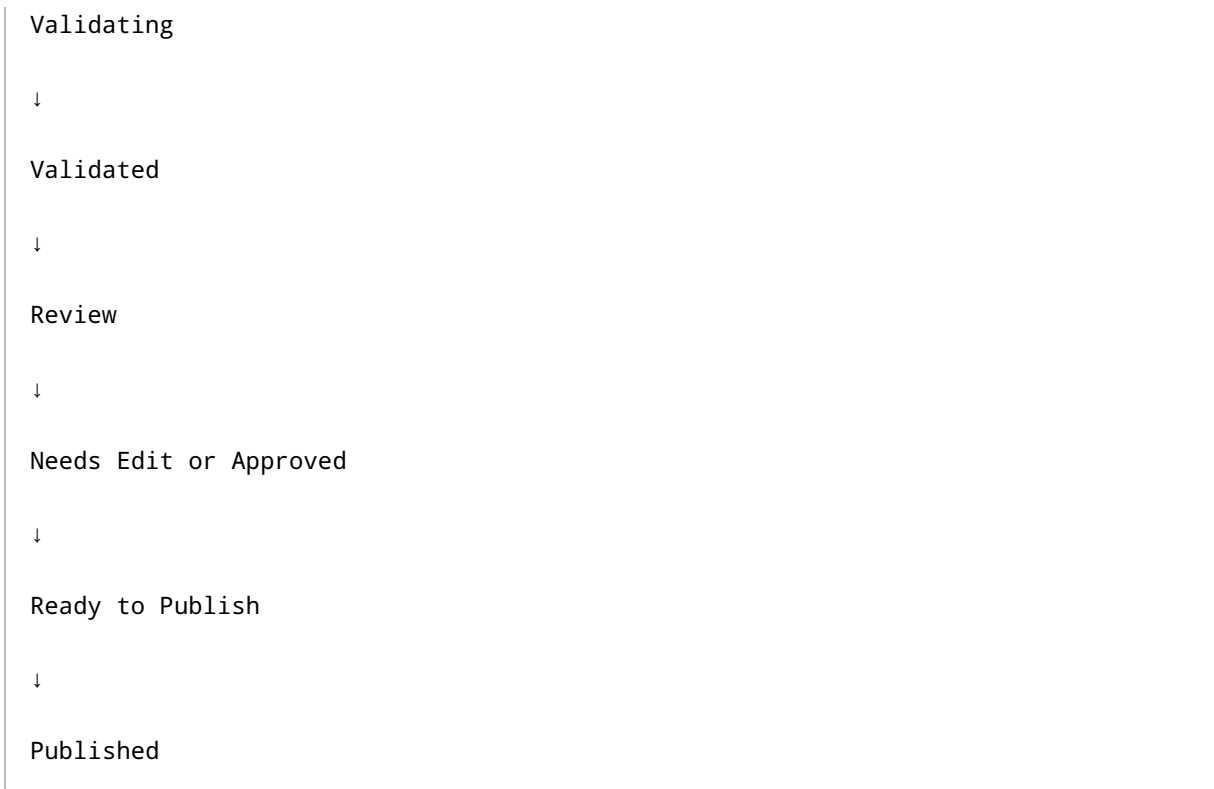
This complete context forms the Case Review Packet.

25. Case Editorial Lifecycle

The current conceptual lifecycle is:

Draft

↓



A case may also be rejected at the appropriate review stage.

25.1 Draft

The case is under authoring or generation.

It may be incomplete and should not be exposed to learners.

25.2 Validating

Automated or structured validation is in progress.

25.3 Validated

The current revision has completed validation successfully.

Validated does not mean approved.

25.4 Review

The case is awaiting or undergoing human editorial review.

25.5 Needs Edit

The case requires revision before approval.

25.6 Approved

The current revision has satisfied editorial review.

Approved does not mean published.

25.7 Ready to Publish

The case has satisfied the operational and diagnosis-level requirements required for publication.

25.8 Published

The approved case revision is authorised for learner exposure.

25.9 Rejected

The case has not been accepted into governed case inventory.

Rejection should preserve its history.

26. Playability

26.1 Definition

Playability means that the case can function correctly as a Wardle learning experience.

A playable case should possess:

- a governed diagnosis
 - sufficient clues
 - coherent clue order
 - valid answer matching
 - learner-safe content
 - a complete explanation
 - appropriate differential reasoning
 - no blocking technical defect
-

26.2 Playability and quality

A technically playable case may still be educationally poor.

A clinically sound case may still be technically unplayable.

Playability must therefore remain separate from:

- approval
 - educational quality
 - diagnosis activation
 - publication
-

26.3 Diagnosis dependency

A case cannot be operationally playable where its diagnosis is not permitted as a playable answer.

Both diagnosis-level and case-level playability must be satisfied.

27. Publication Eligibility

A case is publication-eligible when:

- its current revision is approved
- required validation has passed
- diagnosis mapping is resolved
- required differentials are governed
- blocking AI drafts are resolved
- clue revision drafts are resolved or explicitly deferred
- required explanation content is complete
- diagnosis-level publication dependencies are satisfied
- a permitted publication track is selected where required

Publication eligibility should be computed and explained.

28. Publication Tracks

A case may be intended for:

- Daily
- Premium

- Practice

Publication track affects distribution.

It should not alter the core clinical quality standard.

A Premium case should not be considered inherently more advanced.

A Daily case should not receive reduced governance because it is freely available.

29. AI Contribution to Cases

AI may contribute to:

- initial case drafts
- clue revisions
- differential suggestions
- explanation drafts
- reasoning analysis
- quality recommendations
- generation-context application

AI contribution should preserve:

- action type
- source issue
- input context
- generated output
- affected artifact
- review status
- reviewer decision
- review note

AI-generated changes must not modify the governed case silently.

30. Case Maintenance

Published cases should remain subject to review when:

- clinical guidance changes
- learner behaviour reveals ambiguity
- explanation errors are identified
- answer mapping changes

- diagnosis terminology changes
- new differentials become important
- clue progression is found to leak early
- management becomes outdated
- evidence relationships change
- quality standards evolve

Maintenance should create a new revision where the learner-facing case changes materially.

31. Case Retirement and Supersession

A case may be withdrawn, retired, or superseded where it is:

- clinically outdated
- educationally redundant
- persistently ambiguous
- based on deprecated terminology
- replaced by a stronger case
- inconsistent with current diagnosis governance
- unsafe for learner exposure

Retirement should not erase:

- learner history
 - publication history
 - review decisions
 - revision history
 - AI provenance
 - differential relationships
-

32. Quality Standard for a Governed Clinical Case

A high-quality Wardle case should be:

Clinically coherent

Every finding is compatible with the diagnosis, timeline, severity, and patient context.

Educationally intentional

The case serves an explicit learning and reasoning purpose.

Progressively discriminating

The clue sequence meaningfully changes diagnostic probability.

Appropriately uncertain

The case preserves useful ambiguity without becoming indeterminate.

Differentially grounded

Important mimics are plausible and distinguishable.

Explanation-aligned

The breakdown accurately interprets the clues actually presented.

Appropriately difficult

Difficulty arises from reasoning complexity rather than obscure trivia.

Learner-safe

The case avoids misinformation, unsupported certainty, and unsafe guidance.

Governed

The current revision has been validated, reviewed, and explicitly approved.

Maintainable

The case can evolve without erasing history.

33. Normative Statements

- Every clinical case **MUST** possess a stable case identity.
- Every material case change **MUST** be represented by an identifiable revision.
- Every case **MUST** resolve to a governed Diagnosis Registry entry before publication.
- Every case **MUST** possess an explicit educational or reasoning purpose.
- Every clue **MUST** contribute meaningfully to clinical interpretation, differential reasoning, severity, escalation, or management.
- Clues **MUST NOT** contain their own explanatory breakdown.
- Case explanations **MUST** interpret the clues actually presented.

- Case explanations **MUST NOT** invent findings or misattribute clue significance.
 - Important differentials **SHOULD** resolve to governed diagnosis identities.
 - Primary mimics **SHOULD** possess explicit discriminator coverage.
 - Clue progression **MUST** be evaluated across the complete sequence.
 - Premature diagnostic leakage **MUST** block approval where it defeats the intended reasoning task.
 - Unresolved final ambiguity **MUST** block approval unless uncertainty is the explicit educational objective.
 - Validation **MUST NOT** be treated as equivalent to human approval.
 - Approval **MUST** apply to a specific case revision.
 - Approved cases **MUST NOT** be treated as automatically publishable.
 - Playability **MUST** remain distinct from approval and publication.
 - AI-generated case content **MUST** undergo human editorial review.
 - Accepted clue drafts **MUST NOT** be treated as applied until the case revision changes.
 - Publication **MUST** identify the approved revision exposed to learners.
 - Published cases **MUST** remain maintainable and historically traceable.
 - Rejected, retired, and superseded cases **SHOULD** preserve their governance history.
-

Closing Principle

A Wardle case is not successful merely because the learner can guess the diagnosis.

It is successful when every clue contributes to a clinically defensible progression from uncertainty to understanding, when important alternatives are meaningfully addressed, and when the final explanation teaches the reasoning that made the diagnosis credible.

Clinical Case Governance ensures that Wardle does not simply publish questions.

It publishes structured diagnostic reasoning experiences.

Wardle Editorial Canon

Standards for Governed Editorial Artifacts

Part IV - Diagnosis Education Standards

WEOS-CANON-004 | Version 0.1 | Draft for approval

1. Purpose

Diagnosis Education Standards define how diagnosis-level learner education is structured, authored, generated, revised, evaluated, approved, published, and maintained within the Wardle Editorial Operating System.

Diagnosis Education is the governed learner-facing explanation of a diagnosis outside the progression of any single clinical case.

Its purpose is to help learners consolidate the clinical pattern, interpret important findings, distinguish mimics, choose investigations, understand management priorities, avoid common errors, and retain the knowledge through active recall.

Diagnosis Education must therefore be more than a collection of correct medical facts.

It must be clinically defensible, educationally intentional, diagnosis-specific, internally coherent, appropriately evidenced, and aligned with the approved educational foundation of the diagnosis.

These standards govern the complete Diagnosis Education artifact and every section or recall unit contained within it.

2. Diagnosis Education as a Governed Artifact

Diagnosis Education is a first-class governed editorial artifact associated with one Diagnosis Registry identity.

It translates approved diagnosis-level educational intent into a structured resource that learners can study after gameplay, during review, or through recall practice.

Diagnosis Education may include:

- Summary;
- Clinical Pattern;
- Key Symptoms;
- Key Signs;
- Examination Pearls;

- Scoring Systems or Diagnostic Criteria;
- Investigations;
- Differentials;
- Management;
- Complications;
- Pitfalls;
- Recall Prompts;
- References.

The artifact is governed as a coherent whole, while individual sections may also possess their own drafts, reviews, provenance, and revision history.

A section that is individually acceptable must still remain consistent with the rest of the Diagnosis Education artifact.

3. Foundational Distinctions

3.1 Diagnosis Education and Case Explanation

A Case Explanation explains the reasoning within one Clinical Case.

Diagnosis Education explains the diagnosis beyond that case.

A Case Explanation may discuss why particular clues changed probability, how a mimic was weakened, and where diagnostic lock-in occurred. Diagnosis Education should generalise governed knowledge across clinically appropriate presentations without implying that one case represents the entire disease.

Diagnosis Education should not copy a case explanation and remove the patient details. It should provide a diagnosis-level learning resource with broader scope, explicit structure, and controlled clinical generalisation.

3.2 Educational Intent and Learner-Facing Education

The Editorial Brief, Learning Goals, Teaching Rules, required mimics, discriminators, investigation anchors, management anchors, and education guidance define what Wardle intends to teach.

Diagnosis Education communicates that intent to learners.

Learner-facing education should be evaluated against approved intent. It should not create new educational priorities accidentally or contradict the educational foundation from which it was produced.

3.3 Diagnosis-Level Knowledge and Case-Specific Knowledge

Diagnosis-level knowledge describes principles that remain defensible across an appropriately defined diagnosis scope.

Case-specific knowledge describes the interpretation of one presentation, severity, population, setting, or reasoning path.

Diagnosis Education may use examples, but it must not generalise a finding, investigation result, or management choice beyond the context supported by clinical knowledge and evidence.

3.4 Section and Artifact

A Diagnosis Education section is a governed component of the complete Diagnosis Education artifact.

Section-level review may determine whether a section is acceptable in isolation. Artifact-level review must also determine whether sections agree, use compatible scope, avoid duplication, and form a coherent learner resource.

3.5 Correctness and Educational Usefulness

A statement may be medically correct yet educationally weak because it is generic, irrelevant to the target learner, poorly prioritised, unsupported by context, or disconnected from diagnostic reasoning.

Diagnosis Education requires both clinical correctness and educational usefulness.

3.6 Reference and Evidence Authority

A reference identifies an external source available to support or extend a claim.

An Evidence Relationship records how a governed claim is supported within the WEOS knowledge system.

The presence of a reference does not independently establish that every statement in a section is correct, applicable, or institutionally approved.

4. Educational Purpose and Scope

Every Diagnosis Education artifact must have a defined educational purpose.

Its scope should be derived from the diagnosis identity, learner level, editorial brief, curriculum role, and intended Wardle use.

The educational purpose should identify:

- what learners should understand after studying the resource;
- which diagnostic pattern they should recognise;
- which mimics they should distinguish;
- which investigations they should interpret;
- which management or escalation principles they should retain;
- which misconceptions or unsafe shortcuts the education should prevent;
- which concepts should be reinforced through recall.

Scope should be sufficient to support diagnostic education without attempting to reproduce a comprehensive specialty textbook.

Depth should be determined by educational need rather than by the maximum amount of content that can be generated.

5. Artifact Identity and Diagnosis Relationship

5.1 Stable identity

Every Diagnosis Education artifact must possess a stable identity independent of its current wording, section content, or publication status.

Stable identity enables revision history, review continuity, publication history, recall linkage, knowledge extraction, supersession, and auditability.

5.2 Diagnosis mapping

Every Diagnosis Education artifact must resolve to one governed Diagnosis Registry entry.

The diagnosis relationship must identify the canonical diagnosis rather than infer identity from a learner-facing title or generated summary.

Education with unresolved diagnosis identity must not become approved or published.

5.3 Scope qualifiers

Where the education applies only to a subtype, stage, age group, pregnancy state, setting, or severity, that scope must be explicit.

Scope qualifiers must not be hidden in one section while other sections imply universal applicability.

6. Source and Provenance

Diagnosis Education may originate through manual authoring, AI generation, section generation, administrative revision, migration, restoration, evidence-led maintenance, or controlled adaptation from earlier educational material.

Source describes origin. It does not establish quality or authority.

The artifact and each material revision should preserve:

- original producer;
- production method;
- generation or authoring purpose;
- diagnosis context;
- source Editorial Brief;
- source Learning Goals and Teaching Rules;
- source artifacts and versions;
- evidence context;
- review history;
- application history where a draft was accepted and applied.

Provenance should make it possible to reconstruct why a section exists, what context shaped it, and how it became governed learner-facing knowledge.

7. Diagnosis Education Revision

7.1 Revision purpose

A Diagnosis Education Revision is a versioned snapshot of the complete learner-facing education artifact at a particular point in editorial history.

Approval and publication must apply to an identified revision rather than to an indefinitely mutable record.

7.2 Material change

A new revision should be created when a material change affects:

- clinical meaning;
- diagnostic scope;
- educational emphasis;
- section structure;
- investigation guidance;
- differential reasoning;
- management guidance;
- scoring criteria;
- complications;
- high-risk pitfalls;
- recall answers;
- references supporting material claims.

Minor typographical correction may follow implementation policy, but no correction should erase a previously reviewed or published state.

7.3 Revision comparison

Reviewers should be able to determine what changed, why it changed, which source issue prompted the change, whether the change affects other sections, and which review must be repeated.

Approval of an earlier revision does not automatically approve a materially changed revision.

8. Canonical Structure

The canonical Diagnosis Education structure is modular but coherent.

Not every diagnosis requires identical depth in every section. However, section omission must be deliberate and explainable rather than the result of incomplete generation or authoring.

Section	Primary educational question
Summary	What is this diagnosis and why does it matter?
Clinical Pattern	What presentation should the learner recognise?
Key Symptoms	Which reported features meaningfully affect suspicion?
Key Signs	Which observable findings support, discriminate, or indicate severity?
Examination Pearls	How should examination be performed and interpreted?
Scoring Systems	Which criteria or scores assist diagnosis, severity, or action?
Investigations	What should be tested, what findings are expected, and what are the limitations?
Differentials	Which diagnoses are confused with this one and how are they separated?
Management	What management, escalation, and follow-up principles are educationally essential?
Complications	What important consequences should be recognised or prevented?
Pitfalls	Which reasoning or management errors should the learner avoid?
Recall Prompts	Which governed concepts should the learner retrieve actively?
References	Which external sources support or extend the education?

Implementations may present these sections under learner-friendly labels, but the canonical editorial meaning must remain preserved.

9. Summary

9.1 Purpose

The Summary provides a concise diagnosis-level orientation.

It should establish the diagnosis identity, essential clinical meaning, typical pattern, and educational importance without attempting to duplicate every later section.

9.2 Required qualities

A strong Summary should be:

- diagnosis-specific;
- clinically accurate;
- brief enough for orientation;
- consistent with the diagnosis scope;
- free from decorative language;
- free from unsupported certainty;
- aligned with the approved Learning Goals.

Where a concise definition is educationally necessary, it should be incorporated into the Summary or represented as a clearly governed component rather than generated as isolated dictionary text.

9.3 Prohibited patterns

The Summary should not:

- repeat the canonical name without explaining the clinical concept;
- use a case narrative as the diagnosis definition;
- present one manifestation as universal;
- include detailed management that belongs later;
- rely on vague phrases such as “a condition with several symptoms”;
- make epidemiological or prognostic claims without appropriate support.

10. Clinical Pattern

10.1 Purpose

The Clinical Pattern describes the recognisable configuration of context, chronology, symptoms, signs, investigations, and severity that should organise learner understanding.

It should teach pattern recognition without reducing diagnosis to a memorised slogan.

10.2 Content standard

The Clinical Pattern may identify:

- typical patient or risk context;
- onset and time course;
- dominant symptom pattern;
- important examination pattern;
- expected investigation pattern;
- features indicating severity or complication;
- common atypical or population-specific variation where educationally important.

The section should distinguish between common, characteristic, supportive, and pathognomonic findings where those distinctions matter.

10.3 Avoiding overfitting

The Clinical Pattern must not overfit to the presentation used in one Wardle case.

A learner should not leave the section believing that the diagnosis requires every clue used in a flagship case or that absence of one classic feature excludes the diagnosis.

11. Key Symptoms

11.1 Purpose

Key Symptoms identify patient-reported features that meaningfully contribute to diagnosis, severity assessment, differential discrimination, or escalation.

11.2 Symptom structure

A symptom entry should communicate more than a label.

Where relevant, it should identify:

- quality;
- location;
- timing;
- duration;
- progression;
- triggers;
- relieving factors;
- associated symptoms;
- clinical significance;
- which mimic it helps distinguish.

Symptoms should be prioritised. Long undifferentiated lists weaken recognition and obscure the diagnostic pattern.

11.3 Negative symptoms

Important absent symptoms may be included where their absence meaningfully changes probability or helps distinguish a dangerous mimic.

Negative findings should not be presented as absolute exclusion rules unless clinically justified.

12. Key Signs

12.1 Purpose

Key Signs identify observable examination or bedside findings that support diagnosis, discriminate alternatives, indicate severity, reveal complications, or change management.

12.2 Sign interpretation

A sign should be connected to its clinical meaning.

Education should explain whether the sign is:

- common;
- characteristic;
- late;
- severity-related;
- a complication signal;
- a discriminator;
- population-dependent;
- limited by poor sensitivity or specificity.

A correct list of signs without interpretation does not satisfy the educational standard.

13. Examination Pearls

13.1 Purpose

Examination Pearls teach how to look for, elicit, interpret, or avoid missing clinically meaningful examination findings.

They should strengthen bedside reasoning rather than repeat the Key Signs list.

13.2 Appropriate content

An Examination Pearl may address:

- examination sequence;
- technique;
- comparison with the opposite side;
- patient positioning;
- interpretation of a subtle finding;
- limitations of a sign;
- a finding that changes urgency;
- a commonly omitted part of examination.

Technique should not be described in procedural detail beyond the educational scope or in a way that creates unsafe unsupervised practice.

14. Scoring Systems and Diagnostic Criteria

14.1 Purpose

This section governs formal scores, classifications, staging systems, and diagnostic criteria used to support diagnosis, severity assessment, prognosis, or management decisions.

14.2 Required elements

A scoring-system entry should identify:

- official or accepted name;
- intended purpose;
- applicable population and setting;
- components;
- interpretation;
- validated use;
- important limitations;
- action thresholds where appropriate;
- source or authority;
- version where criteria have changed.

A score should not be included merely because it is associated with the diagnosis.

Its educational value and correct use must be explicit.

14.3 Safety and fidelity

Criteria, cut-offs, units, and point assignments must be reproduced accurately and verified against an appropriate source.

WEOS must not invent, approximate, combine, or silently simplify scoring systems in a way that changes interpretation.

Where a score does not replace clinical judgement, that limitation should be stated clearly.

14.4 Mnemonics and memory aids

Mnemonics are not a required canonical Diagnosis Education section.

Where used, a mnemonic should support accurate recall of governed content, avoid offensive or misleading language, and not replace understanding of clinical meaning.

A mnemonic that omits important qualifiers or creates an unsafe rule should not be published.

15. Investigations

15.1 Purpose

The Investigations section teaches what to investigate, why the investigation is used, what findings are expected, how results should be interpreted, and what limitations remain.

15.2 Investigation hierarchy

Investigations should be organised according to clinical purpose rather than presented as an unprioritised list.

Possible purposes include:

- initial bedside assessment;
- confirmatory testing;
- excluding dangerous alternatives;
- severity assessment;
- complication assessment;
- pre-treatment assessment;
- monitoring;
- specialist or second-line investigation.

15.3 Result interpretation

Where educationally relevant, each investigation should identify:

- expected finding;
- diagnostic role;
- sensitivity or specificity limitations;
- timing;
- false-negative or false-positive context;
- population limitations;
- how the result changes probability or action.

A test result should not be represented as diagnostic in isolation where clinical interpretation remains necessary.

15.4 Imaging and pathology

Imaging and pathology guidance should use accurate descriptive terminology and distinguish a supportive appearance from a definitive diagnosis.

Image acquisition, specialist reporting, and clinical correlation limitations should be stated where necessary.

16. Differentials

16.1 Purpose

The Differentials section teaches which diagnoses are plausibly confused with the target diagnosis, why the confusion occurs, and how they should be distinguished.

It should support diagnostic reasoning rather than provide a generic list of alternatives.

16.2 Differential entry standard

Each important differential should identify:

- resolved Diagnosis Registry identity where available;
- shared presentation;
- reason for confusion;
- finding or context favouring the target;
- finding or context favouring the alternative;
- decisive or high-value discriminator;
- important limitation or overlap;
- risk if the alternative is missed.

16.3 Differential priority

Differentials should be prioritised by clinical plausibility, danger, educational importance, and curriculum role.

A rare alternative should not displace a common or dangerous mimic merely because it is more distinctive.

16.4 Relationship governance

Differential text should resolve to governed diagnosis identities where possible.

The presence of a differential in education may support a candidate Teaching Relationship or Differential Link, but authored text does not automatically create a governed graph relationship.

17. Management

17.1 Purpose

The Management section teaches the diagnosis-level principles that a learner should recognise after diagnosis or when urgent action is required.

Management content should remain appropriate to Wardle's educational scope, learner level, clinical context, and evidence base.

17.2 Management structure

Where applicable, management should distinguish:

- immediate stabilisation;
- first-line treatment;
- definitive treatment;
- supportive care;
- management by severity or subtype;
- referral or specialist involvement;
- monitoring;
- follow-up;
- prevention or risk modification;
- management of complications.

17.3 Context and qualification

Management recommendations should identify the context in which they apply.

Statements should not imply universal treatment where decisions depend upon severity, age, pregnancy, comorbidity, local resistance, contraindications, resources, or specialist assessment.

17.4 Drug and dose content

Drug names, routes, doses, intervals, duration, adjustments, and monitoring requirements are high-risk knowledge.

When included, they require exactness, appropriate evidence, population context, unit verification, contraindication awareness, and heightened clinical review.

Dose content should not be generated or copied into learner-facing education without verification against an appropriate current source.

17.5 Local practice and universal principles

Where management varies by jurisdiction, institution, or resource setting, education should distinguish broadly accepted principles from local protocols.

Local practice should not be presented as universal without qualification.

18. Complications

18.1 Purpose

The Complications section teaches important adverse consequences of the disease, delayed recognition, progression, or treatment.

18.2 Complication classification

Where relevant, the section should distinguish:

- direct disease complication;
- severity consequence;
- treatment complication;
- associated condition;
- late sequela;
- recurrence or chronic impairment.

Complications should be prioritised according to seriousness, frequency, preventability, diagnostic value, and educational importance.

18.3 Recognition and prevention

A complication entry should explain how the complication may be recognised, which warning signs matter, and how timely diagnosis or management may reduce risk where appropriate.

Rare catastrophic complications should not be used to sensationalise the resource.

19. Pitfalls

19.1 Purpose

Pitfalls identify plausible learner or clinical reasoning errors and provide a safer governing principle.

A Pitfall should teach an error to avoid rather than merely repeat a fact in negative form.

19.2 Pitfall structure

A strong Pitfall should identify:

- the mistaken assumption or action;
- why it is plausible;
- the consequence;
- the finding or principle that corrects it;
- the safer reasoning or management response.

19.3 Appropriate pitfall types

Pitfalls may address:

- premature closure;
- over-reliance on a single test;
- misinterpretation of a classic sign;
- failure to consider a dangerous mimic;
- incorrect exclusion based on an absent feature;
- unsafe treatment generalisation;
- missed escalation;
- misuse of a scoring system;
- terminology confusion.

Pitfalls should be diagnosis-specific. Generic warnings such as “always correlate clinically” do not satisfy the standard without explaining the relevant decision error.

20. Recall Prompts

20.1 Purpose

Recall Prompts convert governed Diagnosis Education into active-recall units that reinforce clinically meaningful knowledge.

A Recall Prompt is not merely a question appended to a section. It is a governed educational artifact with an identifiable source concept and intended retrieval outcome.

20.2 Prompt standard

A Recall Prompt should:

- test one meaningful concept or a deliberately bounded set of linked concepts;
- derive from governed education;
- have a concise defensible answer;
- include an explanatory layer;
- identify relevant qualifiers;
- match the declared learner level or difficulty;
- avoid ambiguity;
- avoid unsupported extension beyond its source content.

20.3 Prompt families

Recall may test:

- pattern recognition;
- a discriminator;
- an investigation interpretation;
- a scoring-system threshold;
- a management priority;
- a complication warning;
- a pitfall correction;
- a key comparison with a mimic.

20.4 Answer quality

An answer should be specific enough to establish what the learner was expected to retrieve while allowing clinically equivalent wording where appropriate.

Recall answers must not use absolute language where the governed source contains necessary uncertainty or context.

21. References

21.1 Purpose

References identify external sources that support, qualify, or extend Diagnosis Education.

References strengthen traceability and maintenance. They do not substitute for clinical review.

21.2 Reference standard

A reference record should preserve, where available:

- title;
- author or responsible body;
- publisher or journal;
- publication or update date;
- edition or version;
- persistent identifier or stable location;
- date accessed where relevant;
- applicable claim or section;
- jurisdiction or population limitations;
- review status.

21.3 Source selection

Source selection should be proportionate to the claim.

High-risk management, diagnostic criteria, scoring systems, and changing guidelines require stronger source authority and freshness than stable introductory definitions.

21.4 Citation fidelity

A source must not be cited for a claim it does not support.

References must not be fabricated, approximated, or represented as verified when only suggested by an automated system.

Where credible sources disagree, the education should not conceal the disagreement through false certainty.

22. Supplementary Educational Elements

Implementations may support supplementary elements such as definitions, classic-pattern labels, clinical pearls, key concepts, memory aids, or “take-home” summaries.

Supplementary elements do not escape governance because they are brief or visually separated from the principal sections.

They must:

- derive from governed content;
- preserve diagnosis scope;
- avoid contradiction;
- avoid misleading simplification;
- identify evidence needs where clinically material;
- remain versioned with the Diagnosis Education artifact.

A visually prominent callout may carry greater educational influence than a longer paragraph and should be reviewed proportionately.

23. Cross-Section Coherence

Diagnosis Education should read as one governed educational resource rather than a collection of independently generated sections.

Cross-section review should evaluate:

- consistent diagnosis scope;
- consistent terminology;
- compatible severity framing;
- agreement between investigations and differentials;
- agreement between complications and management;
- alignment of pitfalls with the actual content;
- recall prompts that match governed sections;
- references that support the claims they are associated with;
- absence of unnecessary repetition;
- absence of contradictions.

A section change may make another section stale even when that other section has not been edited directly.

Dependency examples

Changing diagnostic criteria may affect the Clinical Pattern, Investigations, Differentials, Pitfalls, and Recall Prompts.

Changing Management may affect Complications, Pitfalls, and high-risk recall content.

Changing diagnosis scope may require review of every section.

24. Diagnosis Specificity and Educational Reasoning

Diagnosis Education must contain meaning that is specific to the governed diagnosis.

Generic phrases may be grammatically acceptable but educationally empty.

Weak content includes:

- “patients may present with a range of symptoms”;
- “investigations are required to confirm the diagnosis”;
- “management depends on severity”;
- “early diagnosis is important”;
- “complications may be serious”;
- “clinical correlation is advised”.

Such statements require diagnosis-specific explanation.

Education should explain why findings matter, how they change suspicion, what they distinguish, what action they prompt, and which error they prevent.

25. Clinical Uncertainty and Qualification

Diagnosis Education should communicate uncertainty accurately.

It should distinguish:

- typical from required;
- supportive from confirmatory;
- common from possible;
- sensitive from specific;
- association from causation;
- diagnosis from severity;
- general principle from local protocol;
- evidence-based recommendation from expert practice.

Absolute terms such as “always,” “never,” “diagnostic,” or “rules out” should be used only where their meaning is clinically defensible within the stated scope.

Uncertainty should not be expressed through vague disclaimers that avoid teaching a clear principle.

26. Learner Level and Cognitive Scope

Diagnosis Education should be appropriate to the intended learner level.

Learner level may influence:

- terminology;
- depth;
- assumed prerequisite knowledge;
- complexity of differentials;
- investigation interpretation;
- management detail;

- use of scoring systems;
- recall difficulty.

Simplification should reduce unnecessary complexity without making the clinical concept inaccurate.

Advanced detail should not obscure the high-yield reasoning principle for a basic learner.

Where the same Diagnosis Education artifact serves multiple levels, layered presentation should preserve a clear core while allowing deeper expansion.

27. Population, Setting, and Equity

Education should represent the populations and settings in which the diagnosis is taught accurately and respectfully.

Where clinically relevant, content should consider:

- age;
- pregnancy;
- sex-related clinical variation;
- comorbidity;
- immunocompromise;
- resource setting;
- geography and epidemiology;
- differences in access to investigation or treatment.

Population descriptors should be used because they change clinical interpretation, not as decorative demographic variation.

Resource limitations should not be ignored, but constrained practice should not be misrepresented as the universal standard of care.

28. Terminology, Units, and Numerical Accuracy

Terminology should align with the governed Diagnosis Registry and accepted clinical usage.

Outdated, ambiguous, stigmatising, or locally idiosyncratic terminology should be avoided or explained.

Numerical content requires particular care.

Values should preserve:

- correct units;
- reference context;
- age or population variation;
- threshold meaning;
- conversion accuracy;
- significant clinical qualifiers;

- source version where values may change.

A unit or decimal error may transform otherwise correct education into high-risk misinformation and must be treated accordingly.

29. Completeness and Non-Applicability

Completeness means that the artifact addresses the educational requirements applicable to the diagnosis and intended use.

Completeness does not require meaningless text in every possible section.

A section may be marked not applicable where:

- the concept genuinely does not apply;
- the artifact scope excludes it deliberately;
- the learner level does not require it;
- another governed artifact carries the content and the relationship is explicit.

Non-applicability should be an editorial conclusion, not a substitute for missing authoring or failed generation.

Generic filler must not be used to create the appearance of completeness.

30. Diagnosis Education Lifecycle

The canonical Diagnosis Education lifecycle should preserve the distinction between production, review, approval, publication, maintenance, and archival states.

Current conceptual progression:

Lifecycle

Draft or Generated -> Needs Review -> Needs Edit or Approved -> Published -> Maintained -> Archived, Superseded, or Withdrawn

Lifecycle labels may differ by implementation, but their editorial meaning must remain explicit.

30.1 Draft

The artifact is under human authoring or controlled revision and remains candidate knowledge.

30.2 Generated

The artifact or section has been produced primarily by an automated system and has not yet received the required human review.

Generated status must not be treated as evidence of completeness, correctness, or readiness.

30.3 Needs Review

The artifact is sufficiently formed for editorial evaluation but no final approval has been granted.

30.4 Needs Edit

Material concerns prevent approval. Required changes should remain linked to explicit review findings.

30.5 Approved

The identified revision has satisfied its required editorial review.

Approval does not automatically establish publication, learner exposure, or continued validity after later material change.

30.6 Published

An approved revision has received explicit publication authorisation and is available within its intended learner-facing context.

30.7 Rejected

The proposed artifact or revision has not been accepted for governed use.

Rejection should preserve rationale and should not erase the draft or its provenance.

30.8 Archived, Superseded, or Withdrawn

The artifact is no longer the active learner-facing resource but remains historically preserved according to governance policy.

31. Review Standard

Diagnosis Education review should evaluate the complete artifact and, where necessary, its material sections separately.

Review dimensions include:

- diagnosis identity and scope;
- clinical correctness;
- educational purpose;
- diagnosis specificity;
- cross-section coherence;
- differential reasoning;
- investigation interpretation;
- management safety;
- evidence sufficiency;
- learner level;
- terminology;
- recall quality;

- publication readiness.

Review should identify findings and rationale rather than silently editing the artifact into an approved state.

32. Clinical Review

Clinical review should determine whether Diagnosis Education is accurate, appropriately qualified, safe, and consistent with accepted practice within its stated context.

Clinical review should pay particular attention to:

- diagnostic criteria;
- exclusion statements;
- investigation limitations;
- drug doses;
- emergency management;
- paediatric and pregnancy-related content;
- procedural guidance;
- contraindications;
- escalation thresholds;
- evolving guidelines.

Review intensity should reflect the consequence of error.

33. Educational Review

Educational review should determine whether the artifact teaches the intended knowledge clearly, efficiently, and coherently.

It should assess:

- alignment with Learning Goals;
- prioritisation;
- clarity;
- reasoning value;
- avoidance of trivia;
- appropriate cognitive load;
- usefulness of comparisons;
- quality of pitfalls;
- quality and difficulty of recall;
- duplication across sections.

An artifact should not be approved merely because every section contains text.

34. Evidence Review

Evidence review should determine whether material claims are appropriately supported and whether sources remain relevant to the stated diagnosis, population, setting, and time.

Evidence review should be heightened for:

- drug doses;
- treatment duration;
- diagnostic criteria;
- scoring systems;
- time-sensitive interventions;
- screening recommendations;
- guideline-dependent management;
- claims likely to change with emerging evidence.

Where evidence is incomplete or contested, the education should communicate a defensible bounded conclusion rather than imply certainty.

35. Approval Standard

Approval applies to an identified Diagnosis Education revision.

Approval should require:

- resolved diagnosis identity;
- defined scope;
- required sections complete or deliberately non-applicable;
- no unresolved blocking clinical defects;
- no unresolved high-risk unsupported claims;
- appropriate clinical and educational review;
- acceptable cross-section coherence;
- adequate evidence for material claims;
- complete provenance and review records.

Conditional approval may be used only where governance policy permits and where remaining obligations do not create unsafe learner exposure.

36. Publication Standard

Publication is separate from approval.

A Diagnosis Education revision may be approved but not published because diagnosis-level dependencies, publication strategy, learner scope, or operational conditions remain unresolved.

Publication assessment should consider:

- approved revision;
- publication channel;
- intended learner population;
- readiness blockers;
- evidence freshness;
- consistency with published cases;
- maintenance ownership;
- effect of replacing an earlier published revision.

Technical availability must not be interpreted as publication authority.

37. Section-Level Drafting, Repair, and Regeneration

A specific section may be drafted, repaired, or regenerated without replacing the complete Diagnosis Education artifact.

Section-level work should preserve:

- target section;
- source artifact revision;
- original section;
- reason for change;
- input context;
- proposed replacement;
- affected dependencies;
- review decision;
- resulting artifact revision.

A section must not be applied solely because it is locally improved. Review must consider whether it remains coherent with the complete artifact.

Published content should not be overwritten directly by a generated replacement.

38. Relationship to Clinical Cases

Clinical Cases and Diagnosis Education should reinforce the same governed diagnosis foundation while serving different educational functions.

Cases may contribute examples, reasoning insights, discriminators, and observed misconceptions to Diagnosis Education maintenance.

Diagnosis Education may provide context, differentials, investigations, management, and recall concepts that cases reinforce.

The relationship should not require every case to teach every section or every education section to appear within a case explanation.

Material contradiction between a published case and published Diagnosis Education should create a governance issue requiring review.

39. Relationship to Knowledge and Reasoning Governance

Diagnosis Education both consumes and produces knowledge-governance signals.

Governed Graph Facts, Teaching Relationships, Evidence Relationships, Reasoning Paths, and Differential Links may inform education authoring and review.

Education may also produce Graph Candidates such as findings, investigations, mimics, pitfalls, management principles, complications, recall concepts, and reasoning statements.

Extraction preserves provenance. It does not automatically transform learner-facing prose into a governed fact.

40. Relationship to Artificial Intelligence

Artificial Intelligence may generate complete Diagnosis Education drafts, draft specific sections, identify omissions, compare revisions, propose repairs, generate recall prompts, and assist maintenance.

AI-generated education remains candidate knowledge until human review and controlled application.

AI generation should use governed context including:

- diagnosis identity;
- Editorial Brief;
- Learning Goals;
- Teaching Rules;
- required mimics and discriminators;
- investigation and management anchors;
- complications and pitfalls;
- learner level;
- evidence requirements;
- current education revision.

AI must not create institutional trust, approve its own output, fabricate references, or publish education independently.

Acceptance of an AI Draft approves the proposal for controlled application. It does not itself approve the resulting Diagnosis Education revision.

41. Validation

Automated and structured validation may support Diagnosis Education governance by detecting defects that can be evaluated deterministically or consistently.

Validation may assess:

- required structure;
- diagnosis mapping;
- empty sections;
- generic filler;
- duplicate content;
- terminology consistency;
- unit format;
- broken references;
- unsupported absolute language;
- recall answer completeness;
- cross-section conflicts;
- stale dependencies;
- publication prerequisites.

Validation does not establish clinical correctness or educational quality independently of human review.

Validation results should remain version-specific and become stale after material change.

42. Maintenance and Revalidation

Published Diagnosis Education remains living governed knowledge.

Maintenance may be triggered by:

- new evidence;
- updated guideline;
- changed diagnostic criteria;
- changed scoring system;
- terminology change;
- new treatment safety information;
- learner feedback;
- case contradiction;
- quality audit;
- specialty review;
- scheduled revalidation.

Maintenance should assess both the changed claim and the sections that depend upon it.

High-risk or rapidly evolving content should have shorter review intervals than stable foundational content.

Failure to complete required maintenance may block continued publication or require temporary withdrawal.

43. Withdrawal, Supersession, and Archival

Diagnosis Education may be withdrawn where continued learner exposure is unsafe, misleading, outdated, or no longer institutionally supported.

A revised artifact may supersede an earlier published revision while preserving historical access for governance and audit.

Archival should preserve:

- artifact and revision identity;
- publication history;
- reason for withdrawal or supersession;
- replacement artifact where applicable;
- responsible authority;
- affected recall prompts and extracted knowledge;
- learner-facing correction requirements where applicable.

Removal from active use must not erase governance history.

44. Quality Assessment

Diagnosis Education quality should be assessed through a combination of editorial review, validation, maintenance signals, and learner-informed evidence.

Quality indicators may include:

- clinical defect rate;
- unsupported-claim rate;
- cross-section contradiction rate;
- generic-content rate;
- learning-goal coverage;
- differential and discriminator coverage;
- evidence freshness;
- recall ambiguity;
- post-publication correction rate;
- learner-reported confusion;
- maintenance backlog.

Learner performance may identify possible educational weakness. It must not independently determine clinical correctness.

45. Implementation Requirements

Implementation should preserve the canonical distinctions established in this Part regardless of database design or interface layout.

The system should support:

- stable Diagnosis Education identity;
- versioned revisions;
- section-level identity where material governance requires it;
- diagnosis mapping;
- source provenance;
- draft separation;
- human review;
- artifact-level and section-level findings;
- cross-section dependency tracking;
- explicit approval;
- separate publication decision;
- withdrawal and supersession;
- recall linkage;
- reference linkage;
- knowledge extraction provenance;
- maintenance triggers;
- audit history.

Where current implementation differs from the Canon, the divergence should be documented explicitly rather than allowed to redefine the artifact silently.

46. Normative Statements

The following statements summarise the mandatory requirements of this Part.

- Every Diagnosis Education artifact **MUST** resolve to a governed Diagnosis Registry identity.
- Diagnosis Education **MUST** remain distinct from a Case Explanation.
- Approval and publication **MUST** apply to an identified artifact revision.
- Generated education **MUST** remain candidate knowledge until human review and controlled application.
- A section marked non-applicable **MUST** represent an editorial conclusion rather than missing work.
- Diagnosis Education **MUST** be evaluated for clinical correctness and educational usefulness.
- Material sections **MUST** remain coherent with the complete artifact.

- High-risk management, dose, criteria, and scoring content MUST receive proportionate clinical and evidence review.
 - Differentials MUST teach clinically meaningful comparison rather than provide an unprioritised list.
 - Recall Prompts MUST derive from governed education and preserve necessary qualifiers.
 - References MUST NOT be fabricated or cited for claims they do not support.
 - Automated validation MUST NOT substitute for human clinical or educational judgement.
 - Publication MUST remain separate from approval and technical availability.
 - Material change MUST create a new governed revision or equivalent version-specific record.
 - Withdrawal, supersession, and archival MUST preserve governance history.
 - Current implementation divergence from this Canon MUST be recorded explicitly.
-

47. Relationship to the Wider Canon

Part I defines the foundational distinctions and universal artifact standard governing Diagnosis Education.

Part II defines the diagnosis identity, Editorial Brief, Learning Goals, Teaching Rules, and education guidance from which Diagnosis Education derives its purpose.

Part III defines the Clinical Cases and Case Explanations that Diagnosis Education complements without duplicating.

Part V defines the Graph Facts, Evidence Relationships, Teaching Relationships, Differential Links, Reasoning Paths, and extraction governance that support or emerge from Diagnosis Education.

Part VI defines how Artificial Intelligence may generate or revise Diagnosis Education while remaining subordinate to human authority.

Part VII defines the assessments, reviews, decisions, validation results, publication records, and governance history through which Diagnosis Education becomes institutionally trusted.

Together, these Parts ensure that learner-facing diagnosis education is not merely produced, but clinically governed, educationally coherent, explainable, maintainable, and safe for institutional publication.

Closing Principle

Diagnosis Education should not be judged by how much information it contains.

It should be judged by whether it helps learners recognise the diagnosis, interpret evidence, distinguish important alternatives, act safely, avoid predictable errors, and retain the knowledge that matters.

Within WEOS:

- Educational intent defines what should be taught.
- Diagnosis Education communicates that intent.
- Clinical and educational review establish confidence.
- Evidence supports material claims.
- Publication authorises learner exposure.
- Maintenance preserves trust as knowledge evolves.

Wardle Editorial Canon

Standards for Governed Editorial Artifacts

Part V — Knowledge and Reasoning Governance

1. Purpose

Knowledge and Reasoning Governance defines how Wardle represents, evaluates, connects, and maintains the clinical knowledge underlying diagnostic education.

Cases and Diagnosis Education communicate knowledge to learners.

The Knowledge and Reasoning layer represents the clinical relationships that make those educational artifacts coherent.

It enables WEOS to govern:

- findings associated with diagnoses
- discriminating evidence
- important mimics
- diagnostic contrasts
- complication signals
- management signals
- escalation triggers
- reasoning pathways
- differential relationships
- extracted knowledge candidates
- unsupported claims
- knowledge provenance

Its purpose is not to build a graph for technical sophistication.

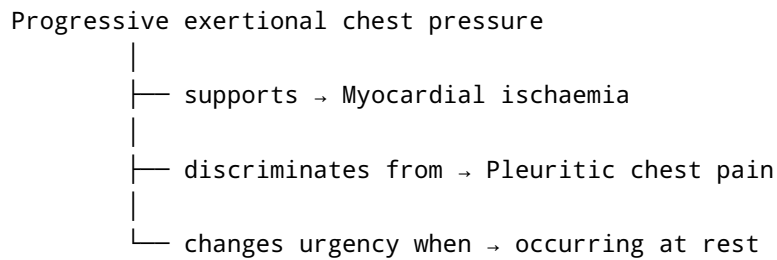
Its purpose is to make clinical reasoning explicit, reusable, reviewable, and governable.

2. Knowledge Governance Philosophy

Clinical knowledge should not remain trapped inside isolated paragraphs, case explanations, or generation prompts.

Where clinically meaningful relationships can be represented explicitly, WEOS should preserve them as governed knowledge.

For example:



This representation allows the same knowledge to support:

- case generation
- case review
- Diagnosis Education
- differential analysis
- recall prompts
- quality assessment
- curriculum coverage
- maintenance

Explicit representation does not replace clinical narrative.

It provides the governed reasoning structure beneath it.

3. Foundational Distinctions

Knowledge Governance depends upon several distinctions.

3.1 Extracted statement and governed fact

An extracted statement is knowledge identified within an existing artifact.

A governed fact is knowledge that has completed the required editorial decision process.

Extraction establishes provenance.

It does not establish authority.

3.2 Association and reasoning relationship

An association indicates that two concepts are connected.

A reasoning relationship explains how the connection affects clinical interpretation.

For example:

Association:

Hyperpigmentation is associated with Addison disease.

Reasoning relationship:

Diffuse hyperpigmentation, particularly involving palmar creases and buccal mucosa, supports primary adrenal insufficiency over secondary adrenal insufficiency because elevated adrenocorticotrophic hormone is expected in primary disease.

The second representation provides greater educational value because it explains discrimination.

3.3 Graph fact and teaching relationship

A Graph Fact represents a governed diagnosis-level knowledge statement.

A Teaching Relationship represents how two diagnoses should be compared or connected for educational purposes.

A graph fact may state:

Haemoptysis may occur in chronic pulmonary aspergillosis.

A teaching relationship may state:

Chronic pulmonary aspergillosis is commonly confused with pulmonary tuberculosis in patients with chronic constitutional and respiratory symptoms; pre-existing cavities with fungal evidence support aspergillosis.

These artifacts may support one another but should not be treated as identical.

3.4 Evidence and evidence source

An Evidence Node represents a clinical finding or concept.

An external reference supports the scientific authority of a claim.

A case may demonstrate an evidence relationship.

A case does not independently establish clinical evidence authority.

WEOS must therefore distinguish:

- clinical evidence concept
- diagnosis–evidence relationship
- supporting Wardle artifact
- external supporting source

3.5 Differential text and governed differential relationship

A differential name written in a case or education section is authored content.

A differential mapping resolves that text to a Diagnosis Registry identity.

A differential link records a governed relationship between the source artifact and the resolved diagnosis.

A teaching relationship may then explain why the diagnoses are confused and how they should be distinguished.

Each layer serves a different function.

4. Knowledge Artifact Families

Knowledge and Reasoning Governance includes:

- Graph Candidates
- Graph Facts
- Evidence Nodes
- Diagnosis–Evidence Relationships
- Teaching Relationships
- Differential Mappings
- Differential Links
- Reasoning Paths
- Diagnostic Comparisons
- Reasoning Draft Validation
- Unsupported-Claim Signals
- Knowledge-Governance Decisions

These artifacts collectively form the governed reasoning foundation of WEOS.

5. Graph Candidate

5.1 Purpose

A Graph Candidate is a proposed unit of structured knowledge extracted from or suggested by an existing artifact.

Graph Candidates allow Wardle to identify potentially reusable knowledge without automatically treating the source text as institutionally trusted fact.

5.2 Sources

A Graph Candidate may originate from:

- a Clinical Case
- a Case Explanation
- Diagnosis Education
- a generated educational section
- an editor suggestion
- a future imported source

The candidate should preserve the exact artifact and version from which it originated.

5.3 Candidate types

Current canonical Graph Candidate types include:

- Finding
- Investigation
- Mimic
- Pitfall
- Management
- Complication
- Recall Prompt
- Case Reasoning

Each type should have an artifact-specific quality standard.

5.4 Finding candidate

A Finding Candidate represents a clinically meaningful symptom, sign, risk factor, historical feature, or observed pattern.

It should identify:

- the finding
 - its diagnosis context
 - its clinical significance
 - its source
 - any target diagnosis where relevant
-

5.5 Investigation candidate

An Investigation Candidate represents a diagnostic test, result, interpretation, or testing principle.

It should distinguish between:

- test name
 - expected finding
 - diagnostic role
 - limitation
 - discriminator value
-

5.6 Mimic candidate

A Mimic Candidate proposes that another diagnosis is educationally relevant as a source of confusion or differential consideration.

A mimic candidate should not be approved solely because the alternative diagnosis appears in a differential list.

Review should determine:

- whether confusion is clinically plausible
 - which presentation is shared
 - which discriminator is educationally important
 - whether the relationship is directional or reciprocal
-

5.7 Pitfall candidate

A Pitfall Candidate represents a possible learner or clinical reasoning error.

It should state:

- the mistaken assumption or action
 - why the error is plausible
 - the consequence
 - the safer reasoning principle
-

5.8 Management candidate

A Management Candidate represents a diagnosis-level management principle.

It should not be promoted without sufficient context regarding:

- indication
 - timing
 - severity
 - contraindication
 - escalation
 - patient population
-

5.9 Complication candidate

A Complication Candidate represents an important adverse consequence of the diagnosis or its delayed treatment.

It should distinguish:

- direct disease complication
 - treatment complication
 - associated condition
 - severity marker
-

5.10 Recall Prompt candidate

A Recall Prompt Candidate represents potentially reusable active-recall content.

Promotion into governed recall content should require compliance with the Recall Prompt standard in Part IV.

5.11 Case Reasoning candidate

A Case Reasoning Candidate represents a reasoning insight derived from case clue progression or explanation.

It may capture:

- a discriminator
- differential elimination
- escalation recognition
- clue-order principle
- diagnostic turning point

Case-specific reasoning should not automatically be generalised to all presentations of the diagnosis.

6. Graph Candidate Provenance

Every Graph Candidate should preserve:

- diagnosis
- candidate type
- source artifact type
- source artifact identity
- source version
- source path
- raw source text
- normalised text
- structured payload where relevant
- target diagnosis where relevant
- unresolved target text where necessary
- extraction confidence
- reviewer
- review date
- review note
- merge destination where applicable
- promoted Graph Fact where applicable

Provenance should make it possible to trace every governed Graph Fact back to the content from which it emerged.

7. Graph Candidate Normalisation

Graph Candidates should undergo controlled normalisation to support:

- duplicate detection
- clustering
- consistent labels
- target diagnosis resolution
- merge decisions

Normalisation may address:

- letter case
- punctuation
- superficial phrasing differences
- accepted terminology variation

Normalisation must not remove clinically meaningful qualifiers.

Examples include:

- acute versus chronic
- unilateral versus bilateral
- painless versus painful
- proximal versus distal
- exertional versus resting
- early versus late

These qualifiers may determine the clinical meaning of the candidate.

8. Graph Candidate Deduplication

Candidates may be duplicates where they express the same clinical knowledge from different sources.

Duplicate analysis should consider:

- diagnosis
- candidate type
- normalised meaning
- target diagnosis
- clinical qualifiers
- source context
- structured payload

Candidates should not be merged merely because they share keywords.

For example:

- “Chest pain” and “pleuritic chest pain” are not equivalent.
- “Normal troponin” and “serially normal troponin” may carry different diagnostic meaning.
- “Weakness” and “progressive proximal muscle weakness” should not be collapsed.

9. Graph Candidate Lifecycle

The canonical Graph Candidate lifecycle is:

Candidate

↓

Editorial Review

↓

Approved, Rejected, or Merged

↓

Promoted to Graph Fact where appropriate

9.1 Candidate

The proposed knowledge exists but remains ungoverned.

9.2 Approved

The candidate has been accepted as valid structured knowledge.

Approval should identify whether promotion into a Graph Fact is immediate or separately controlled.

9.3 Rejected

The candidate has not been accepted.

Rejection may occur because it is:

- clinically inaccurate
- too generic

- duplicative
- improperly scoped
- unsupported
- not educationally useful
- incorrectly classified

9.4 Merged

The candidate has been consolidated into another candidate or existing fact.

The original candidate and source provenance should remain traceable.

10. Graph Fact

10.1 Purpose

A Graph Fact is a governed diagnosis-level knowledge statement available for use across WEOS.

It represents accepted structured knowledge rather than unreviewed extraction.

10.2 Required structure

A Graph Fact should preserve:

- diagnosis
 - fact type
 - label
 - normalised label
 - structured payload
 - target diagnosis where relevant
 - operational status
 - source candidate
 - provenance
 - creation and maintenance history
-

10.3 Active status

An Active Graph Fact may influence:

- editorial workspace projections
- generation context

- case validation
- education assessment
- reasoning paths
- teaching relationships
- recall generation
- coverage analysis

Active status must follow governance.

It should not result merely from successful extraction.

10.4 Archived status

A Graph Fact may be archived where it is:

- outdated
- superseded
- duplicated
- no longer educationally relevant
- inconsistent with current evidence
- based on a deprecated diagnosis relationship

Archival should preserve provenance and prior use.

10.5 Fact scope

Every Graph Fact should clearly identify its scope.

Possible scope qualifiers include:

- typical
- early
- late
- severe
- uncommon
- paediatric
- adult
- pregnancy-associated
- complication-specific
- setting-specific

A fact that is true only in a subgroup must not be represented as universally applicable.

11. Evidence Node

11.1 Purpose

An Evidence Node represents a reusable clinical concept that may participate in reasoning across multiple diagnoses.

Examples include:

- exertional chest pressure
- hyperreflexia
- elevated serum lipase
- cavitary lung lesion
- recent antibiotic exposure
- haemodynamic instability
- painless haematuria

The Evidence Node defines the finding.

It does not determine what the finding means for every diagnosis.

Meaning is established through Diagnosis–Evidence Relationships.

11.2 Evidence types

Canonical Evidence Types include:

- Symptom
- Examination
- Laboratory
- Imaging
- Risk Factor
- History
- Management
- Complication
- Epidemiology

Evidence type should describe the nature of the concept rather than how it happened to be presented in one case.

11.3 Clinical category

Evidence Nodes may also belong to broad clinical categories such as:

- pain
- bleeding
- infection
- neurological
- respiratory
- cardiovascular
- gastrointestinal
- endocrine
- renal
- trauma
- other

Clinical categories support organisation and reasoning analysis.

They should not replace evidence type.

11.4 Node identity

Every Evidence Node should possess:

- stable identity
- normalised key
- display label
- evidence type
- clinical category
- accepted synonyms
- governance status

The display label should remain clinically meaningful outside one source sentence.

11.5 Evidence Node lifecycle

An Evidence Node may be:

- Candidate
- Active
- Rejected
- Deprecated

An Active Evidence Node is available for governed relationships.

Activation does not imply that the node supports every diagnosis to which it has been tentatively connected.

Each relationship requires independent governance.

12. Diagnosis–Evidence Relationship

12.1 Purpose

A Diagnosis–Evidence Relationship defines how an Evidence Node affects reasoning about a diagnosis.

It transforms a clinical finding into explicit diagnostic meaning.

12.2 Relationship types

Current canonical relationship types include:

- Supports
 - Discriminates
 - Escalates
 - Rules Out
 - Complication Signal
 - Management Signal
-

12.3 Supports

A supporting relationship indicates that the evidence increases the plausibility of the diagnosis.

Support should be appropriately qualified.

A common non-specific symptom may provide weak support.

A characteristic pattern may provide strong support.

Support does not necessarily establish diagnosis.

12.4 Discriminates

A discriminating relationship indicates that the evidence helps distinguish the diagnosis from one or more plausible alternatives.

A discriminator should identify:

- the target diagnosis
 - relevant competing diagnosis or group
 - direction of discrimination
 - clinical reasoning summary
 - strength
-

12.5 Escalates

An escalating relationship indicates that the evidence changes urgency, level of care, monitoring, or intervention.

Examples include:

- haemodynamic instability
- altered mental state
- respiratory failure
- peritonism
- rapidly progressive neurological deficit

Escalation evidence may support management even when it does not strongly identify the diagnosis.

12.6 Rules Out

A rules-out relationship indicates that evidence materially weakens or excludes the diagnosis.

This relationship should be used cautiously.

Most clinical findings do not absolutely rule out a diagnosis.

Where exclusion is incomplete, the relationship should reflect reduced probability rather than certainty.

12.7 Complication Signal

A complication-signal relationship indicates that the evidence suggests a recognised complication of the diagnosis.

The relationship should distinguish between:

- presence of the primary diagnosis
 - severity of the diagnosis
 - presence of a complication
-

12.8 Management Signal

A management-signal relationship indicates that the evidence changes treatment, referral, monitoring, or intervention.

It should preserve the clinical context in which the management implication applies.

13. Relationship Strength

Diagnosis–Evidence Relationships may carry strength.

Strength should reflect the clinical and educational weight of the relationship.

It may consider:

- sensitivity
- specificity
- prevalence
- clinical context
- discriminator value
- severity relevance
- evidence quality

Strength must not create false mathematical precision where evidence is qualitative.

14. Discriminator Weight

Discriminator weight represents the relative value of the evidence in distinguishing the diagnosis from important alternatives.

A finding may strongly support several diagnoses while possessing low discriminator value.

For example:

Fever may support infection but may not distinguish between several infectious diagnoses.

A narrower finding may possess greater discriminator value.

Discriminator weight should therefore remain separate from general relationship strength.

15. Contradictory Diagnosis Context

A Diagnosis–Evidence Relationship may identify diagnoses contradicted or weakened by the evidence.

This should support:

- differential elimination
- clue-progression analysis
- explanation generation
- diagnostic comparison
- case validation

Contradictory diagnosis references should resolve to governed registry identities where possible.

16. Supporting Artifacts

A Diagnosis–Evidence Relationship may be supported by:

- a Teaching Relationship
- a Teaching Rule
- a Clinical Case
- a Graph Fact
- external evidence

Supporting artifacts explain why the relationship exists or how it is taught.

They do not automatically establish scientific authority.

17. Diagnosis–Evidence Relationship Lifecycle

The canonical lifecycle is:

Candidate

↓

Review

↓

Active, Rejected, or Deprecated

17.1 Candidate

The relationship has been proposed but remains ungoverned.

17.2 Active

The relationship has been accepted for operational use.

17.3 Rejected

The proposed relationship has not been accepted.

17.4 Deprecated

The relationship is preserved historically but should not influence current editorial work.

18. Teaching Relationship

18.1 Purpose

A Diagnosis Teaching Relationship defines how two diagnoses should be compared, contrasted, clustered, or taught together.

It represents an educational relationship rather than a simple clinical association.

18.2 Directionality

Teaching Relationships are directional.

The source diagnosis is the diagnosis being taught.

The target diagnosis is the diagnosis used for contrast or connection.

For example:

Source: Chronic Pulmonary Aspergillosis
Target: Pulmonary Tuberculosis
Purpose: Teach discriminator

The reverse direction may require different teaching content and should not be assumed automatically.

18.3 Relationship types

Canonical Teaching Relationship types include:

- Differential Discriminator
 - Mimic Confusion
 - Shared Presentation
 - Escalation Contrast
 - Management Contrast
 - Investigation Contrast
 - Complication Relationship
-

18.4 Differential discriminator

Identifies the key findings that help distinguish two plausible diagnoses.

The relationship should state:

- why both diagnoses are considered
 - which finding favours the source diagnosis
 - which finding favours the target
 - what uncertainty remains
-

18.5 Mimic confusion

Describes a common learner or clinical confusion between diagnoses.

It should identify:

- shared presentation
 - reason for confusion
 - harmful assumption
 - corrective discriminator
-

18.6 Shared presentation

Identifies a presentation shared by the diagnoses.

A shared-presentation relationship should not imply equivalence.

Its value lies in teaching how the same presenting problem can arise from different diseases.

18.7 Escalation contrast

Compares how urgency, deterioration, or referral differs between diagnoses.

It may teach:

- which red flags demand urgent action
 - why similar symptoms carry different risks
 - what deterioration means in each diagnosis
-

18.8 Management contrast

Compares important treatment differences.

This relationship is especially valuable where confusing the diagnoses would lead to:

- inappropriate treatment
 - delayed treatment
 - contraindicated treatment
 - missed definitive intervention
-

18.9 Investigation contrast

Compares how tests are selected or interpreted across diagnoses.

It may identify:

- different first-line tests
 - characteristic findings
 - limitations
 - false reassurance
 - confirmatory strategies
-

18.10 Complication relationship

Describes a meaningful relationship between diagnosis complications.

It should clarify whether:

- both diagnoses cause similar complications
 - one diagnosis is a complication of another
 - complication patterns help distinguish them
 - delayed diagnosis produces different outcomes
-

19. Teaching Purpose

Every Teaching Relationship should possess an explicit teaching purpose.

Canonical purposes include:

- Teach Discriminator
- Prevent Common Error
- Build Differential Cluster
- Support Case Generation
- Support Education
- Support Recall

A relationship may be clinically true but unsuitable for a particular teaching purpose.

Purpose should therefore be independently reviewed.

20. Teaching Relationship Content

A Teaching Relationship may contain:

- discriminator summary
- common confusion reason
- learner pitfall
- suggested teaching-rule key
- supporting graph fact
- supporting differential link
- supporting teaching rule
- relationship strength
- governance status

Each field should contribute to how the relationship is used.

21. Teaching Relationship Lifecycle

The canonical lifecycle is:

Candidate

↓

Needs Review

↓

Active, Rejected, or Deprecated

An Active Teaching Relationship may influence learner-facing content and generation.

Activation should require human review.

22. Differential Mapping

22.1 Purpose

Differential Mapping resolves authored differential text to a governed Diagnosis Registry identity.

Mappings may originate from:

- Clinical Cases
 - Case Revisions
 - Diagnosis Education
 - Education Revisions
-

22.2 Mapping states

A mapping may be:

- Resolved
- Ambiguous
- Unresolved
- Rejected

These states describe identity resolution.

They do not determine the clinical quality of the differential itself.

22.3 Resolution standard

A mapping should consider:

- canonical name
- accepted aliases
- abbreviation
- normalised terminology
- clinical context
- subtype
- source section
- authorial intent

Automatic string similarity should not override clinically meaningful distinctions.

22.4 Ambiguous mapping

A mapping is ambiguous where more than one registry diagnosis could reasonably match the text.

Ambiguity should be surfaced for editorial review.

The system must not select a diagnosis silently where the choice would alter clinical meaning.

22.5 Unresolved mapping

A mapping is unresolved where no suitable registry entry can be identified.

Resolution may require:

- clarification of the source text
 - selection of an existing diagnosis
 - creation of a registry candidate
 - rejection of the differential
 - preservation as non-registry text where policy permits
-

23. Differential Link

23.1 Purpose

A Differential Link represents the governed connection between a source artifact and a resolved differential diagnosis.

It allows WEOS to use the relationship in:

- reasoning analysis
 - graph development
 - coverage assessment
 - case comparison
 - education governance
 - generation context
-

23.2 Differential roles

Canonical roles include:

- Primary Mimic
 - Differential
 - Important Exclusion
 - Teaching Differential
-

23.3 Primary mimic

A Primary Mimic is a diagnosis central to the reasoning challenge.

It normally requires:

- shared presentation
 - explicit discriminator
 - explanation coverage
 - appropriate clue-progression treatment
-

23.4 Important exclusion

An Important Exclusion may be less likely but clinically dangerous to miss.

It should be governed according to its safety relevance.

23.5 Teaching differential

A Teaching Differential is included primarily because it reinforces an important educational comparison.

It should still remain clinically plausible.

24. Diagnostic Comparison

24.1 Purpose

A Diagnostic Comparison is the structured educational interpretation of differences between diagnoses.

It may be assembled from:

- Teaching Relationships
 - Evidence Relationships
 - Differential Links
 - Teaching Rules
 - Graph Facts
 - Reasoning Paths
-

24.2 Comparison dimensions

A comparison may address:

- epidemiology
- onset
- time course
- symptoms
- examination
- laboratory findings
- imaging
- discriminator
- management
- escalation
- complications

Only dimensions that meaningfully distinguish the diagnoses should be included.

24.3 Comparison quality

A strong comparison should:

- begin from genuine shared uncertainty
 - identify the highest-value discriminators
 - avoid artificial binary distinctions
 - acknowledge overlap
 - avoid absolute wording where exceptions exist
 - remain applicable beyond one memorised table
-

25. Reasoning Path

25.1 Purpose

A Reasoning Path is a governed diagnosis-level specification for how a particular reasoning objective should be taught or generated.

It coordinates the knowledge required to produce a coherent case, education section, teaching rule, or recall activity.

25.2 Reasoning goals

Canonical Reasoning Goals include:

- Differential Discrimination
- Escalation Recognition
- Management Contrast
- Complication Recognition
- Early-Presentation Recognition
- Resource-Limited Reasoning

Each path should focus on one primary goal.

25.3 Generation purposes

A Reasoning Path may support:

- Case Generation
- Teaching-Rule Generation
- Education Generation

- Recall Generation

Generation purpose identifies where the path may be used.

It does not determine whether generated content is approved.

25.4 Required structure

A Reasoning Path may define:

- stable key
 - title
 - diagnosis
 - reasoning goal
 - primary differentials
 - supporting Teaching Relationships
 - supporting Evidence Relationships
 - discriminator Evidence Nodes
 - escalation Evidence Nodes
 - contradictory Evidence Nodes
 - required teaching points
 - forbidden evidence patterns
 - recommended clue distribution
 - generation purpose
 - readiness score
 - governance status
 - reviewer
-

26. Primary Differentials

Primary differentials identify the diagnoses that create the central reasoning challenge.

They should be selected because they:

- share an important presentation
- are commonly confused
- are dangerous to miss
- teach a valuable discriminator
- align with the reasoning goal

A reasoning path should not contain an unlimited differential list.

27. Supporting Relationships

Reasoning Paths should reference governed Teaching and Evidence Relationships where available.

This prevents the same reasoning from being recreated inconsistently in every generation request.

Supporting relationships should be active and relevant to the path's purpose.

28. Discriminator Evidence

Discriminator Evidence Nodes identify the findings expected to separate the diagnosis from its primary alternatives.

The path should clarify:

- which diagnosis the evidence favours
 - what it argues against
 - expected strength
 - appropriate clue timing
 - whether it is required or optional
-

29. Escalation Evidence

Escalation Evidence identifies findings that should trigger urgent action, referral, monitoring, or intervention.

It should be included where escalation recognition is part of the reasoning objective.

30. Contradictory Evidence

Contradictory Evidence identifies findings that should weaken the target diagnosis or raise concern about case coherence.

This enables generation and validation to detect when a draft accidentally undermines its intended answer.

Contradictory evidence should not be interpreted as absolute exclusion unless clinically justified.

31. Required Teaching Points

Required teaching points identify the concepts that generated or authored content must address.

They may include:

- a discriminator
- common error
- investigation limitation
- escalation threshold
- management contrast
- complication signal

Required teaching points should remain specific enough for validation.

32. Forbidden Evidence Patterns

Forbidden Evidence Patterns identify evidence use that would make content unsafe or educationally weak.

Examples include:

- revealing the diagnosis in the opening clue
- using a pathognomonic finding before meaningful reasoning
- combining mutually inconsistent findings
- relying on an unavailable specialist test in a resource-limited path
- using a normal result as absolute exclusion
- introducing treatment response that directly names the diagnosis
- contradicting the intended disease stage

Forbidden patterns should include rationale.

33. Recommended Clue Distribution

A Reasoning Path may specify the expected distribution of clue types.

For example:

History → Examination → Basic investigations → Discriminator

or:

Risk factor → Early presentation → Deterioration → Escalation evidence

Recommended distribution guides generation.

It should not force all cases into identical sequences.

34. Reasoning Path Readiness

A Reasoning Path should only become active when its required foundations are sufficiently complete.

Readiness may consider:

- resolved diagnosis
- defined reasoning goal
- resolved primary differentials
- active supporting relationships
- sufficient discriminator evidence
- required teaching points
- absence of critical contradictions
- appropriate generation purpose
- human review

A readiness score is an assessment.

It is not an approval decision.

35. Reasoning Path Lifecycle

The canonical lifecycle is:

Candidate

↓

Review

↓

Active, Rejected, or Deprecated

35.1 Candidate

The path is under development.

35.2 Active

The path may influence generation and assessment.

35.3 Rejected

The path has not been accepted.

35.4 Deprecated

The path should no longer guide new work but remains historically traceable.

36. Reasoning Draft Validation

36.1 Purpose

Reasoning Draft Validation assesses whether generated or authored artifacts correctly apply governed reasoning context.

It may evaluate:

- Cases
 - Teaching Rules
 - Diagnosis Education
 - Individual Education Sections
-

36.2 Validation outputs

Reasoning Draft Validation may produce:

- trust score
- trust tier
- validation status
- blockers
- warnings
- strengths
- hallucination-risk signals

- reasoning coverage
 - evidence coverage
 - discriminator coverage
 - unsupported-claim signals
 - recommendations
-

36.3 Trust tiers

Canonical Trust Tiers include:

- High Trust
- Review Required
- Low Trust
- Blocked

Trust tier communicates assessment confidence and risk.

It does not authorise approval or publication.

36.4 Validation statuses

Canonical Validation Statuses include:

- Passed
- Needs Review
- Failed

A passed validation means the automated assessment found no defined blocking failure.

It does not establish full clinical correctness.

37. Unsupported-Claim Signal

37.1 Purpose

An Unsupported-Claim Signal identifies a statement that may lack sufficient support from the available governed context or external evidence.

It is a quality and governance signal.

It is not automatically proof that the claim is false.

37.2 Signal sources

Unsupported-claim signals may arise from:

- contradiction with Teaching Rules
 - absence from active Evidence Relationships
 - mismatch with Reasoning Path
 - lack of supporting reference
 - overstatement of certainty
 - management claim without context
 - invented discriminator
 - ungrounded AI output
 - inconsistency with diagnosis metadata
-

37.3 Claim classification

Signals should distinguish:

- factual clinical claim
- diagnostic interpretation
- management recommendation
- prognostic statement
- scoring-system statement
- epidemiological statement
- educational generalisation

Different claim types require different forms of support.

37.4 Signal severity

Severity may consider:

- potential learner harm
- clinical consequence
- degree of contradiction
- confidence of detection
- publication status
- visibility
- repetition across artifacts

A minor unsupported educational phrase should not be treated identically to unsafe management guidance.

37.5 Resolution

An Unsupported-Claim Signal may be resolved by:

- adding supporting evidence
- revising the claim
- narrowing its scope
- adding qualification
- linking an active relationship
- rejecting the draft
- documenting an accepted editorial rationale
- removing the claim

Resolution should preserve the original signal and decision history.

38. Knowledge Contradiction

A Knowledge Contradiction exists where governed or candidate artifacts make clinically incompatible claims.

Contradiction analysis may compare:

- Case and Diagnosis Education
- Teaching Rule and Case
- Graph Fact and Reference
- Evidence Relationship and Explanation
- Management guidance across sections
- Scoring-system use across diagnoses
- active and deprecated knowledge

Contradictions should be surfaced for review.

The system should not silently choose one version as correct without governance.

39. Knowledge Coverage

Knowledge Coverage evaluates whether the diagnosis possesses sufficient governed relationships for its intended editorial use.

Coverage may assess:

- findings
- investigations

- mimics
- discriminators
- escalation signals
- management signals
- complications
- pitfalls
- reasoning paths
- recall relationships

Coverage should be purpose-specific.

A diagnosis may possess enough knowledge for education generation but not enough for advanced case generation.

40. Knowledge Readiness

Knowledge Readiness should identify the operation being evaluated.

Examples include:

- ready for graph review
- ready for reasoning-path creation
- ready for case generation
- ready for differential teaching
- ready for recall generation
- ready for publication assessment

Readiness should identify:

- available foundations
 - missing relationships
 - blockers
 - warnings
 - recommended next actions
-

41. Knowledge-Governance Review

Review of knowledge artifacts should assess:

- clinical accuracy
- appropriate scope
- diagnosis identity
- relationship direction

- relationship type
- teaching purpose
- source provenance
- generalisability
- contradiction risk
- evidence support
- educational usefulness
- duplicate risk
- operational consequences

Graph review should not be reduced to accepting or rejecting extracted text.

42. Governance Decisions

Knowledge Governance may produce decisions including:

- approve candidate
- reject candidate
- merge candidate
- promote candidate to fact
- archive fact
- activate evidence node
- reject evidence node
- activate evidence relationship
- deprecate evidence relationship
- activate teaching relationship
- request relationship changes
- reject teaching relationship
- activate reasoning path
- deprecate reasoning path
- resolve differential mapping
- reject differential mapping
- approve differential link
- resolve unsupported claim
- request evidence
- repair contradiction

Every decision should preserve:

- reviewer
- rationale
- affected artifact
- affected version where relevant
- source context
- resulting state
- date and time

43. Human Authority

Automated systems may:

- extract Graph Candidates
- suggest target diagnoses
- estimate confidence
- detect duplicates
- propose Evidence Relationships
- identify Teaching Relationships
- generate Reasoning Paths
- calculate readiness
- detect contradictions
- flag unsupported claims
- recommend editorial action

Automated systems must not independently:

- promote candidate knowledge into institutional fact
- activate diagnosis relationships
- resolve clinically ambiguous diagnosis mappings
- approve reasoning paths
- dismiss high-risk unsupported claims
- publish learner-facing consequences

Human reviewers retain authority over governed knowledge.

44. Knowledge Maintenance

Knowledge maintenance may be triggered by:

- new clinical evidence
- guideline changes
- diagnosis deprecation
- terminology changes
- repeated learner confusion
- case inconsistency
- education revision
- duplicate detection
- graph contradiction
- new important mimics
- changed management
- changed scoring systems
- specialty review

Maintenance should assess downstream impact.

Changing one relationship may affect:

- cases
 - education
 - recall prompts
 - reasoning paths
 - validation rules
 - publication readiness
-

45. Provenance Preservation

Every active knowledge artifact should remain traceable to its origin and governance history.

Provenance should answer:

- Where did this knowledge originate?
- Which artifact and version contained it?
- Was it extracted, generated, imported, or manually authored?
- Who reviewed it?
- Which decision activated it?
- Has it been revised, merged, or deprecated?
- Which learner-facing artifacts depend upon it?

Provenance is essential to institutional trust.

46. Quality Standard for Governed Knowledge

High-quality governed knowledge should be:

Clinically accurate

The knowledge is defensible within its stated context.

Explicitly scoped

Qualifiers, population, stage, and setting are preserved.

Diagnosis-resolved

Diagnosis relationships use governed identities.

Reasoning-centred

Relationships explain how evidence changes clinical interpretation.

Differentially useful

Important mimics and discriminators are represented meaningfully.

Provenance-preserving

The source and review history remain traceable.

Non-duplicative

Equivalent knowledge is consolidated without erasing source history.

Educationally purposeful

Relationships support a defined teaching or reasoning function.

Operationally safe

Only governed knowledge influences generation, validation, and learner-facing publication.

Maintainable

Knowledge may be revised, deprecated, merged, or archived without erasing history.

47. Normative Statements

- Extracted knowledge **MUST** begin as candidate knowledge.
- Candidate knowledge **MUST NOT** be treated as a governed fact before editorial approval.
- Every Graph Candidate **MUST** preserve its source artifact, source version, and source location.
- Candidate normalisation **MUST NOT** remove clinically meaningful qualifiers.
- Candidate merging **MUST** preserve the provenance of all contributing candidates.
- Every active Graph Fact **MUST** possess governed provenance.
- Evidence Nodes **MUST** remain distinct from Diagnosis–Evidence Relationships.
- An active Evidence Node **MUST NOT** imply an active relationship with any diagnosis.
- Diagnosis–Evidence Relationships **MUST** state how the evidence affects reasoning.
- `RULES_OUT` relationships **MUST** be used conservatively and clinically defensibly.
- Relationship strength **MUST NOT** be treated as equivalent to discriminator value.
- Teaching Relationships **MUST** preserve source and target diagnosis direction.

- Every Teaching Relationship **MUST** possess an explicit teaching purpose.
 - Active Teaching Relationships **MUST** undergo human review.
 - Differential mappings **MUST NOT** silently resolve clinically ambiguous text.
 - Primary mimics **SHOULD** possess explicit discriminator coverage.
 - Reasoning Paths **MUST** define a primary reasoning goal.
 - Reasoning Paths used for generation **MUST** identify their generation purpose.
 - Forbidden Evidence Patterns **MUST** include clinically or educationally meaningful rationale.
 - Reasoning readiness scores **MUST NOT** be interpreted as approval.
 - Automated validation **MUST NOT** replace clinical editorial review.
 - Unsupported-Claim Signals **MUST** remain distinguishable from confirmed factual errors.
 - High-risk unsupported management claims **MUST** block publication until resolved.
 - Knowledge contradictions **MUST** be surfaced rather than silently overwritten.
 - Only governed knowledge **SHOULD** influence production generation context.
 - Deprecated and archived knowledge **MUST** remain historically traceable.
 - Knowledge-governance decisions **MUST** preserve reviewer accountability and rationale.
 - AI-generated relationships and reasoning paths **MUST NOT** become active without human governance.
-

Closing Principle

Clinical reasoning cannot be governed if it remains implicit.

Wardle must be able to identify which findings support a diagnosis, which findings distinguish it from its mimics, which signals change urgency, which relationships shape learner understanding, and where every claim originated.

Knowledge and Reasoning Governance transforms isolated educational content into a connected, explainable, and maintainable clinical knowledge system.

Its objective is not merely to connect diagnoses and findings.

Its objective is to ensure that every connection used to teach, generate, assess, or publish diagnostic knowledge is clinically meaningful, educationally purposeful, and institutionally governed.

Wardle Editorial Canon

Standards for Governed Editorial Artifacts

Part VI — AI Draft Standards

1. Purpose

AI Draft Standards define how artificial intelligence may contribute to the creation, analysis, revision, and maintenance of governed clinical knowledge within the Wardle Editorial Operating System.

Artificial intelligence may increase editorial capacity by:

- producing initial drafts
- proposing revisions
- identifying missing knowledge
- generating structured educational content
- suggesting relationships
- analysing reasoning quality
- detecting inconsistencies
- recommending editorial actions

Artificial intelligence does not possess editorial authority.

Its outputs begin as candidate knowledge and must pass through the same clinical, educational, and governance standards applied to manually authored work.

The purpose of AI governance is therefore not to restrict useful generation.

It is to ensure that AI contribution remains:

- clinically grounded
 - educationally purposeful
 - transparent
 - reviewable
 - reversible
 - attributable
 - subordinate to human judgement
-

2. Foundational Principle

AI may contribute to editorial production.

AI may not confer institutional trust.

The canonical flow is:

Governed Editorial Context

↓

AI Generation or Analysis

↓

Candidate Draft

↓

Automated Validation

↓

Human Editorial Review

↓

Accept, Reject, Request Changes, or Supersede

↓

Controlled Application

↓

Governed Artifact Revision

No stage before human editorial decision creates approved knowledge.

3. AI as an Editorial Contributor

Within WEOS, AI should be treated as a contributor with defined capabilities and limitations.

AI may:

- synthesise provided context
- structure content
- generate controlled variants
- detect likely omissions
- compare artifacts
- propose corrections
- identify possible contradictions
- generate candidate relationships
- assist with maintenance

AI must not be treated as:

- a clinical authority
- an autonomous reviewer
- a publisher
- an approver
- an evidence source
- a substitute for specialty judgement

The credibility of AI output depends on its governed context, validation, and human review.

4. AI Artifact Families

AI may generate or contribute to:

- Clinical Case Drafts
- Case Explanation Drafts
- Clue Revision Drafts
- Diagnosis Education Drafts
- Education Section Drafts
- Recall Prompt Drafts
- Teaching Rule Drafts
- Editorial Brief Drafts
- Graph Candidates
- Teaching Relationship Candidates
- Evidence Relationship Candidates
- Reasoning Path Drafts
- Differential Suggestions
- Unsupported-Claim Repairs
- Editorial Recommendations

Each AI output should be governed according to the standard of the artifact it proposes to create or revise.

There is no separate reduced clinical standard for AI-generated content.

5. AI Draft Definition

An AI Draft is a proposed artifact or proposed change produced with material artificial-intelligence involvement.

An AI Draft may be:

- a complete new artifact
- a replacement section
- a focused revision
- an added clue
- a relationship proposal
- a correction to an unsupported claim
- a structured recommendation
- an alternative formulation

An AI Draft is not the governed artifact itself.

It remains separate until accepted and applied through an explicit editorial action.

6. AI Provenance

6.1 Purpose

AI provenance records how and why a draft was produced.

It allows reviewers to understand:

- what prompted generation
- what context was supplied
- what the system produced
- which artifact was affected
- what decision followed
- whether the output was applied

Without provenance, AI-assisted editorial change becomes difficult to audit or reproduce.

6.2 Required provenance

Every material AI Draft should preserve:

- diagnosis identity
 - generation purpose
 - action type
 - source issue or identified need
 - input context
 - source artifacts
 - source artifact versions
 - applied teaching rules
 - reasoning path where relevant
 - generated output
 - affected artifact type
 - affected artifact identity
 - generation time
 - model or generator version where available
 - validation outcome
 - reviewer decision
 - reviewer identity
 - review note
 - application status
 - resulting artifact revision where applied
-

6.3 Source issue

The source issue explains why AI generation occurred.

Examples include:

- missing Diagnosis Education
- weak investigation section
- incomplete differential coverage
- premature clue leakage
- unsupported management claim
- absent recall prompts
- insufficient case inventory
- missing escalation evidence
- inconsistent explanation
- outdated scoring-system guidance

AI generation should respond to an identifiable editorial need.

Unbounded generation without purpose should be avoided.

6.4 Input context

Input context is the governed knowledge supplied to the AI system.

It may contain:

- diagnosis metadata
- editorial brief
- learning goals
- teaching rules
- reasoning paths
- active graph facts
- evidence relationships
- approved cases
- approved Diagnosis Education
- required mimics
- discriminators
- pitfalls
- investigations
- scoring systems
- management anchors
- difficulty guidance
- reviewer instructions

The exact context used should remain reconstructable.

7. Generation Context

7.1 Purpose

Generation Context defines the governed editorial boundaries within which AI may draft content.

It should reduce the likelihood that the AI:

- invents educational priorities
 - ignores required mimics
 - contradicts teaching rules
 - introduces unsupported findings
 - reveals a diagnosis too early
 - creates generic education
 - recommends unsafe management
 - generates at the wrong difficulty
-

7.2 Context components

A generation context may include:

- diagnosis
 - concise clinical context
 - learning goals
 - mandatory content
 - prohibited or discouraged content
 - mimics
 - discriminators
 - pitfalls
 - investigations
 - scoring systems
 - management anchors
 - required teaching units
 - acceptable manifestations
 - editorial brief
 - difficulty strategy
 - source-availability summary
-

7.3 Diagnosis context

The diagnosis context should identify:

- canonical diagnosis
- display label
- specialty
- category
- body system
- clinical setting
- baseline difficulty
- age group where relevant
- urgency where relevant

AI should not infer diagnosis identity from an ungoverned descriptive title where a registry identity exists.

7.4 Learning goals

Learning goals define what the generated artifact should teach.

The AI should not introduce a competing educational objective unless explicitly permitted.

For example, a case intended to teach differential discrimination should not become primarily a management-recall case.

7.5 Mandatory content

Mandatory content may include:

- required findings
- required teaching points
- required mimics
- required investigations
- required management principles
- required complication recognition
- required escalation evidence

Mandatory content should be relevant to the artifact type.

Not every item must appear verbatim.

Its educational meaning must be preserved.

7.6 Avoidance guidance

Avoidance guidance identifies content or patterns the AI should not use.

Examples include:

- forbidden early clues
- pathognomonic wording in the opening
- unsupported diagnostic certainty
- generic educational phrases
- inappropriate investigation dependence
- unsafe management generalisation
- repetitive manifestations
- prohibited evidence patterns
- outdated terminology

Avoidance guidance should include reasons where possible.

7.7 Mimics and discriminators

Where diagnostic reasoning is required, generation context should identify:

- important mimics
- why they are confused
- key separators
- findings that favour the target
- findings that weaken the target
- mimics that should remain plausible until a later stage

AI should not manufacture artificial differentials merely to populate a section.

7.8 Difficulty strategy

Difficulty guidance may define:

- target difficulty
- expected diagnostic lock-in point
- findings to avoid too early
- differentials to keep alive
- acceptable alternative manifestations
- depth of investigation
- reasoning complexity

Difficulty should arise from controlled uncertainty and discrimination.

It should not arise from obscure terminology or withheld essential information.

7.9 Context sufficiency

Generation should not proceed automatically where required context is critically incomplete.

Possible blockers include:

- unresolved diagnosis identity
- diagnosis prohibited from generation
- absent educational scope
- contradictory teaching rules
- unresolved required mimic
- insufficient reasoning foundation
- no reliable management anchor for a management-focused artifact

Where generation proceeds with incomplete non-critical context, the limitation should be disclosed to reviewers.

8. Generation Purpose

Every AI generation request must identify its purpose.

Canonical purposes include:

- Case Generation
- Diagnosis Education Generation
- Education Section Regeneration
- Teaching Rule Generation
- Editorial Brief Generation
- Graph Candidate Generation
- Teaching Relationship Generation
- Evidence Relationship Generation
- Recall Prompt Generation
- Clue Revision
- Unsupported-Claim Repair
- Maintenance Update
- Quality Recommendation

The purpose determines:

- required context
 - output schema
 - validation standard
 - review packet
 - permitted application action
-

9. Structured Output

9.1 Purpose

AI output should use a defined structure appropriate to the target artifact.

Structured output improves:

- validation
- review
- traceability
- completeness assessment

- controlled application
 - revision comparison
-

9.2 Schema compliance

An AI Draft should satisfy the expected artifact schema before entering substantive review.

Schema failure may include:

- missing required fields
- invalid types
- invalid lifecycle value
- malformed identifiers
- duplicate stable keys
- incomplete structured entries
- content outside the permitted artifact shape

Schema compliance does not establish clinical quality.

9.3 No hidden narrative

Important claims should not be hidden within unstructured text where the artifact schema expects explicit fields.

For example:

- a discriminator should appear as a discriminator
 - a management caution should appear as a caution
 - an escalation implication should be explicitly represented
 - a scoring-system limitation should not be buried in a general paragraph
-

10. AI-Generated Clinical Case Draft

10.1 Purpose

An AI-Generated Clinical Case Draft proposes a new reasoning experience for a governed diagnosis.

It must comply with the Clinical Case Standards in Part III.

10.2 Required generation context

Case generation should consider:

- diagnosis identity
 - case purpose
 - learning goal
 - reasoning objective
 - target difficulty
 - required mimics
 - discriminators
 - teaching rules
 - forbidden evidence patterns
 - preferred clue types
 - excluded clue types
 - expected clue distribution
 - escalation requirements
 - management anchors where relevant
-

10.3 Case draft requirements

The draft should include:

- clinically plausible patient context
 - coherent chronology
 - appropriate opening uncertainty
 - structured clue sequence
 - meaningful differentials
 - defensible final diagnosis
 - aligned explanation
 - intended difficulty
 - source and generation provenance
-

10.4 Case-specific AI risks

Review should specifically assess for:

- premature leakage
- internally inconsistent findings
- invented laboratory values
- unrealistic chronology
- descriptive rather than canonical diagnosis
- incorrect clue explanations
- artificial differentials

- unsupported management
 - excessive narrative
 - clue duplication
 - failure to meet the intended learning goal
-

10.5 Case draft status

An AI-generated case should begin as a draft.

It must not enter approved, playable, ready-to-publish, or published states directly from generation.

11. AI-Generated Case Explanation

An AI-generated explanation should:

- interpret the actual clues
- identify how probability changed
- explain the major discriminator
- compare relevant mimics
- avoid invented findings
- preserve appropriate uncertainty
- use the canonical diagnosis
- align with the case revision reviewed

An explanation generated from an outdated case revision must not be applied to a newer revision without renewed assessment.

12. AI-Generated Clue Revision

12.1 Purpose

An AI-Generated Clue Revision proposes a focused repair to an existing case.

It may:

- rewrite a clue
- remove a clue
- add a clue
- reposition a clue
- narrow an over-specific clue
- strengthen a discriminator

- correct clinical inconsistency
-

12.2 Required content

A Clue Revision Draft should preserve:

- original clue
 - proposed clue
 - clue position
 - source issue
 - rationale
 - expected effect
 - case revision
 - validation findings
 - review status
-

12.3 Expected effect

The expected effect should be testable.

Examples include:

- reduce premature leakage
 - maintain pulmonary tuberculosis as a plausible mimic until clue four
 - correct the mismatch between the fever clue and its explanation
 - add evidence of neurological localisation
 - remove an unsupported laboratory abnormality
 - improve escalation recognition
-

12.4 Application

Accepting a Clue Revision Draft should not silently change the governed case.

Application should:

- identify the accepted draft
 - create or update an appropriate case revision
 - preserve the original clue
 - record the responsible editor
 - trigger revalidation where required
 - require renewed review where the reasoning changed materially
-

13. AI-Generated Diagnosis Education

13.1 Purpose

An AI-Generated Diagnosis Education Draft proposes a structured learner-facing resource for one diagnosis.

It must comply with the Diagnosis Education Standards in Part IV.

13.2 Required context

Education generation should consider:

- diagnosis identity
 - editorial brief
 - learning goals
 - teaching rules
 - clinical pattern
 - required findings
 - required mimics
 - discriminators
 - investigations
 - scoring systems
 - management anchors
 - complications
 - required pitfalls
 - recall concepts
 - references or evidence requirements
-

13.3 Section completeness

A complete generated education draft should address required sections through the expected structured schema.

Missing non-applicable content should be represented deliberately rather than concealed through generic filler.

13.4 Diagnosis specificity

AI-generated education must be checked for generic content that could apply unchanged to multiple unrelated diagnoses.

Weak content includes statements such as:

- may present with several symptoms
- diagnosis requires clinical correlation
- management depends on severity
- early diagnosis is important
- complications can be serious

These statements require diagnosis-specific clinical meaning.

13.5 Educational reasoning

Generated entries should explain:

- why a finding matters
- how it changes suspicion
- what it distinguishes
- what action it prompts
- which trap it prevents

Listing correct facts is insufficient.

14. AI-Generated Education Section

14.1 Purpose

Section generation allows a specific Diagnosis Education section to be drafted or repaired without regenerating the entire artifact.

Appropriate targets include:

- Summary
- Clinical Pattern
- Key Symptoms
- Key Signs
- Examination Pearls
- Scoring Systems
- Investigations
- Differentials
- Management
- Complications
- Pitfalls
- Recall Prompts
- References

14.2 Section context

Section generation should receive enough surrounding context to preserve coherence.

For example:

- Management generation should consider complications, urgency, and pitfalls.
- Differential generation should consider the clinical pattern and discriminators.
- Recall generation should use approved source sections.
- Scoring-system generation should consider investigation and management use.
- Pitfall generation should consider known mimics and teaching rules.

14.3 Regeneration safeguards

Section regeneration must preserve:

- source education version
- original section
- generated replacement
- reason for regeneration
- cross-section dependencies
- reviewer decision
- applied education revision

A regenerated section should not replace published content without controlled review and application.

15. AI-Generated Recall Prompt

An AI-generated Recall Prompt should:

- derive from governed education
- identify its source section
- test one meaningful concept
- provide a concise answer
- include an explanatory layer
- match the declared difficulty
- avoid unsupported extension beyond source content

AI should not produce recall prompts from unapproved or contradictory content without clearly marking the source as candidate knowledge.

16. AI-Generated Teaching Rule

16.1 Purpose

An AI-Generated Teaching Rule proposes a reusable diagnosis-specific instructional constraint.

It should not be created merely by converting arbitrary prose into rule form.

16.2 Required structure

The draft should propose:

- stable-key candidate
 - title
 - category
 - importance
 - rationale
 - acceptable manifestations
 - required differentials
 - expected evidence
 - difficulty guidance
 - timing constraint
 - applicability
-

16.3 Review focus

Reviewers should determine:

- whether the rule is truly reusable
- whether it reflects clinical and educational intent
- whether it duplicates an existing rule
- whether its importance is justified
- whether it should block generation or publication
- whether it applies to cases, education, graph work, or more than one domain

AI-generated teaching rules must not become active automatically.

17. AI-Generated Editorial Brief

An AI may propose a Diagnosis Editorial Brief where no brief exists or where significant repair is required.

The draft may include:

- editorial summary
- learning goals
- required teaching rules
- required mimics
- required pitfalls
- key investigations
- management anchors
- difficulty guidance
- case-generation guidance
- education guidance
- graph guidance

Because the Editorial Brief defines downstream educational intent, its review should occur before it is used as authoritative generation context.

18. AI-Generated Graph Candidate

18.1 Purpose

AI may extract or propose structured knowledge from cases and education.

All generated graph knowledge must begin as a Graph Candidate.

18.2 Required provenance

The candidate should preserve:

- source artifact
 - source version
 - source path
 - raw text
 - normalised text
 - candidate type
 - target diagnosis where relevant
 - confidence
 - structured payload
-

18.3 Prohibited automatic promotion

AI must not automatically convert an extracted statement into an active Graph Fact.

The candidate must undergo:

- identity resolution
 - duplicate assessment
 - scope assessment
 - clinical review
 - provenance confirmation
 - promotion decision
-

19. AI-Generated Teaching Relationship

AI may propose a relationship between two diagnoses.

The proposal should identify:

- source diagnosis
- target diagnosis
- relationship type
- teaching purpose
- shared presentation
- discriminator
- confusion reason
- learner pitfall
- supporting artifacts
- confidence

Reviewers must assess directionality.

A relationship that is useful from Diagnosis A to Diagnosis B may require different content in the reverse direction.

20. AI-Generated Evidence Relationship

AI may propose how an Evidence Node relates to a diagnosis.

The proposal should identify:

- evidence node
- diagnosis
- relationship type
- strength
- discriminator weight
- reasoning summary

- contradictory diagnoses
- source support

AI should not overuse absolute relationships such as `RULES_OUT`.

All proposed relationships begin as candidates.

21. AI-Generated Reasoning Path

An AI-generated Reasoning Path may propose:

- reasoning goal
- primary differentials
- supporting relationships
- discriminator evidence
- escalation evidence
- contradictory evidence
- required teaching points
- forbidden evidence patterns
- clue distribution
- generation purpose

A generated Reasoning Path must not guide production generation until reviewed and activated.

22. AI-Generated Differential Suggestion

AI may suggest differentials for:

- a clinical case
- Diagnosis Education
- an Editorial Brief
- a Reasoning Path
- a Teaching Relationship

Each suggestion should be evaluated for:

- clinical plausibility
- diagnosis identity
- educational relevance
- role
- discriminator availability
- safety significance
- duplicate risk

A suggested diagnosis should not be linked automatically where identity is ambiguous.

23. Unsupported-Claim Repair

23.1 Purpose

AI may propose a repair where a claim has been flagged as unsupported, unsafe, contradictory, or overstated.

23.2 Repair strategies

A repair may:

- remove the claim
 - narrow its scope
 - qualify certainty
 - add missing context
 - align it with governed evidence
 - replace unsafe management guidance
 - add a supporting reference candidate
 - separate universal from subgroup-specific claims
-

23.3 Review requirement

The repair must be reviewed against:

- original claim
- source issue
- evidence context
- dependent sections
- intended learner meaning
- potential clinical consequence

AI-generated repair does not prove the revised claim is correct.

24. AI Draft Validation

24.1 Purpose

AI Draft Validation evaluates whether a generated draft satisfies defined structural, clinical-reasoning, educational, and governance requirements before or during human review.

Validation should reduce reviewer burden.

It should not replace reviewer judgement.

24.2 Validation dimensions

Validation may assess:

- schema compliance
 - diagnosis alignment
 - teaching-rule compliance
 - reasoning coverage
 - evidence coverage
 - discriminator coverage
 - differential quality
 - hallucination risk
 - unsupported claims
 - contradiction risk
 - genericity
 - management safety
 - difficulty alignment
 - clue progression
 - cross-section consistency
 - publication blockers
-

24.3 Validation outputs

Validation may produce:

- validation status
- trust score
- trust tier
- blockers
- warnings
- strengths
- hallucination-risk signals

- reasoning-coverage analysis
 - evidence-coverage analysis
 - discriminator-coverage analysis
 - unsupported-claim signals
 - recommendations
 - validator version
-

24.4 Trust score

A Trust Score is a structured assessment of draft reliability relative to known context and rules.

It is not a probability that the content is medically correct.

It must not be presented as equivalent to clinical approval.

24.5 Trust tiers

Canonical Trust Tiers include:

High Trust

The draft demonstrates strong alignment with known context and no detected critical blocker.

Human review remains required.

Review Required

The draft is potentially usable but contains issues or uncertainty requiring focused human judgement.

Low Trust

The draft contains substantial weakness, missing support, or reasoning inconsistency.

It should not be applied without major revision.

Blocked

The draft contains a critical failure preventing safe progression.

Examples include:

- unresolved diagnosis identity
- unsafe management guidance

- major contradiction
 - insufficient required context
 - severe hallucination risk
 - failed schema
 - prohibited content pattern
-

24.6 Validation status

Canonical statuses include:

- Passed
- Needs Review
- Failed

A passed status means no configured blocking failure was detected.

It does not mean the draft is approved.

25. Hallucination-Risk Signals

25.1 Purpose

Hallucination-risk signals identify content that may have been invented, misapplied, overstated, or disconnected from the supplied context.

25.2 Common signals

Signals may include:

- finding absent from source context
 - invented scoring system
 - fabricated threshold
 - unsupported drug dose
 - nonexistent relationship
 - false diagnostic exclusivity
 - invented reference
 - incorrect guideline attribution
 - explanation of a clue that does not exist
 - management recommendation outside the diagnosis scope
 - contradictory disease mechanism
-

25.3 Signal interpretation

A hallucination-risk signal is not always proof of error.

It should guide human verification.

High-risk claims with potential learner harm should block application until resolved.

26. AI Draft Review Packet

Every AI Draft review should provide the reviewer with a complete packet containing:

- draft purpose
- source issue
- target artifact
- diagnosis context
- original content where revising
- generated content
- generation context
- applied rules
- reasoning path where relevant
- validation outcome
- blockers
- warnings
- unsupported-claim signals
- differences from current content
- expected effect
- available decisions
- downstream impact

Reviewers should not be required to infer why the draft exists.

27. AI Draft Review

27.1 Purpose

AI Draft Review is the human editorial evaluation of an AI-generated proposal.

It determines whether the proposal should be:

- accepted
- rejected

- revised
 - superseded
 - applied
 - deferred
-

27.2 Review dimensions

The reviewer should assess:

- clinical accuracy
 - diagnosis alignment
 - educational value
 - artifact-standard compliance
 - reasoning quality
 - source adequacy
 - evidence sufficiency
 - specificity
 - internal consistency
 - learner safety
 - downstream effects
 - AI validation findings
-

27.3 Review independence

Reviewers should not assume that an AI draft is reliable because:

- it is well written
- it follows the schema
- validation passed
- confidence is high
- the model is advanced
- similar content exists elsewhere

Fluency must not be confused with authority.

28. AI Draft Lifecycle

The canonical AI Draft lifecycle is:

Draft

↓

Review Required

↓

Pending Review

↓

Accepted, Rejected, or Needs Changes

↓

Applied or Superseded

28.1 Draft

The AI output has been created but may not yet satisfy minimum review requirements.

28.2 Review Required

The draft has been identified as needing human editorial evaluation.

28.3 Pending Review

The draft is queued or prepared for a reviewer.

28.4 Accepted

The reviewer has determined that the proposal is suitable for controlled application.

Acceptance does not itself mean that application has occurred.

28.5 Rejected

The proposal has not been accepted.

The draft and rationale should remain traceable.

28.6 Needs Changes

The proposal may become acceptable after revision.

The requested changes should be actionable.

28.7 Superseded

The proposal has been replaced by another draft, direct editorial change, or newer artifact version.

Supersession should identify the replacement where possible.

29. Acceptance and Application

Acceptance and application are distinct.

Acceptance

The reviewer approves the proposed content or change.

Application

The accepted proposal is incorporated into the governed artifact through a controlled revision.

The separation is necessary because:

- the underlying artifact may have changed since generation
 - dependencies may have become outdated
 - multiple accepted drafts may conflict
 - application may require a new revision
 - revalidation may be required
-

30. Application Preconditions

Before applying an accepted AI Draft, WEOS should verify:

- target artifact still exists
- target version matches the reviewed version
- diagnosis identity remains unchanged
- no conflicting revision has occurred
- required permissions are present
- critical blockers remain resolved
- dependent artifacts are known
- accepted output has not been superseded
- application will produce an auditable revision

Where the target has changed materially, the draft should return for comparison or regeneration.

31. Controlled Application

Application should preserve:

- accepted draft identity
- reviewer
- application actor
- target artifact
- previous version
- resulting version
- applied content
- application time
- validation triggered
- downstream review requirements

Application must not overwrite historical content invisibly.

32. Revalidation After Application

An applied AI Draft may require revalidation where it changes:

- clinical claims
- diagnosis mapping
- case clues
- clue order
- differential relationships
- explanation
- management
- scoring systems
- reasoning path
- learner-facing content

Acceptance of the draft does not waive validation of the resulting artifact.

33. Downstream Impact

AI-assisted changes may affect related artifacts.

Examples include:

- changing a differential may affect Teaching Relationships
- changing a clue may affect Clue Progression Analysis
- changing Management may affect Pitfalls and Recall Prompts
- changing a scoring system may affect Investigations
- changing a Teaching Rule may affect existing cases
- changing a Graph Fact may affect generation context
- changing the Editorial Brief may affect diagnosis readiness

The application process should surface known downstream effects.

34. AI Draft Conflict

A conflict occurs when:

- two drafts propose incompatible changes
- the target artifact has changed
- a newer draft supersedes an older one
- the draft contradicts an approved Teaching Rule
- an accepted draft conflicts with updated evidence
- applying one draft invalidates another

Conflicts should be resolved explicitly.

The system should not apply drafts in creation order without evaluating compatibility.

35. AI Draft Supersession

An AI Draft should be superseded where:

- a newer draft replaces it
- the underlying issue has been resolved manually
- the target artifact has materially changed
- the generation context is outdated
- evidence or guidance has changed
- the draft no longer reflects current editorial intent

Superseded drafts should remain historically traceable but unavailable for ordinary application.

36. AI Feedback and Learning

Reviewer decisions may inform future AI-assistance quality.

Useful feedback includes:

- accepted without change
- accepted with modification
- rejected for clinical inaccuracy
- rejected for genericity
- rejected for duplication
- rejected for unsafe management
- rejected for reasoning mismatch
- rejected for poor difficulty
- superseded by newer context

This feedback may improve prompts, validators, context construction, and recommendation policies.

Reviewer decisions should not be used to create hidden autonomous policy changes without governance.

37. Human Authority

Human editors retain authority over:

- clinical correctness
- educational intent
- artifact approval
- relationship activation
- diagnosis mapping in ambiguous cases
- acceptance of evidence
- management safety
- publication
- retirement
- institutional trust

AI may recommend.

Human reviewers decide.

38. Role and Permission Governance

AI Draft actions should be restricted according to editorial role.

Examples include:

- contributors may request generation
- editors may review ordinary drafts
- specialty reviewers may approve high-risk clinical content
- senior editors may approve publication-affecting repairs
- administrators may apply controlled migrations
- only authorised roles may publish resulting artifacts

Permission should reflect the consequence of the decision rather than the apparent simplicity of the draft.

39. High-Risk AI Content

Some AI outputs require heightened review.

High-risk content includes:

- drug doses
- emergency management
- procedural guidance
- contraindications
- paediatric treatment
- pregnancy-related management
- critical-care thresholds
- scoring-system cut-offs
- antimicrobial recommendations
- claims that a finding rules out a diagnosis
- guideline-dependent recommendations

High-risk content may require:

- explicit reference support
 - specialty review
 - dual review
 - stronger validation
 - publication blocking until resolved
-

40. AI and Evidence

AI is not an evidence source.

An AI-generated claim may be supported by:

- governed evidence relationships
- approved references
- recognised clinical guidelines
- verified primary or secondary evidence

The statement “generated by AI” provides provenance but no clinical authority.

AI should not fabricate, approximate, or imply references that were not verified.

41. AI and Publication

No AI-generated artifact should become learner-facing solely because:

- it passed schema validation
- it received a high trust score
- it was generated from approved context
- it resembles existing content
- no automated blocker was detected

Publication requires:

- artifact-level human approval
 - resolved blockers
 - appropriate evidence review
 - controlled application
 - review of the resulting artifact version
 - explicit publication decision
-

42. AI Operational Failure

AI generation may fail because of:

- provider unavailability
- timeout
- malformed output
- schema failure
- insufficient context
- safety restriction
- conflicting generation request
- output truncation
- validation failure

Operational failure should not alter the target artifact.

The system should preserve:

- attempted action
- failure type
- time
- diagnosis context
- retry eligibility
- absence of application

A failed generation must not leave an artifact in an ambiguous partially updated state.

43. AI Draft Maintenance

AI Drafts should be reviewed for staleness where:

- the target artifact changes
- teaching rules change
- the Editorial Brief changes
- a Reasoning Path is deprecated
- graph facts change
- references are updated
- diagnosis identity is merged
- generation policy changes
- the AI model or schema changes materially

Old drafts should not remain indefinitely applicable without renewed context validation.

44. Quality Standard for AI-Assisted Editorial Work

High-quality AI-assisted work should be:

Context-grounded

The draft visibly reflects governed diagnosis and educational context.

Purpose-specific

The output solves an identified editorial need.

Structurally valid

The draft follows the target artifact schema.

Clinically defensible

Claims and relationships are suitable for expert review.

Educationally useful

The draft supports a defined learner or editorial outcome.

Transparent

The source issue, context, output, and decisions are preserved.

Reviewable

The reviewer can compare the proposal with current governed content.

Reversible

The draft can be rejected, superseded, or removed without erasing history.

Controlled in application

Accepted content enters the system through an explicit versioned change.

Human-governed

No AI output independently receives institutional authority.

45. Normative Statements

- Every material AI contribution **MUST** preserve AI provenance.
- Every AI generation request **MUST** identify an editorial purpose.
- AI generation **MUST** use a governed diagnosis identity where available.
- Generation Context **SHOULD** include applicable learning goals, teaching rules, mimics, discriminators, pitfalls, and difficulty guidance.
- AI **MUST NOT** infer institutional approval from the presence of existing content.
- AI-generated content **MUST** begin as candidate or draft knowledge.
- AI-generated cases **MUST NOT** enter playable or published states directly.

- AI-generated Diagnosis Education **MUST NOT** enter approved or published states directly.
 - AI-generated Graph Candidates **MUST NOT** be promoted automatically into active Graph Facts.
 - AI-generated Teaching Relationships, Evidence Relationships, and Reasoning Paths **MUST NOT** become active without human review.
 - AI-generated differential mappings **MUST NOT** silently resolve clinically ambiguous diagnosis identities.
 - AI-generated clinical claims **MUST** remain distinguishable from verified evidence.
 - AI **MUST NOT** be represented as a clinical evidence source.
 - High-risk management content **MUST** receive heightened clinical review.
 - Validation scores and trust tiers **MUST NOT** be interpreted as approval.
 - A passed AI validation **MUST NOT** replace human editorial review.
 - Hallucination-risk signals **MUST** remain visible until reviewed or resolved.
 - Accepted AI Drafts **MUST NOT** be treated as applied automatically.
 - Application **MUST** produce an auditable artifact change.
 - Materially changed target artifacts **MUST** invalidate or trigger reassessment of pending drafts.
 - Applying an AI Draft **SHOULD** trigger appropriate revalidation.
 - Rejected and superseded AI Drafts **MUST** remain historically traceable.
 - AI Draft conflicts **MUST** be resolved explicitly.
 - Publication **MUST** apply to the governed artifact resulting from AI-assisted work, not to the AI Draft itself.
 - Human editors **MUST** retain final authority over approval and publication.
 - AI operational failure **MUST NOT** partially modify governed artifacts.
 - AI Drafts **SHOULD** expire or require reassessment when their generation context becomes stale.
-

Closing Principle

Artificial intelligence can expand Wardle's editorial capacity, but it cannot create institutional trust by itself.

Trust arises when AI contribution is bounded by governed context, expressed as a transparent candidate draft, tested through validation, evaluated by qualified human reviewers, and applied through a controlled and auditable revision.

Within WEOS, AI does not replace editorial judgement.

It makes more editorial work possible while leaving responsibility, authority, and accountability with people.

Wardle Editorial Canon

Standards for Governed Editorial Artifacts

Part VII - Governance Record Standards

WEOS-CANON-007 | Version 0.1 | Draft for approval

1. Purpose

Governance Record Standards define how WEOS preserves the assessments, reviews, validations, decisions, publication actions, exceptions, and historical events through which clinical knowledge becomes institutionally trusted.

Governance records are not secondary administrative metadata.

They are first-class editorial artifacts.

A governed artifact cannot be understood only from its current content and current status. Editors must also be able to determine:

- what was evaluated;
- which artifact version was considered;
- what evidence and context were available;
- which concerns were identified;
- who exercised authority;
- which decision was made;
- why that decision was defensible;
- what changed as a result;
- which obligations remain;
- what learners were permitted to see.

The purpose of this Part is to ensure that every meaningful editorial conclusion remains explicit, attributable, version-specific, explainable, and historically recoverable.

2. Governance Record Philosophy

WEOS governs living clinical knowledge.

Living knowledge changes through repeated interpretation and decision.

If the system preserves only the latest artifact state, it loses the reasoning that created institutional trust.

Governance records therefore preserve the institutional memory of editorial work.

They allow WEOS to reconstruct:

- how candidate knowledge entered the system;
- how clinical and educational confidence developed;
- why an artifact progressed or failed to progress;
- how disagreements were resolved;
- which exceptions were granted;
- why publication occurred;
- why knowledge was later withdrawn, superseded, or retired.

Governance records should not exist merely to demonstrate that an action occurred.

They should preserve enough context to understand whether the action was appropriate.

3. Governance Artifact Family

The canonical Governance Artifact family includes:

- Editorial Assessment;
- Editorial Review;
- Editorial Decision;
- Validation Result;
- Readiness Assessment;
- Publication Assessment;
- Publication Decision;
- Controlled Application Record;
- Withdrawal Record;
- Supersession Record;
- Retirement Record;
- Governance Exception;
- Emergency Correction Record;
- Disagreement and Adjudication Record;
- Conflict-of-Interest Declaration;
- Governance History.

These artifacts serve related but distinct purposes.

Implementations must not collapse them into a single status field or generic activity log where doing so erases editorial meaning.

4. Foundational Distinctions

4.1 Assessment and Decision

An assessment interprets the quality, state, risk, or readiness of an artifact.

A decision changes or confirms its governed standing.

An assessment may conclude that a case is clinically coherent.

A decision may approve the identified case revision.

The assessment supports the decision but is not the decision itself.

4.2 Review and Decision

A review is the governed process through which one or more assessments are completed and a required editorial question is considered.

A decision is the explicit outcome of that review.

Closing a review without a decision should be distinguishable from completing a review with approval, rejection, revision request, deferral, or escalation.

4.3 Validation and Approval

Validation evaluates compliance with defined rules, schemas, completeness expectations, or structured quality requirements.

Approval establishes human editorial confidence in an identifiable artifact version.

A passed validation must not be represented as approval.

A failed validation may block approval where policy identifies the failure as material.

4.4 Readiness and Status

Readiness is a computed or reasoned interpretation of whether an artifact may proceed to a defined next operation.

Status records the current governed or operational state.

Readiness must identify the operation being assessed, such as generation, review, application, playability, publication, maintenance, or retirement.

4.5 Approval and Publication

Approval indicates that an artifact version has satisfied its required editorial review.

Publication indicates that an approved artifact version has been authorised for learner-facing or institutional use.

Every publication must be supported by an explicit Publication Decision.

4.6 Audit Event and Governance Record

An audit event records that an action occurred.

A governance record preserves the editorial meaning, authority, rationale, context, and effect of that action.

A button click may produce an audit event.

The decision created through that action is a Governance Record.

4.7 Correction and Historical Revision

Correcting an inaccurate governance record should not silently replace the historical record.

The correction should preserve:

- the original record;
 - the identified error;
 - the correcting authority;
 - the corrected representation;
 - the reason for correction.
-

5. Universal Governance Record Standard

Every material governance record should preserve a common canonical structure.

5.1 Stable Identity

Every Governance Record must possess a stable identity independent of its display location, workflow screen, or storage implementation.

5.2 Record Type

The record type must identify the editorial meaning of the record.

Examples include Clinical Assessment, Case Review, Request Revision Decision, Publication Decision, or Emergency Withdrawal.

5.3 Artifact Context

Every artifact-scoped record must identify:

- artifact type;
- artifact identity;
- diagnosis context where applicable;
- artifact version or revision;
- relevant source and target roles where more than one artifact is involved.

5.4 Governance Question

The record should identify the question being answered.

Examples include:

- Is this case clinically correct?
- Does this Teaching Relationship possess sufficient discriminatory value?
- May this Diagnosis Education version be published?
- Does this AI Draft remain applicable to the current target revision?
- Should this published artifact be withdrawn?

5.5 Actor and Authority

The record must preserve:

- responsible person or governed system actor;
- editorial role;
- authority exercised;
- institutional or specialty scope where relevant;
- delegation or acting authority where applicable.

Technical identity alone is insufficient.

5.6 Time

The record should preserve:

- creation time;
- decision time where different;
- effective time where relevant;
- closure time;
- expiry or review time where applicable.

5.7 Source Context

The record should preserve the information upon which the conclusion depended.

This may include:

- artifact version;
- review packet;
- previous findings;
- evidence sources;
- applicable policies;
- learning goals;
- teaching rules;
- reasoning paths;
- validation results;
- related governance decisions.

5.8 Findings

Findings should state what the reviewer, validator, or authority observed.

Findings should distinguish:

- confirmed issue;
- possible concern;
- informational observation;
- unresolved uncertainty;
- publication blocker;
- non-blocking warning.

5.9 Rationale

Every material conclusion must preserve rationale proportionate to its consequence.

Rationale should explain why the conclusion follows from the findings, policies, evidence, and artifact purpose.

5.10 Outcome

The record should identify the governed outcome.

Examples include:

- passed;
- failed;
- approved;
- rejected;
- revision required;
- evidence required;
- deferred;
- escalated;
- published;
- withdrawn;
- superseded;
- exception granted.

5.11 Effect

The record should identify what the outcome changed or permitted.

Possible effects include:

- lifecycle transition;
- blocker creation or resolution;
- publication eligibility;
- artifact application;
- operational activation;
- withdrawal from learner exposure;
- maintenance obligation;
- reassignment;
- escalation.

5.12 Remaining Obligations

Where work remains, the record should identify:

- required action;
- responsible role;
- affected artifact;
- expected next state;
- deadline or review trigger where relevant.

5.13 Linkage

Governance Records should be linkable to:

- preceding records;
- resulting records;
- superseded decisions;
- related artifacts;
- resulting revisions;
- publication events;
- maintenance events.

6. Record Scope and Materiality

Not every interaction requires a full Governance Record.

The system should distinguish material governance actions from routine operational activity.

A material action is one that affects:

- clinical meaning;
- educational intent;
- institutional confidence;
- artifact lifecycle;
- evidence sufficiency;
- publication eligibility;
- learner exposure;
- governance authority;
- historical interpretation.

Examples of material actions include:

- approving a case revision;
- rejecting a Teaching Rule;
- requesting evidence for a management claim;
- accepting and applying an AI Draft;
- publishing Diagnosis Education;
- withdrawing a case;
- granting a governance exception;
- adjudicating reviewer disagreement.

Examples of ordinary operational events may include:

- opening a review packet;
- filtering a queue;
- viewing an artifact;
- changing a non-governed display preference.

Operational events may remain auditable without being represented as editorial decisions.

7. Version-Specific Governance

Every approval, rejection, publication, validation, and material assessment must apply to an identifiable artifact version.

Governance must not apply ambiguously to mutable content.

Where an artifact changes materially after review:

- the earlier governance record remains valid for the earlier version;
- the new version does not inherit approval automatically;
- affected assessments should be repeated or explicitly carried forward;
- the system should identify which prior conclusions remain applicable;
- renewed approval should be required where policy determines the change is material.

A governance interface should never present an earlier approval as though it applies to unreviewed current content.

8. Editorial Assessment

8.1 Purpose

An Editorial Assessment is a structured interpretation of one or more dimensions of an artifact, diagnosis, curriculum area, or governance question.

Assessments support decision-making without independently changing the artifact state.

8.2 Assessment Families

Canonical assessment families include:

- Clinical Assessment;
- Educational Assessment;
- Reasoning Assessment;
- Evidence Assessment;
- Knowledge Relationship Assessment;
- Identity and Terminology Assessment;
- Curriculum Assessment;
- Safety and Risk Assessment;
- Readiness Assessment;
- Publication Assessment;
- Maintenance Assessment;
- Retirement Assessment.

8.3 Assessment Scope

Every assessment should identify:

- dimension evaluated;
- applicable standard;
- artifact or portfolio scope;
- evidence and context considered;
- conclusion;
- confidence or uncertainty where useful;
- blockers and warnings;
- recommendation;
- assessor identity.

8.4 Assessment Conclusion

An assessment conclusion may be expressed as:

- satisfactory;
- satisfactory with warning;
- deficient;
- indeterminate;
- not applicable;
- additional expertise required;
- additional evidence required.

Assessment vocabulary should preserve meaning rather than imitate a universal score.

8.5 Assessment Explainability

Every assessment should answer:

- What was evaluated?
- Which standard applied?
- What evidence influenced the conclusion?
- Which findings were material?
- What remains uncertain?
- What action is recommended?

8.6 Assessment Scores

Scores may assist prioritisation or structured review.

A score must not conceal the underlying findings.

A high score must not automatically confer approval.

A low score should identify the deficiencies producing the result.

9. Clinical Assessment Standard

A Clinical Assessment evaluates whether the artifact is clinically defensible within its intended scope.

It may consider:

- factual correctness;
- presentation plausibility;
- chronology;
- diagnostic interpretation;
- investigation meaning;
- management safety;
- severity and escalation;
- population applicability;
- contraindications;
- important exceptions;
- risk of harmful misunderstanding.

The assessment should identify whether concerns are:

- inaccurate;
- incomplete;
- misleading;
- unsafe;
- excessively certain;
- context dependent;
- outside reviewer competence.

High-risk clinical claims should receive proportionately detailed rationale.

10. Educational Assessment Standard

An Educational Assessment evaluates whether the artifact fulfils its intended learning purpose.

It may consider:

- alignment with Learning Goals;
- teaching-rule compliance;
- diagnostic reasoning quality;
- clue progression;
- differential discrimination;
- learner level;
- cognitive load;
- misconception prevention;
- educational usefulness;
- curriculum contribution;
- unnecessary duplication.

A clinically correct artifact may fail educational assessment where it is generic, confusing, poorly sequenced, or misaligned with its intended learner outcome.

11. Evidence Assessment Standard

An Evidence Assessment evaluates whether available sources appropriately support the relevant clinical or educational claim.

It should consider:

- source identity;
- source type;
- authority;
- recency;
- population;
- jurisdiction;
- methodological limitations;
- claim alignment;
- conflicting evidence;
- strength of support;
- expiry or review requirement.

The assessment should distinguish:

- source exists;
- source is credible;
- source is applicable;
- source supports the exact claim;
- editorial interpretation is appropriate.

12. Knowledge Relationship Assessment Standard

A Knowledge Relationship Assessment evaluates whether a proposed or active relationship is clinically meaningful, correctly scoped, educationally useful, and sufficiently supported.

It may assess:

- source diagnosis;
- target diagnosis or evidence concept;
- directionality;
- relationship type;
- clinical qualifiers;
- discriminator value;
- risk of overgeneralisation;
- duplication;
- contradiction;
- provenance;
- generalisability beyond the source artifact.

A relationship should not be approved solely because two concepts appear together in text.

13. Curriculum Assessment Standard

A Curriculum Assessment evaluates collections of governed knowledge rather than one artifact.

It may consider:

- diagnosis coverage;
- Learning Goal coverage;
- specialty balance;
- difficulty distribution;
- case diversity;
- population representation;
- repeated teaching;
- missing mimics;
- longitudinal progression;
- maintenance risk.

Curriculum assessments should preserve the portfolio scope and time at which the conclusion was reached.

14. Readiness Assessment

14.1 Purpose

A Readiness Assessment determines whether an artifact or diagnosis satisfies the requirements for a defined next operation.

14.2 Readiness Types

Examples include:

- Generation Readiness;
- Editorial Review Readiness;
- Controlled Application Readiness;
- Playability Readiness;
- Publication Readiness;
- Maintenance Readiness;
- Withdrawal Readiness;
- Retirement Readiness.

14.3 Required Elements

Every Readiness Assessment should preserve:

- operation assessed;

- applicable policy;
- satisfied dependencies;
- blockers;
- warnings;
- missing information;
- resulting conclusion;
- assessment time;
- recommended next action.

14.4 Readiness Outcomes

Canonical outcomes include:

- ready;
- ready with non-blocking warnings;
- not ready;
- indeterminate;
- assessment not applicable.

Readiness must never be represented without explanation.

15. Editorial Review

15.1 Purpose

An Editorial Review is the governed process through which an artifact or editorial question receives the assessments required for an explicit decision.

15.2 Review Identity

Every review should possess a stable identity separate from the artifact it evaluates.

This allows reviews to be reopened, reassigned, compared, and historically preserved.

15.3 Review Scope

A review should identify:

- artifact and version;
- review type;
- governance question;
- required assessment dimensions;
- assigned reviewer or panel;
- applicable standards;
- review status;
- required decision authority.

15.4 Review Packet

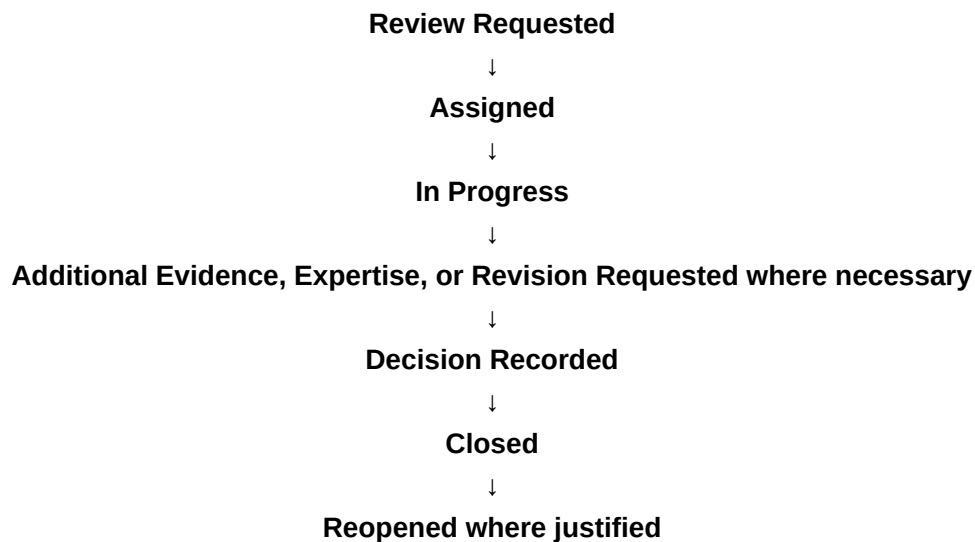
The review should provide sufficient context for safe judgement.

A Review Packet may include:

- current artifact;
- previous version;
- revision comparison;
- diagnosis context;
- Learning Goals;
- Teaching Rules;
- relevant Reasoning Paths;
- evidence;
- validation results;
- previous review findings;
- AI provenance;
- dependency status;
- learner-impact information.

15.5 Review Lifecycle

The canonical review lifecycle is:



15.6 Review Status

Review status describes the progress of the review process.

It must remain separate from the lifecycle status of the artifact being reviewed.

15.7 Review Findings

Findings should be specific, actionable, and linked to the affected content or requirement.

A finding should avoid vague conclusions such as “improve quality” without identifying what is deficient and why it matters.

15.8 Review Completion

A review is complete when:

- required assessments are present;
 - material findings are recorded;
 - required authority has made a decision or explicitly deferred it;
 - resulting obligations are assigned;
 - the review outcome is preserved.
-

16. Multi-Reviewer and Panel Review

Some artifacts require more than one reviewer or more than one form of authority.

A multi-reviewer record should preserve:

- each reviewer;
- role and scope;
- individual findings;
- areas of agreement;
- areas of disagreement;
- consolidated conclusion;
- final decision authority.

The final record must not hide clinically material disagreement merely because a majority conclusion was reached.

Where policy requires dual review, two approvals should remain individually attributable.

17. Editorial Decision

17.1 Purpose

An Editorial Decision is an explicit, accountable outcome that changes, confirms, or limits the governed standing of an artifact or editorial question.

17.2 Canonical Decision Types

Canonical decision types include:

- Approve;
- Reject;
- Request Revision;
- Request Evidence;
- Request Specialty Review;
- Accept Candidate;
- Accept AI Draft;
- Decline Candidate;
- Merge;
- Resolve Mapping;

- Defer;
- Escalate;
- Apply;
- Activate;
- Publish;
- Schedule;
- Restrict;
- Withdraw;
- Supersede;
- Deprecate;
- Archive;
- Retire;
- Restore;
- Grant Exception;
- Deny Exception.

Artifact-specific vocabulary may be retained where it expresses a distinct editorial meaning.

17.3 Decision Preconditions

A decision should identify the conditions that permitted the authority to act.

Examples include:

- required validation completed;
- appropriate reviewer assigned;
- evidence available;
- conflict declared and managed;
- dependencies resolved;
- current version confirmed;
- required dual review completed.

17.4 Decision Rationale

The rationale should explain why the selected outcome is preferable to the available alternatives.

Rationale should be proportionate to risk and consequence.

17.5 Decision Effect

The record must specify the transition, permission, or restriction created by the decision.

A decision should not rely on an implementation to infer its meaning from the label alone.

17.6 Conditional Decisions

A decision may be conditional where policy permits.

Conditions should be explicit, measurable, assigned, and time-limited where appropriate.

A conditional approval must not be presented as unrestricted approval.

17.7 Decision Finality

Decisions may be:

- provisional;
- final within ordinary review;
- subject to senior approval;
- subject to publication approval;
- reopened by defined trigger;
- superseded by later decision.

Finality should be explicit.

18. Request Revision Decision

A Request Revision Decision should identify:

- deficient requirement;
- affected location or artifact component;
- clinical or educational consequence;
- required change;
- whether regeneration is permitted;
- whether evidence is required;
- whether the artifact remains eligible for other work;
- required re-review.

Revision requests should be actionable rather than punitive.

19. Rejection Decision

Rejection indicates that candidate or draft knowledge has not been accepted into governed knowledge.

The record should identify:

- reason for rejection;
- whether the defect is remediable;
- whether a new proposal may be created;
- whether the artifact should be retained, archived, or merged;
- affected dependencies;
- whether a source issue remains unresolved.

Rejection should not erase provenance or review history.

20. Approval Decision

Approval indicates that an identifiable artifact version has satisfied its required editorial review.

Approval should preserve:

- approved artifact version;
- required assessments;
- decision-maker and authority;
- conditions or limitations;
- unresolved non-blocking warnings;
- effective time;
- required future review where relevant.

Approval does not automatically imply:

- publication;
 - playability;
 - generation permission;
 - activation;
 - scheduling;
 - evidence permanence.
-

21. Validation Result

21.1 Purpose

A Validation Result records the outcome of applying one or more defined validation rules to an artifact version or proposed action.

21.2 Validation Sources

Validation may be:

- deterministic;
- schema-based;
- policy-based;
- referential-integrity based;
- rule-assisted;
- AI-assisted;
- manually confirmed.

The validation source should remain visible.

21.3 Validation Record

A Validation Result should preserve:

- validation run identity;
- artifact and version;

- rule set and version;
- rules evaluated;
- pass, fail, warning, or not-applicable result;
- affected fields or relationships;
- severity;
- remediation guidance;
- time;
- validator identity or system version.

21.4 Severity

Canonical severities may include:

- informational;
- warning;
- blocking;
- critical.

Severity should reflect the transition or operation affected.

21.5 Validation Staleness

A Validation Result may become stale when:

- the artifact changes;
- the rule set changes;
- a dependency changes;
- diagnosis identity changes;
- evidence changes;
- applicable policy changes.

Stale validation must not be treated as current proof of compliance.

21.6 Manual Override

Where policy permits override of a validation result, the override must produce a Governance Exception or explicit decision record.

22. Publication Assessment

A Publication Assessment interprets whether an approved artifact and its dependencies are suitable for learner-facing or institutional release.

It should consider:

- approved version;
- clinical safety;
- educational suitability;
- evidence sufficiency;
- operational readiness;

- diagnosis-level dependencies;
- unresolved warnings;
- publication channel;
- target audience;
- maintenance ownership;
- rollback or withdrawal capability.

Publication Assessment supports but does not replace the Publication Decision.

23. Publication Decision

23.1 Purpose

A Publication Decision authorises or declines learner-facing or institutional exposure of an approved artifact version.

23.2 Required Elements

Every Publication Decision should preserve:

- artifact and approved version;
- publication authority;
- Publication Assessment;
- target channel or track;
- intended audience;
- effective or scheduled time;
- unresolved warnings;
- restrictions;
- maintenance owner;
- withdrawal or supersession relationship where applicable.

23.3 Outcomes

Canonical outcomes include:

- publish now;
- schedule publication;
- publish with restriction;
- defer publication;
- deny publication;
- withdraw;
- supersede.

23.4 Technical Release

Technical deployment or scheduling should produce an operational event linked to the Publication Decision.

Technical release must not create publication authority where no decision exists.

24. Controlled Application Record

24.1 Purpose

A Controlled Application Record preserves the governed transformation through which an accepted proposal is applied to a target artifact.

This is particularly important for AI Drafts, generated repairs, imported changes, and structured editorial migrations.

24.2 Required Elements

The record should preserve:

- accepted proposal or source artifact;
- target artifact and pre-application version;
- authorised action;
- applying actor;
- application time;
- resulting artifact version;
- exact changes;
- conflicts or partial application;
- resulting validation status;
- required re-review.

24.3 Acceptance and Application

Accepting a proposal approves the proposal for use.

Application creates or modifies the governed artifact.

The two actions must remain distinguishable.

24.4 Application Failure

A failed or partial application should preserve:

- attempted change;
- failure type;
- unchanged or partially changed target state;
- rollback outcome;
- required remediation.

No target artifact should remain in an ambiguous state after failed application.

25. AI Governance Records

Every material AI contribution should be traceable through linked records that may include:

- AI Draft Audit;
- automated validation result;
- human review;
- acceptance, rejection, or revision decision;
- Controlled Application Record;
- resulting artifact review;
- Publication Decision.

AI provenance should identify:

- purpose;
- generation context;
- source artifacts and versions;
- model or generator version where available;
- output;
- validation;
- reviewer decision;
- application status.

A high AI trust score, passed schema, or generated state must not be represented as human approval.

26. Withdrawal Record

A Withdrawal Record preserves the decision to remove published knowledge from active learner or institutional exposure.

It should identify:

- artifact and published version;
- reason;
- clinical or educational risk;
- urgency;
- authority;
- effective time;
- affected publication channels;
- replacement where available;
- learner notification requirement;
- downstream dependencies;
- restoration conditions;
- follow-up review.

Withdrawal should not erase publication history.

27. Supersession Record

A Supersession Record identifies that a governed artifact or version has been replaced by a preferred successor.

It should preserve:

- superseded artifact or version;
- successor;
- rationale;
- decision authority;
- effective time;
- treatment of existing dependencies;
- whether historical use remains valid;
- maintenance responsibility.

Supersession differs from deletion and withdrawal.

28. Retirement and Deprecation Records

A Retirement or Deprecation Record should identify:

- artifact or diagnosis;
- reason;
- continued permitted uses;
- prohibited new uses;
- replacement where applicable;
- affected mappings and dependencies;
- historical preservation;
- authority;
- effective time.

Deprecated knowledge may remain accessible for history or mapping while no longer being preferred.

29. Governance Exception

29.1 Purpose

A Governance Exception records a deliberate, authorised, limited departure from an ordinary governance requirement.

29.2 Required Elements

An exception should preserve:

- requirement affected;

- reason;
- scope;
- risk;
- approving authority;
- duration;
- compensating controls;
- prohibited extensions;
- review or expiry condition;
- resulting decision.

29.3 Exception Limits

An exception must not silently alter the underlying standard.

Repeated exceptions should trigger policy review.

A technical override without a Governance Exception is not legitimate editorial governance.

30. Emergency Correction Record

Emergency action may be required where learner-facing knowledge creates a credible and significant risk.

The record should preserve:

- identified risk;
- affected artifact and version;
- emergency action;
- authority;
- time;
- immediate containment;
- learner-facing consequence;
- temporary nature where applicable;
- retrospective review requirement;
- permanent corrective action.

Emergency action may shorten the ordinary pathway.

It must not remove accountability.

31. Disagreement Record

A Disagreement Record preserves material differences between reviewers or authorities.

It should identify:

- issue;

- positions;
- evidence and standards cited;
- clinical consequences;
- educational consequences;
- areas of agreement;
- unresolved questions;
- escalation status.

Disagreement should not be reduced to anonymous comments where it affects institutional trust.

32. Adjudication Record

An Adjudication Record preserves the governed resolution of an escalated disagreement.

It should identify:

- disputed question;
- preceding reviews;
- adjudicator and authority;
- independence or conflict management;
- evidence considered;
- final decision;
- rationale;
- minority or alternative position where material;
- conditions;
- future review trigger.

Adjudication resolves the institutional question without erasing the history of disagreement.

33. Conflict-of-Interest Declaration

Where governance policy requires disclosure, the record should preserve:

- declarant;
- relationship or interest;
- affected artifact or decision;
- date;
- assessed significance;
- management action;
- permitted participation;
- excluded authority;
- reviewing authority.

A declaration should be linked to the affected review or decision.

34. Governance History

34.1 Purpose

Governance History is the ordered, interpretable record of how an artifact or diagnosis evolved through governance.

It is not merely an unordered list of events.

34.2 History Contents

Governance History may include:

- creation and provenance;
- revisions;
- validation runs;
- assessments;
- reviews;
- decisions;
- applications;
- publication;
- maintenance;
- disagreement;
- exceptions;
- withdrawal;
- supersession;
- retirement;
- restoration.

34.3 Historical Reconstruction

The history should make it possible to determine:

- what content existed at any governed point;
- which version was approved;
- which version was published;
- which decision was effective at a given time;
- why the current state exists;
- which obligations remain.

34.4 History Views

Implementations may provide summary and detailed history views.

Summary must not replace the underlying records.

35. Record Linkage and Dependency Impact

Governance Records should preserve relationships to affected dependencies.

A decision may affect:

- diagnosis readiness;
- cases;
- Diagnosis Education;
- Teaching Relationships;
- Reasoning Paths;
- generation context;
- publication eligibility;
- scheduled releases;
- learner-facing content;
- maintenance tasks.

Where a change invalidates downstream confidence, the record should identify which artifacts require reassessment.

36. Reopening Governance

A closed review or decision may require reopening when:

- new evidence emerges;
- an artifact version was misidentified;
- a conflict of interest is discovered;
- a material dependency changes;
- learner feedback reveals ambiguity;
- an implementation defect affected the reviewed content;
- the decision was based on inaccurate information;
- policy requires periodic revalidation.

Reopening should create a new governance episode linked to the earlier record.

The earlier decision should not be silently edited away.

37. Record Amendment and Correction

Governance Records may contain clerical or factual recording errors.

Corrections should preserve:

- original record;
- corrected field or statement;
- correcting actor;

- reason;
- correction time;
- whether the substantive decision changed.

Where the decision itself was wrong, a new decision should supersede it rather than merely editing the old record.

38. Immutability, Deletion, and Retention

Material governance records should be immutable in meaning after finalisation.

Implementations may permit append-only clarification or correction while preserving original content.

Deletion should be exceptional.

Records should normally be retained where they support:

- institutional accountability;
- learner-exposure history;
- audit;
- clinical safety;
- supersession;
- dispute resolution;
- accreditation;
- model and workflow evaluation.

Where legal or privacy requirements require removal, the system should preserve the lawful minimum necessary governance trace where permitted.

39. Access and Confidentiality

Governance transparency does not require unrestricted access to every record.

Access may be limited because of:

- reviewer privacy;
- confidential institutional discussion;
- security;
- legal restrictions;
- sensitive operational information.

Access controls must not permit hidden alteration of the governance record.

Learner-facing systems may expose a simplified provenance or review statement while internal systems preserve the complete record.

40. Governance Record Quality

A high-quality Governance Record should be:

Version-specific. It clearly identifies the artifact state considered.

Purpose-specific. It answers a defined editorial question.

Attributable. The responsible actor and authority are identifiable.

Evidence-informed. Relevant support and limitations are visible.

Explainable. Findings and rationale make the outcome understandable.

Actionable. Required follow-up is explicit.

Linked. Related artifacts, decisions, and resulting changes are traceable.

Durable. History remains recoverable after people and systems change.

Proportionate. Detail reflects clinical and educational consequence.

Non-destructive. Corrections and supersession preserve prior governance.

41. Governance Record Validation

Governance Records themselves should be validated.

Possible record validations include:

- missing artifact version;
- missing decision-maker;
- role lacks required authority;
- missing rationale;
- decision lacks required assessment;
- publication decision lacks approved version;
- application lacks accepted source proposal;
- withdrawal lacks affected publication context;
- exception lacks expiry or compensating control;
- unresolved conflict of interest;
- invalid lifecycle transition.

A technically valid record may still be editorially insufficient.

42. Governance Analytics

Governance Records may support institutional quality analysis.

Examples include:

- decision turnaround time;
- revision frequency;
- disagreement rate;
- post-publication correction rate;
- percentage of decisions with complete rationale;
- validation override rate;
- emergency withdrawal rate;
- AI draft acceptance and modification rate;
- overdue revalidation;
- authority and separation-of-duty compliance.

Analytics should improve governance without reducing decisions to performance targets that encourage unsafe approval speed.

43. Relationship to Implementation

Implementations may represent Governance Records through separate entities, event streams, review objects, decision records, or linked histories.

The implementation must preserve the canonical distinctions and information defined in this Part.

Implementation convenience must not erase:

- version specificity;
- authority;
- rationale;
- decision effect;
- historical linkage;
- separation between validation, assessment, review, decision, application, and publication.

Where current implementation lacks a canonical record, the divergence should be documented and assigned for evolution.

44. Normative Statements

Every material governance action **MUST** produce an identifiable governance record.

Every approval **MUST** apply to an identifiable artifact version.

Every Editorial Decision **MUST** preserve responsible authority and rationale.

Every Publication Decision **MUST** identify an approved artifact version.

Publication **MUST NOT** be inferred from technical release, scheduling, or status mutation alone.

Validation **MUST NOT** be represented as human editorial approval.

Readiness MUST identify the operation assessed and explain blockers and warnings.

Material AI contribution MUST preserve provenance, review, decision, and application status.

Acceptance of an AI Draft MUST remain distinct from controlled application.

Controlled application MUST identify the target pre-application version and resulting version.

Rejected, superseded, withdrawn, and retired artifacts SHOULD remain historically traceable.

Governance exceptions MUST identify the modified requirement, authority, scope, risk, and expiry or review condition.

Emergency action MUST receive retrospective review.

Material reviewer disagreement MUST remain visible through resolution.

A governance record MUST NOT be silently altered in a manner that changes its historical meaning.

Corrections MUST preserve the original record and correcting authority.

Governance History MUST make it possible to reconstruct what was approved and what was exposed to learners.

Technical access MUST NOT be treated as editorial authority.

Governance Record validation SHOULD detect missing authority, version context, rationale, required assessments, and invalid transitions.

45. Relationship to the Wider Canon

Part I establishes the universal artifact and governance principles.

Part II defines the diagnosis foundation to which governance records attach.

Part III defines the review, validation, approval, playability, and publication requirements of Clinical Cases.

Part IV defines the governance requirements of Diagnosis Education.

Part V defines the governance of knowledge relationships, evidence, and reasoning artifacts.

Part VI defines the candidate, review, acceptance, application, and publication controls applicable to AI Drafts.

This Part defines the records through which all of those decisions become explicit, auditable, and institutionally durable.

Closing Principle

Governed knowledge is not defined only by what the current artifact says.

It is defined by the transparent history through which people evaluated it, exercised authority, accepted responsibility, and authorised its use.

Within WEOS:

Assessments interpret knowledge.

Reviews organise judgement.

Decisions establish governed outcomes.

Publication expresses institutional trust.

Governance Records preserve why that trust was justified.

Wardle Editorial Operating System (WEOS)

Documentation Completion and Control Plan

WEOS-GOV-PLAN-001 | Version 0.2 | Draft for approval

1. Purpose

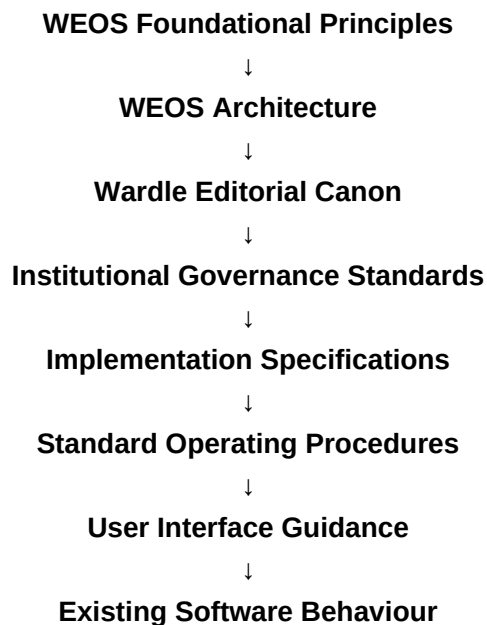
The WEOS Documentation Completion and Control Plan defines how the Wardle Editorial Operating System documentation will be completed, reviewed, approved, maintained, and connected to implementation.

The plan will:

- complete missing Architecture and Canon documents;
 - resolve inconsistencies and terminology drift;
 - establish institutional editorial governance;
 - translate principles into implementable specifications;
 - create a controlled and auditable documentation baseline.
-

2. Documentation Authority

Conflicts should be resolved according to this order:



Where software conflicts with an approved higher-level document, the software behaviour should be recorded as an implementation divergence.

3. Controlled Status Model

Controlled documents may be:

- Proposed;
- Draft;
- In Review;
- Revision Required;
- Approved;
- Effective;
- Superseded;
- Withdrawn;
- Archived.

Approval and effectiveness remain separate.

4. Current Architecture Register

The controlled Architecture sequence is:

- WEOS-ARCH-001 - Vision and Editorial Philosophy;
 - WEOS-ARCH-002 - Editorial Domain Model;
 - WEOS-ARCH-003 - Editorial Capabilities;
 - WEOS-ARCH-004 - Knowledge Evolution and Editorial Lifecycles;
 - WEOS-ARCH-005 - Knowledge Production and Editorial Intelligence;
 - WEOS-ARCH-006 - Editorial Intelligence and Decision Architecture;
 - WEOS-ARCH-007 - Editorial Workspace Architecture;
 - WEOS-ARCH-008 - Institutional Editorial Governance;
 - WEOS-ARCH-009 - Evolution of the Editorial Knowledge Ecosystem.
-

5. Current Canon Register

The controlled Canon sequence is:

- WEOS-CANON-001 - Canon Foundations;
- WEOS-CANON-002 - Diagnosis Governance;
- WEOS-CANON-003 - Clinical Case Standards;
- WEOS-CANON-004 - Diagnosis Education Standards;
- WEOS-CANON-005 - Knowledge and Reasoning Governance;
- WEOS-CANON-006 - AI Draft Standards;
- WEOS-CANON-007 - Governance Record Standards.

The Canon sequence is now structurally complete. Parts IV and VII remain draft standards requiring formal review and approval.

6. Required Governance Standards

The following controlled standards should be created:

- Editorial Roles and Authority Standard;
 - Evidence and Reference Governance Standard;
 - Publication and Change-Control Standard;
 - Editorial Conflict, Appeal, and Adjudication Standard;
 - Editorial Portfolio Operations Standard;
 - Editorial Quality Measurement Standard;
 - Maintenance and Revalidation Standard;
 - Documentation Control Standard.
-

7. Required Implementation Specifications

The following implementation specifications should translate Canon requirements into executable behaviour:

- Canonical Artifact Catalogue;
 - Lifecycle Transition Specification;
 - Editorial Action and Decision Catalogue;
 - Roles and Permission Matrix;
 - Readiness and Publication Blocker Specification;
 - Versioning, Supersession, and Concurrency Specification;
 - Audit Event and Governance History Specification;
 - API and Integration Contract;
 - Migration and Backfill Plan;
 - Acceptance and Conformance Test Catalogue;
 - Current-to-Canonical Implementation Mapping.
-

8. Priority Terminology Decisions

The documentation baseline should adopt:

- Learning Goal as the canonical term, with teaching objective retained as a legacy synonym;
- Educational Framework as the conceptual umbrella;
- Editorial Brief as the governed diagnosis-level specification of intent;
- Educational Scaffold as the approved production foundation;
- acceptance of an AI Draft as distinct from controlled application;

- approval as applicable to the resulting governed artifact version;
 - readiness as a computed and explainable assessment;
 - knowledge relationship as a category containing artifact-specific relationship types.
-

9. Delivery Workstreams

9.1 Structural Completion

Complete missing documents, authoritative names, manifest, and sequence.

9.2 Semantic Alignment

Create glossary, terminology crosswalk, lifecycle crosswalk, contradiction log, and decision register.

9.3 Institutional Governance

Define authority, evidence, publication, conflict management, and adjudication.

9.4 Operational Governance

Define assignments, prioritisation, maintenance, service expectations, and quality measurement.

9.5 Implementation Translation

Define artifacts, transitions, permissions, blockers, events, migrations, and tests.

9.6 Baseline Verification

Complete traceability, divergence review, conformance review, and baseline approval.

10. Completion Gates

Gate 1 - Controlled Source Set

Every existing document has one authoritative identity, version, status, and source file.

Gate 2 - Structural Completeness

All nine Architecture and seven Canon parts exist.

Gate 3 - Semantic Consistency

Material terms and lifecycle concepts possess approved canonical meanings.

Gate 4 - Institutional Governability

Every high-impact action has defined authority, review, evidence, publication, and exception controls.

Gate 5 - Implementation Readiness

Critical normative requirements map to system behaviour and tests.

Gate 6 - Baseline Approval

The complete set receives documented approval and becomes effective.

11. Definition of Completeness

WEOS documentation will be complete when:

- all Architecture and Canon parts are approved;
- every governed artifact has a purpose, identity, lifecycle, and standard;
- every material transition has preconditions and authority;
- every decision has a governance record;
- evidence and publication requirements are defined;
- terminology conflicts are resolved;
- readiness and blockers are explainable;
- every critical requirement is traceable to implementation and testing;
- implementation divergences are recorded;
- the full set has baseline approval.

Completeness does not mean every future capability has been designed.

It means that current and planned work can evolve through a coherent, controlled foundation.

12. Immediate Next Actions

- Review and approve Canon Part IV - Diagnosis Education Standards.
 - Create the authoritative glossary and lifecycle crosswalk.
 - Draft the Evidence and Reference Governance Standard.
 - Draft the Publication and Change-Control Standard.
 - Create the Canonical Artifact Catalogue.
 - Create lifecycle transition and permission matrices.
 - Build the requirements traceability matrix.
 - Map current implementation divergences.
-

13. Document Control

Every controlled document should preserve:

- document ID;
- title;
- version;
- status;
- owner;

- approvers;
- effective date;
- review date;
- change history;
- supersession relationship;
- implementation alignment status.

Material changes require formal review and version increment.

Foundational changes require baseline-level approval.

Closing Principle

The Architecture explains what WEOS is.

The Canon defines what its artifacts mean.

Governance Standards define who may make decisions.

Operational Standards define how work is performed.

Implementation Specifications define how rules are enforced.

Conformance Tests prove that implementation respects the Canon.

Document control preserves that authority over time.